Cloudera on premises / CDP Private Cloud (PvC)
Installation & Setup

:: Openshift Setup Guide ::

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In partnership with:

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install-config.yaml
agent-config.yaml
Create VMs for OpenShift Cluster (OpenShift Virtualization)
pvcocp-master01.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-master02.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-master03.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker01.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker02.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker03.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker04.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker05.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker06.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker07.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker08.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker09.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-worker10.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-vms.yaml
pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-vms.yaml
NTP

ALTERNATIVE CONFIGURATION STEPS
OPTIONAL Method to create VMs
ADD NODES TO THE CLUSTER
BADFISH
NodeNetworkConfigurationPolicy
OPTIONAL METHOD TO CREATE VM YAML

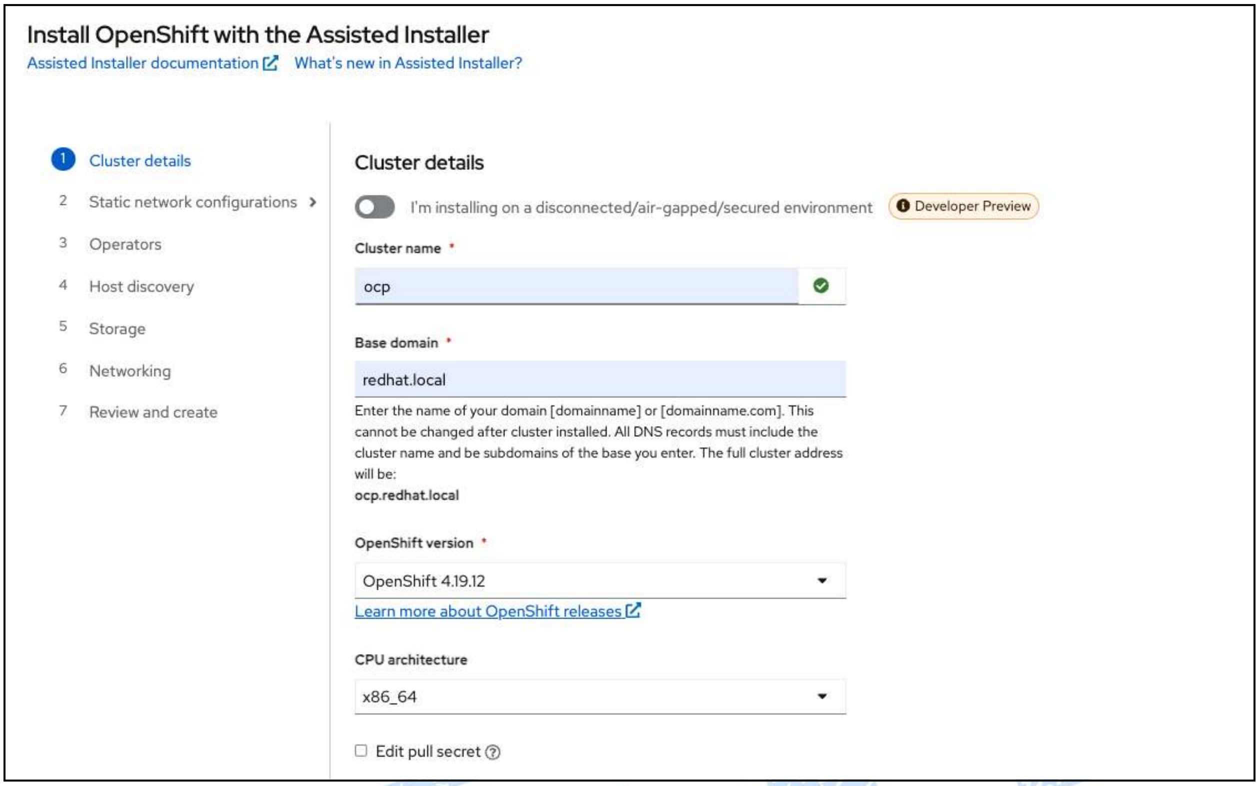
INSTALL/CONFIGURE OPENSHIFT

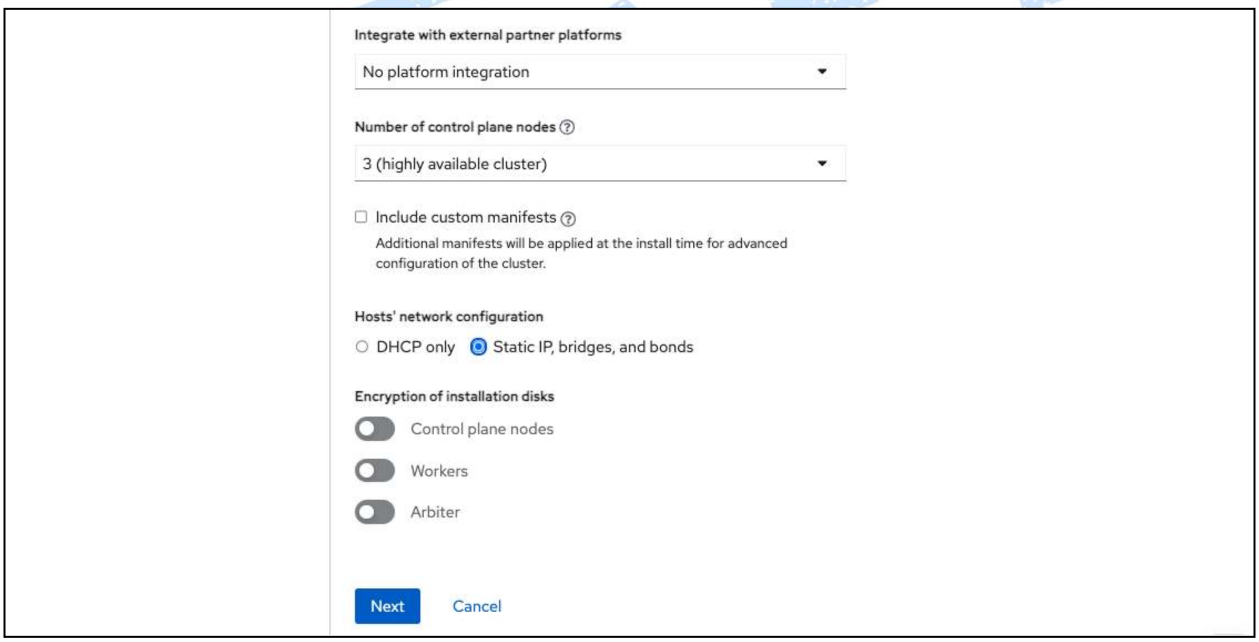
Install OpenShift using the Assisted Installer

Installation parameters:

Parameter	Setting/Value
Cluster name	ocp
Base domain	redhat.local
OpenShift version	4.19.12
CPU architecture	x86_64
Integrate with external partner platforms	No platform integration
Number of control plane nodes	3 (highly available cluster)
Hosts' network configuration	Static IP, bridges, and bonds
Configure via	YAML view
Networking stack type	IPv4
DNS	192.168.1.210
Machine network	192.168.2.0/24
Default gateway	192.168.2.1
api.ocp.redhat.local (API IP)	192.168.2.183
*.apps.ocp.redhat.local (Ingress IP)	192.168.2.184

STEP 1. Create a new OpenShift cluster using the OpenShift Assisted Installer from the Red Hat Hybrid Cloud Console:
https://console.redhat.com/openshift/assisted-installer/clusters/~new
Enter the cluster details using the installation parameters provided, then click ***Next***.


Install OpenShift with the Assisted Installer
Assisted Installer documentation
What's new in Assisted Installer?
1 Cluster details
2 Static network configurations
3 Operators
4 Host discovery
5 Storage
6 Networking
7 Review and create
Cluster details
I'm installing on a disconnected/air-gapped/secured environment
Developer Preview
Cluster name *
ocp
Base domain *
redhat.local
Enter the name of your domain [domainname] or [domainname.com]. This cannot be changed after cluster installed. All DNS records must include the cluster name and be subdomains of the base you enter. The full cluster address will be:
ocp.redhat.local
OpenShift version *
OpenShift 4.19.12
Learn more about OpenShift releases
CPU architecture
x86_64
Edit pull secret


Integrate with external partner platforms
No platform integration
Number of control plane nodes
3 (highly available cluster)
Include custom manifests
Additional manifests will be applied at the install time for advanced configuration of the cluster.
Hosts' network configuration
DHCP only
Static IP, bridges, and bonds
Encryption of installation disks
Control plane nodes
Workers
Arbiter
Next
Cancel

NETWORK CONFIGURATION NOTE:

The three host servers used here have a dual-port 10Gb network adapter. Two YAML NMState configuration examples are provided to utilize both network adapter ports using a network bond:

- ***Balance-rr aggregate mode:*** both network adapters/ports are used simultaneously for network traffic. This does require a static EtherChannel to be configured/enabled on the Ethernet switch (not LACP-negotiated).

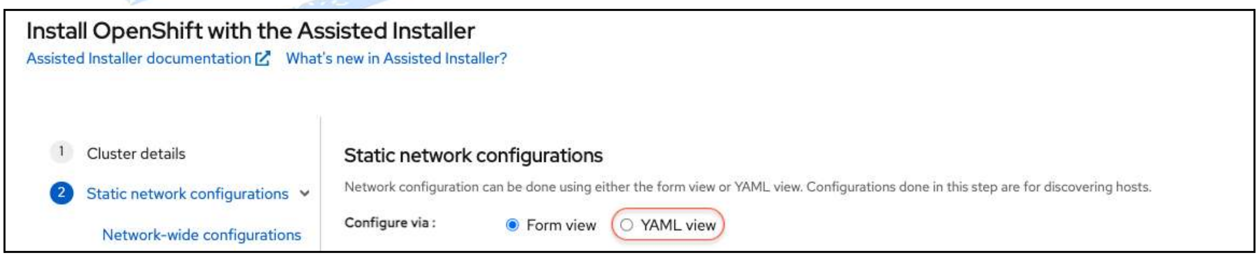
- ***Active-Backup fail-over mode:*** only one network adapter/port will be used at a time for network traffic. This does not require any special network switch configuration.

In my environment, I will be using the **Balance-rr** network bonding mode, and I have configured static EtherChannels for the two ports for each server on my Ethernet switch. If you do not have an Ethernet switch that is capable, you may use the **Active-Backup** network bonding mode, as it does not require any special network switch configuration.

You may also optionally configure OpenShift to use only a single network adapter/port without using a network bond. In this case, it is also possible to assign the second adapter/port for selected traffic types, such as only for storage access, only for use by VMs, or only for VM live migration traffic.

Both configurations provided also set the network mtu to 9000, which requires that jumbo Ethernet frames be enabled on the Ethernet switch for the server network ports in use.

STEP 2. Select the button to configure via the ***YAML view***.



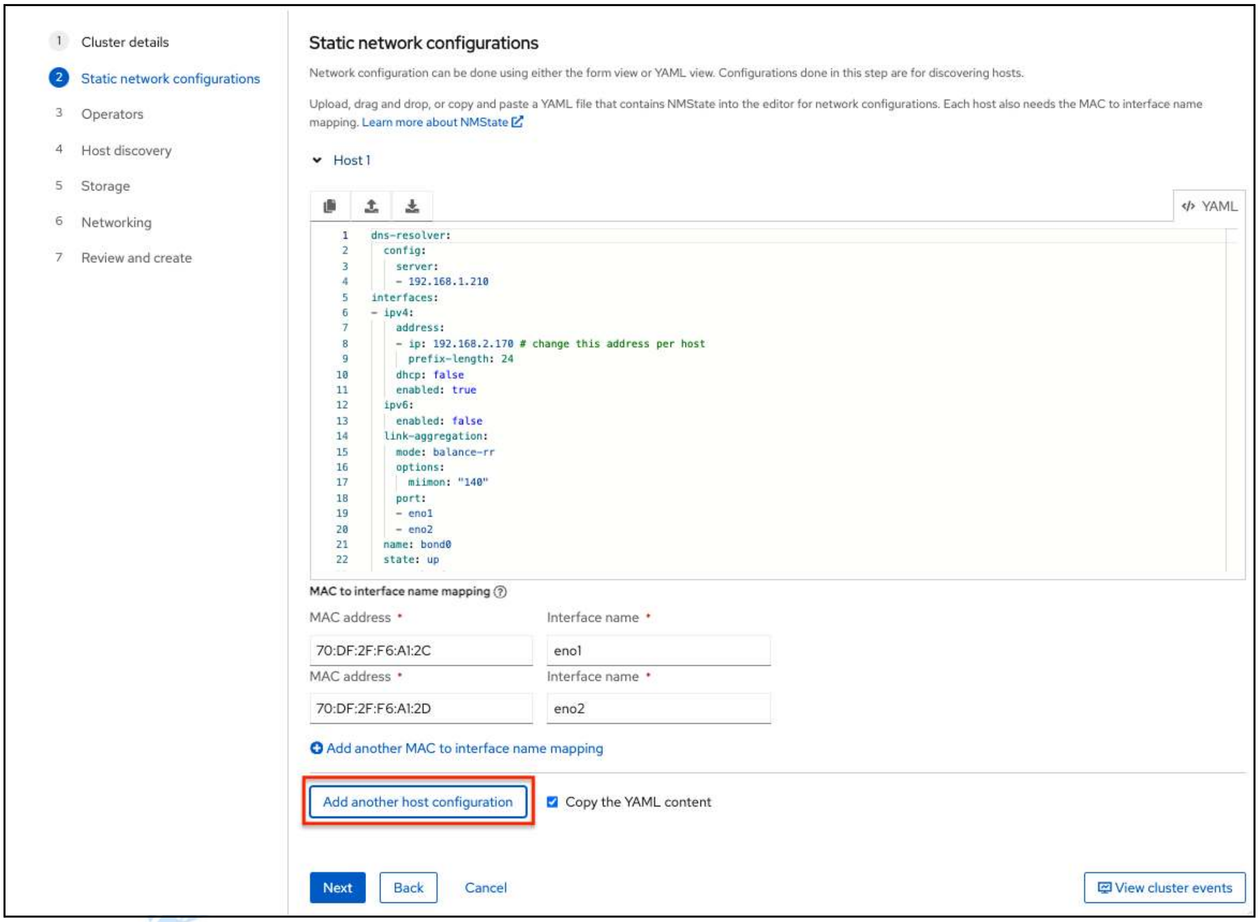
STEP 3. <u>**Enter the static network configuration for Host 1**</u> using the following YAML code example along with the information provided in the table.

NOTE: You will need to provide the MAC addresses for your server's network adapters, as they will be different from those provided.

From the YAML view, click on "***Start from scratch***" and then copy/paste the YAML NMState configuration provided below, while editing the IP address to use for the host.
Enter the MAC address and Interface name for the first network adapter in the host.

Click on “***Add another MAC to interface name mapping***” and then enter the MAC address and Interface name for the second network adapter in the host.

When you have finished entering the static network configuration for Host 1, select the check box for “***Copy the YAML content***” and click on “***Add another host configuration***”.



Parameter	Setting/Value
Host 1 (c240m4-01.redhat.local)	
IP address (IPv4)	192.168.2.170
MAC Address: 70:DF:2F:F6:A1:2C	**Interface name:** eno1
MAC Address: 70:DF:2F:F6:A1:2D	**Interface name:** eno2
Host 2 (c240m4-02.redhat.local)	
IP address (IPv4)	192.168.2.171

MAC Address: 70:DF:2F:F6:A2:2C	**Interface name:** eno1
MAC Address: 70:DF:2F:F6:A2:2D	**Interface name:** eno2
Host 3 (C240m4-03.redhat.local)	
IP address (IPv4)	192.168.2.172
MAC Address: 70:DF:2F:F6:A3:2C	**Interface name:** eno1
MAC Address: 70:DF:2F:F6:A3:2D	**Interface name:** eno2

The following NMState configuration examples are provided for reference only.

These are also in the OpenShift Assisted Installer documentation: [10.4.1. NMState configuration]() and [10.4.3. Additional NMState configuration examples]().

YAML NMState configuration using Balance-rr network bonding mode 0.

```
dns-resolver:
  config:
    server:
    - 192.168.1.210
interfaces:
- ipv4:
    address:
    - ip: 192.168.2.170 # change this address per host
      prefix-length: 24
    dhcp: false
    enabled: true
  ipv6:
    enabled: false
  link-aggregation:
    mode: balance-rr
    options:
      miimon: "140"
    port:
    - eno1
    - eno2
  name: bond0
  state: up
  type: bond
  mtu: 9000
routes:
```

```
  config:
  - destination: 0.0.0.0/0
    next-hop-address: 192.168.2.1
    next-hop-interface: bond0
    table-id: 254
```

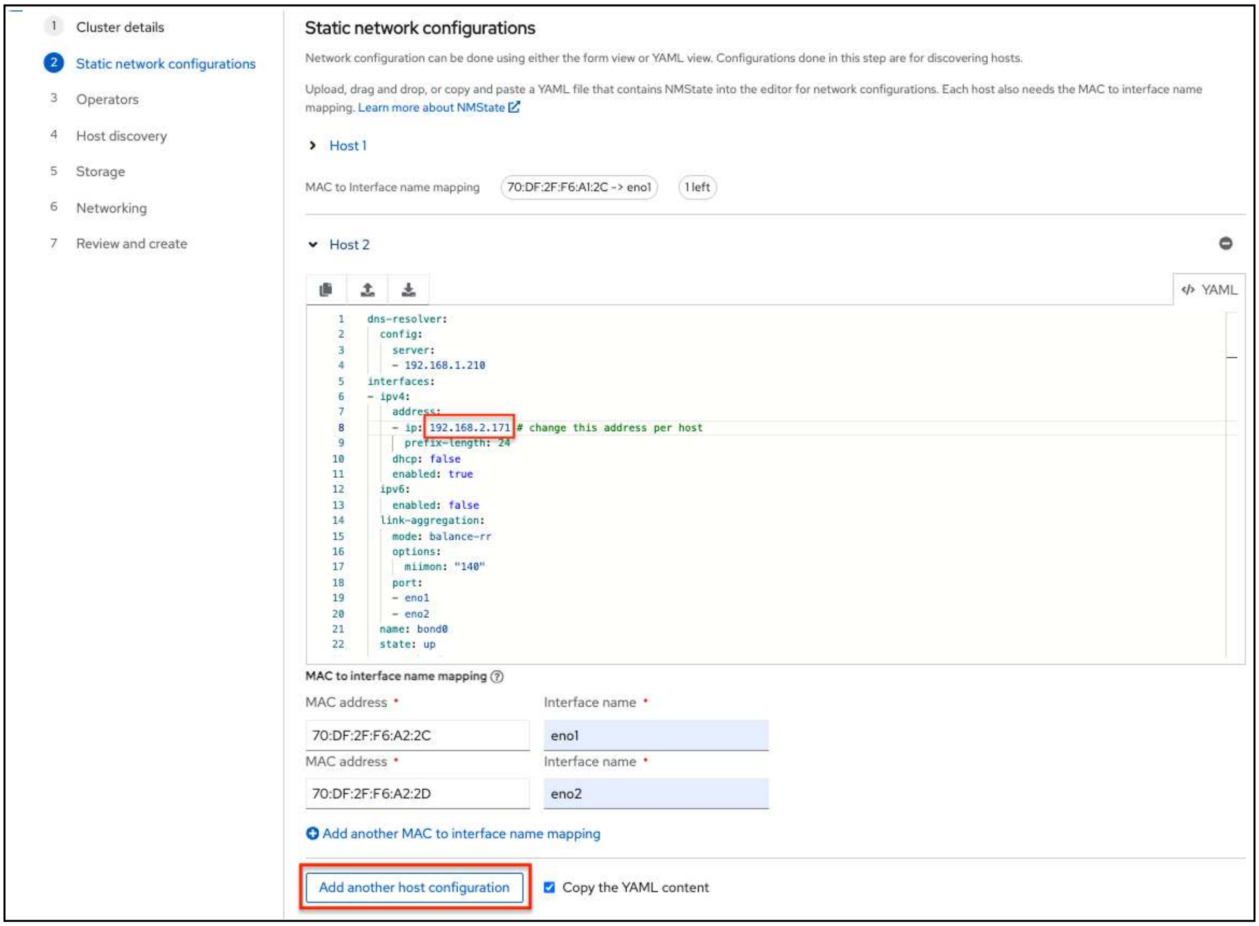
YAML NMState configuration using Active-Backup network bonding mode 1

```
dns-resolver:
  config:
    server:
    - 192.168.1.210
interfaces:
- ipv4:
    address:
    - ip: 192.168.2.170 # change this address per host
      prefix-length: 24
    dhcp: false
    enabled: true
  ipv6:
    enabled: false
  link-aggregation:
    mode: active-backup
    options:
      miimon: "140"
    port:
    - eno1
    - eno2
  name: bond0
  state: up
  type: bond
  mtu: 9000
routes:
  config:
  - destination: 0.0.0.0/0
    next-hop-address: 192.168.2.1
    next-hop-interface: bond0
    table-id: 254
```

STEP 4. **Enter the static network configuration for Host 2** using the following.

Edit the IP address, and enter the MAC address and Interface name for the first network adapter in the host.
Click on “***Add another MAC to interface name mapping***” and then enter the MAC address and Interface name for the second network adapter in the host.

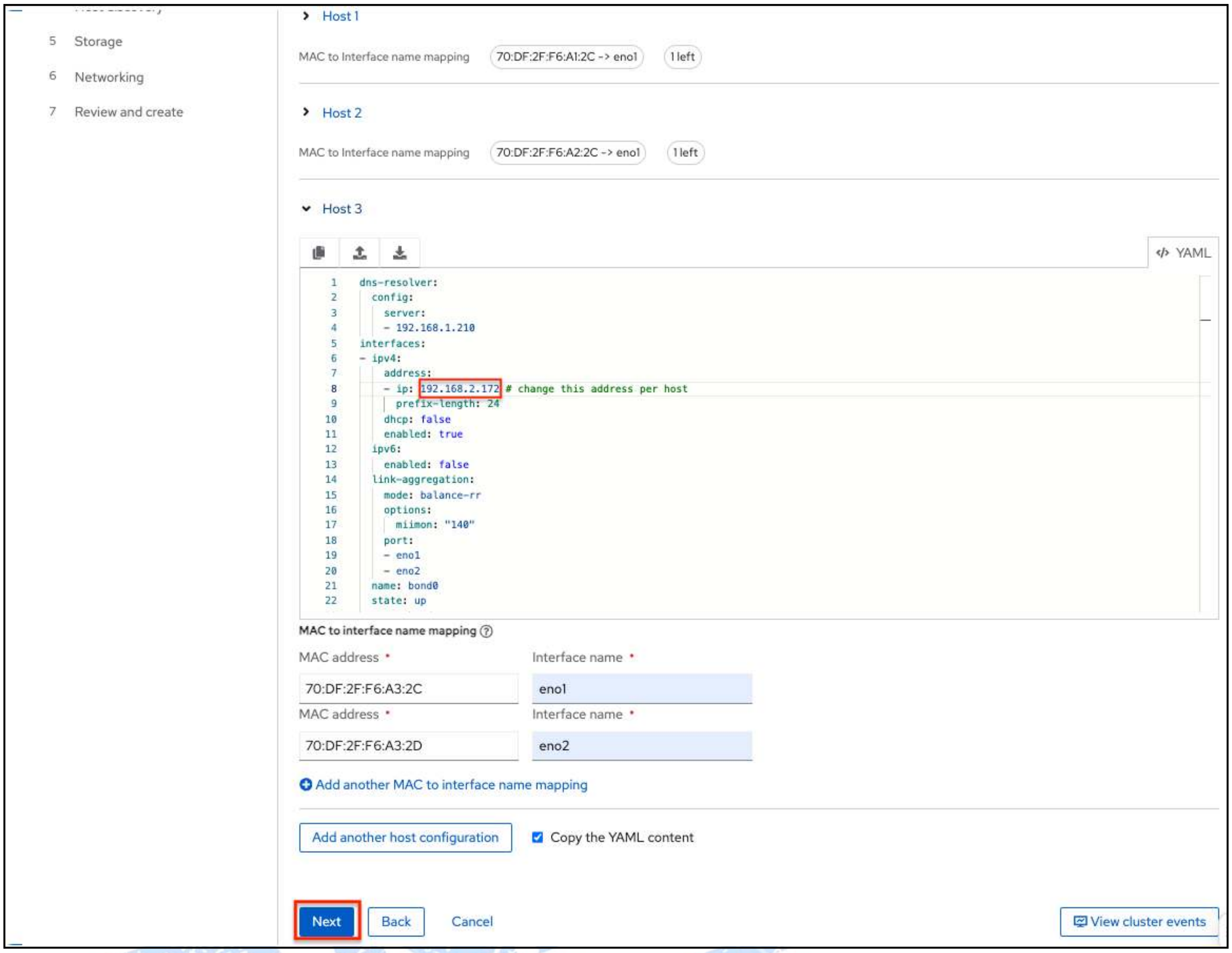
When you have finished entering the static network configuration for Host 2, select the check box for “***Copy the YAML content***” and click on “***Add another host configuration***”.



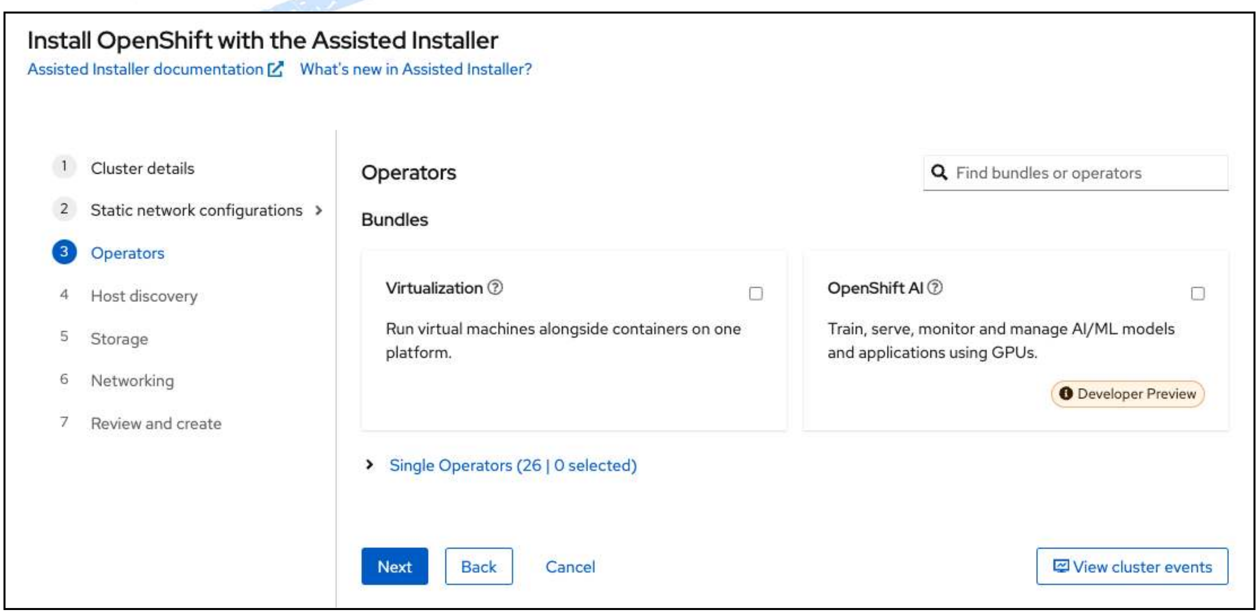
STEP 5. Enter the static network configuration for Host 3.

Enter the MAC address and Interface name for the first network adapter in the host.
Click on “***Add another MAC to interface name mapping***” and then enter the MAC address and Interface name for the second network adapter in the host.

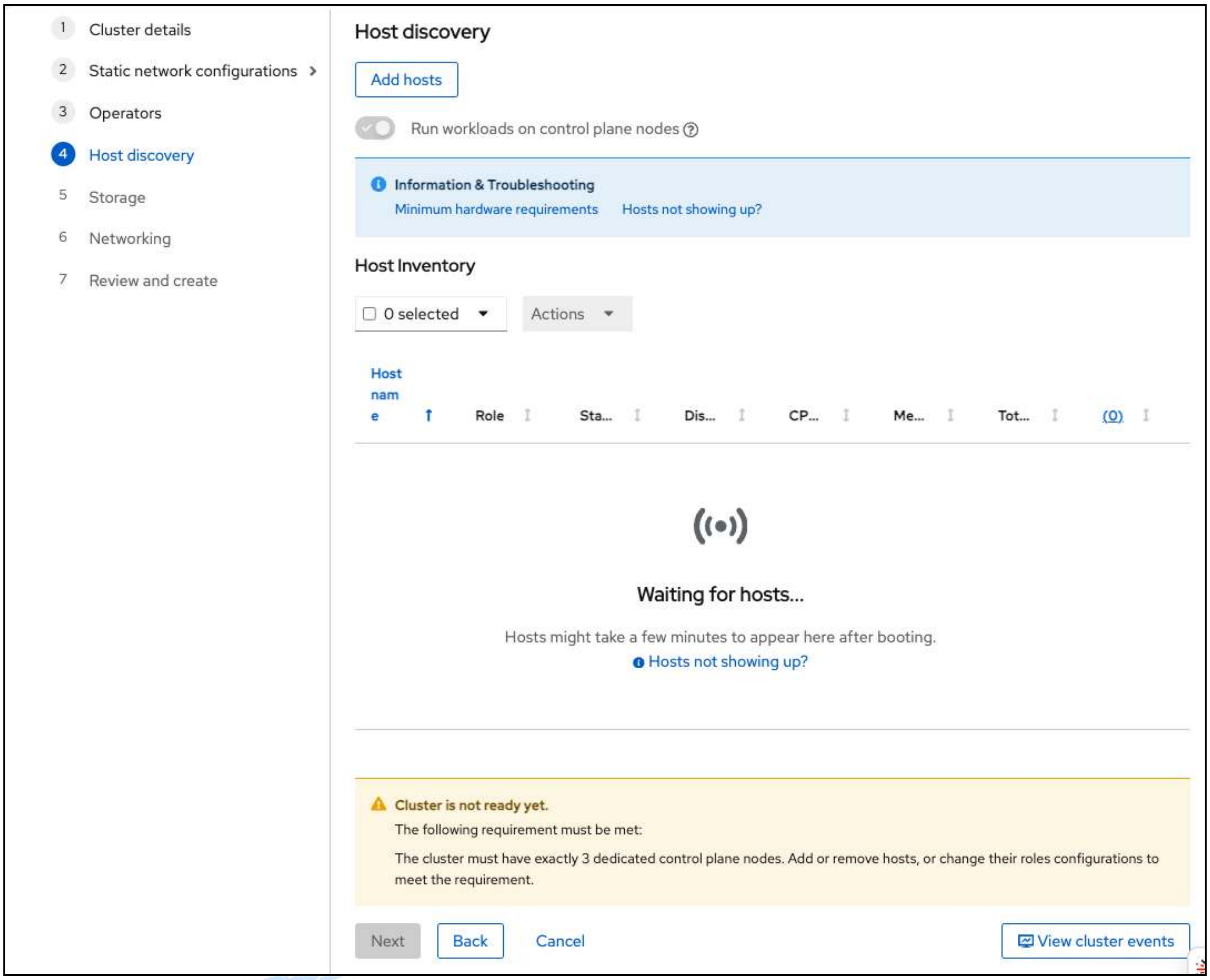
When you have finished entering the static network configuration for Host 3, click “***Next***”.



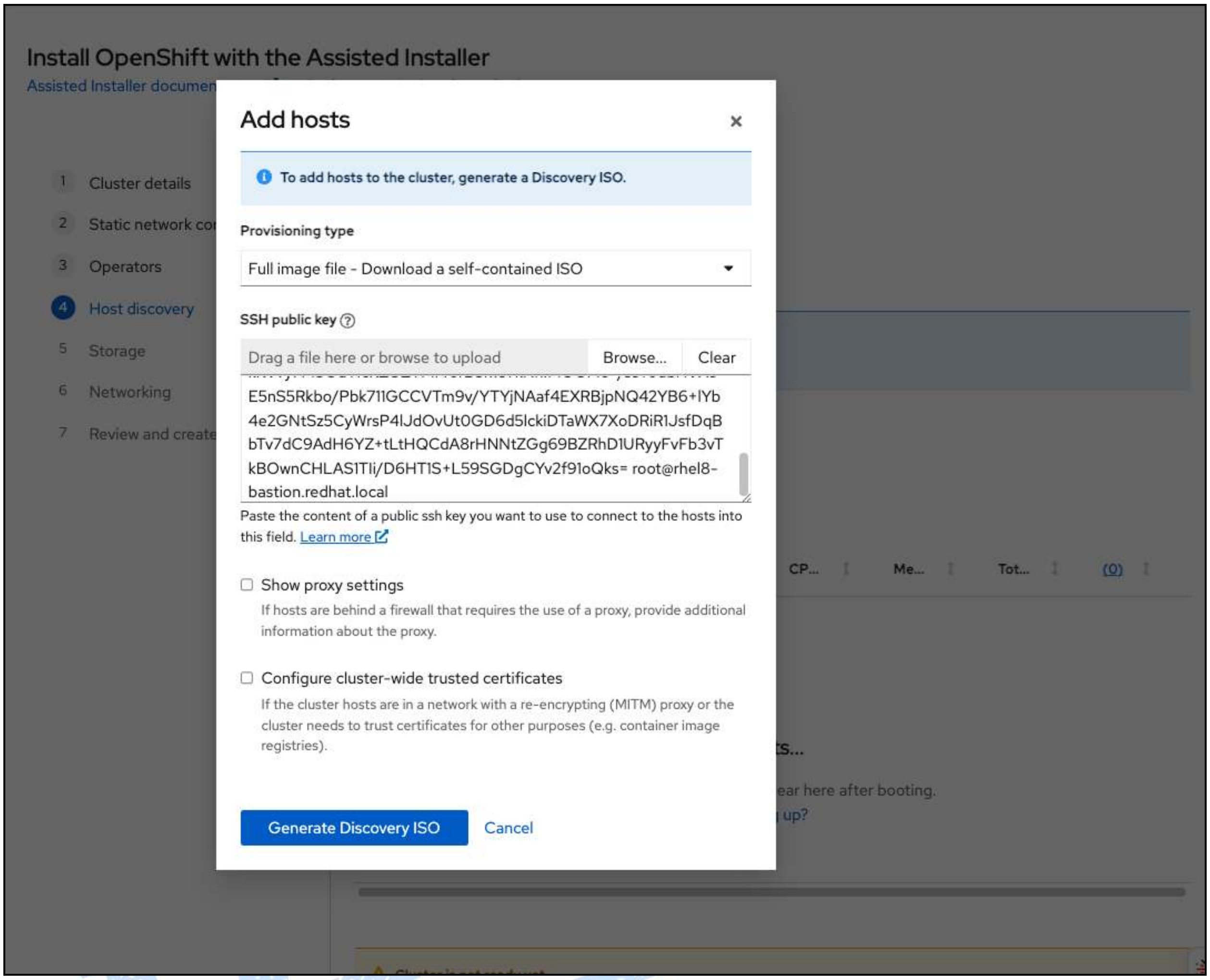
STEP 6. For Operators to install, we will install these separately. So, without selecting any here, click "***Next***".



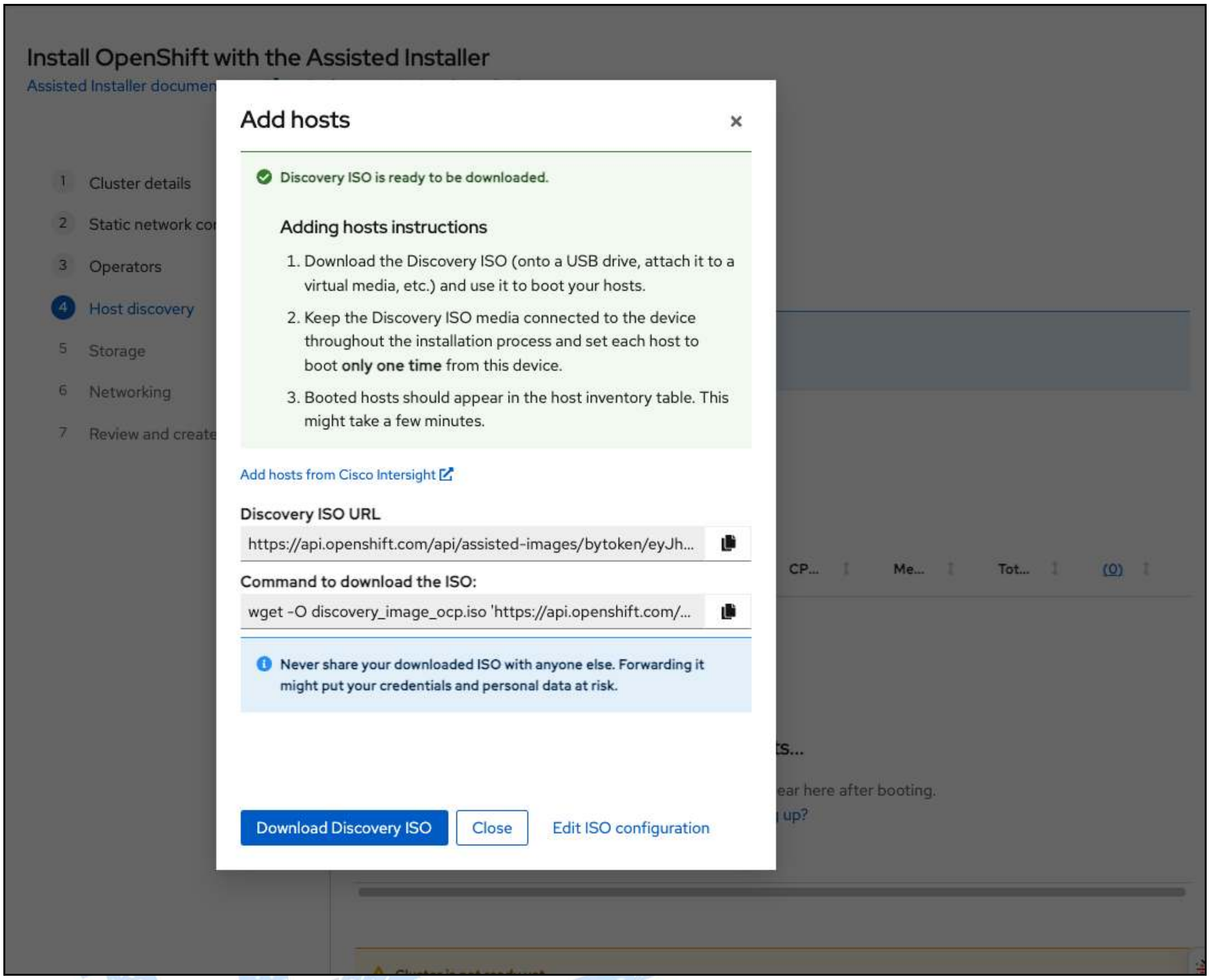
STEP 7. From the Host discovery screen, select "***Add hosts***" to generate a discovery ISO image.



STEP 8. From the "Add hosts" screen, select the ***Full image file***, and paste your SSH public key (from ~/.ssh/id_rsa.pub file), then click on "***Generate Discovery ISO***".



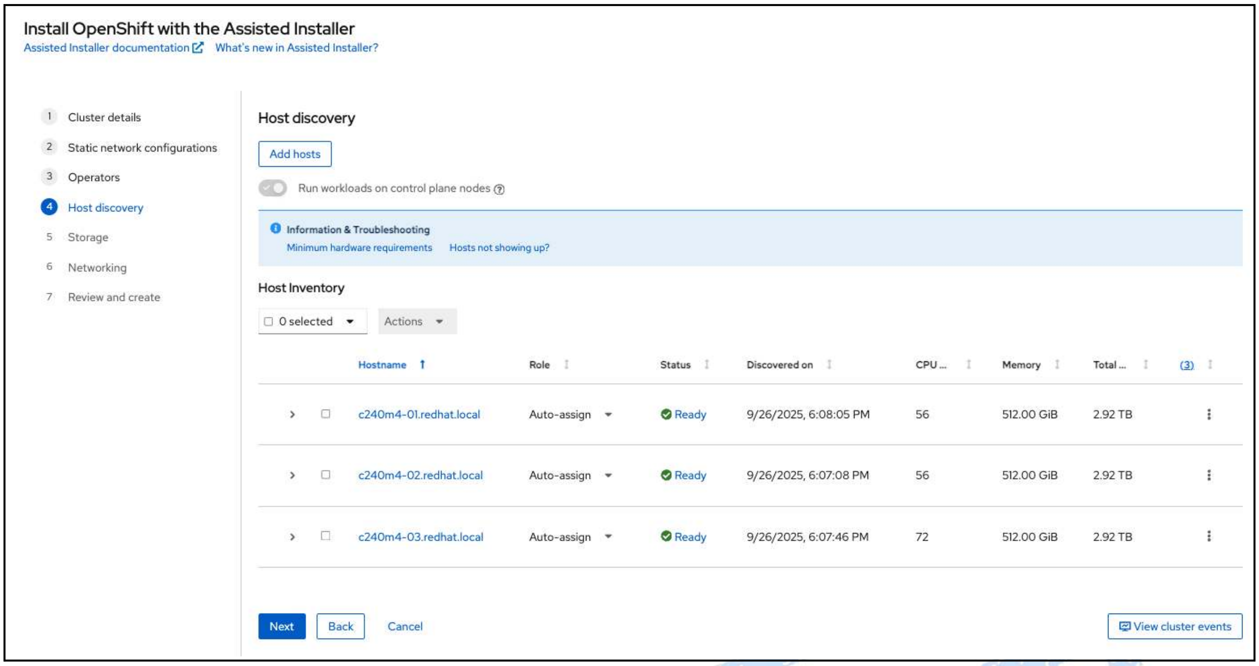
STEP 9. Next, click on the "***Download Discovery ISO***" to download the discovery ISO image. Each server host will boot this image to start the server discovery process. When you have finished downloading the ISO image, click on "***Close***".



STEP 10. Use one of several methods to boot each server host from this ISO image. For my lab, I will be using this image with the Virtual Media function provided by the server's system management processor. You can also write this ISO image to a USB flash drive, or DVD, in order to boot from physical media types.

For my lab, I will be copying this ISO image file to a web server so that each server's system management processor can mount it as virtual media. Each server is then configured to boot from this virtual media as a one-time boot device.

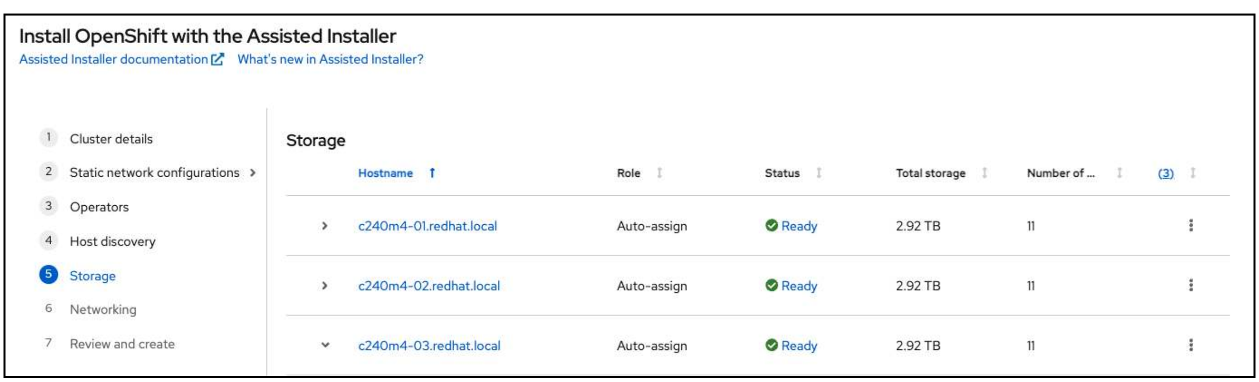
Once all three servers show up on the Host Inventory list, and the cluster is ready, click on "***Next***".

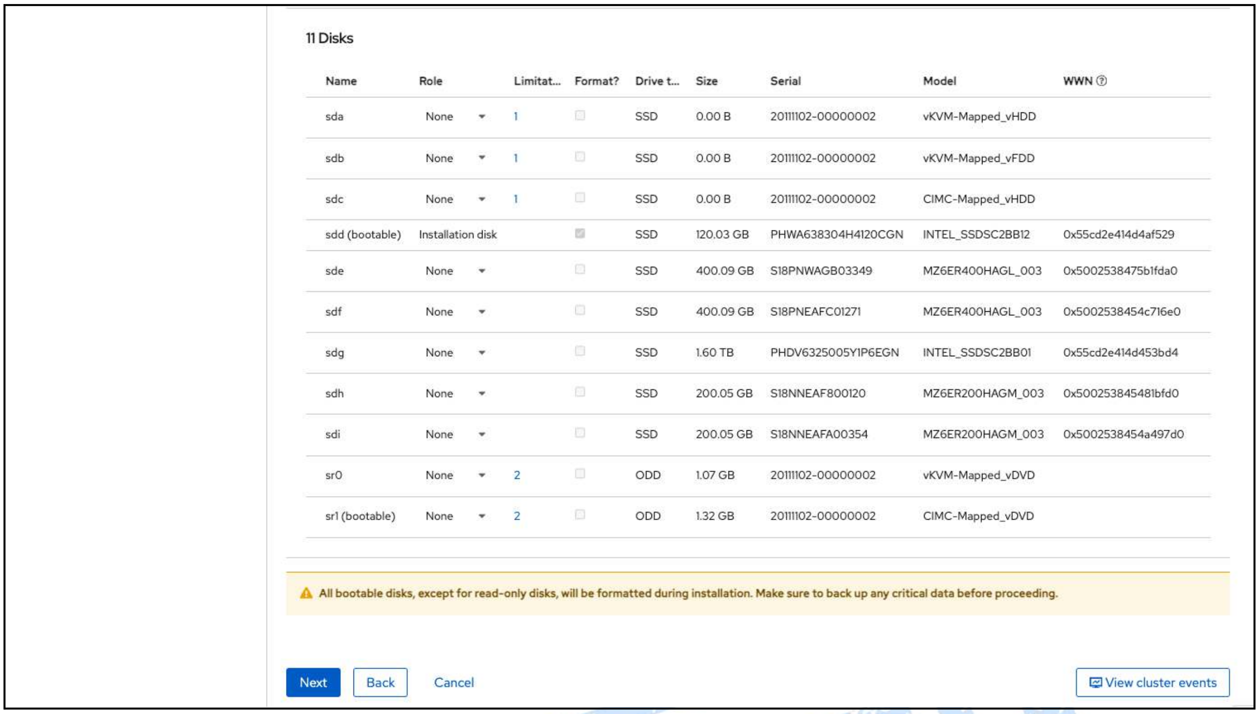


sudo ip link set dev eno1 mtu 9000
sudo ip link set dev eno2 mtu 9000
sudo ip link set dev bond0 mtu 9000
sudo nmcli

sudo sgdisk -Z /dev/sdX
sudo wipefs -a /dev/sdX

STEP 11. From the Storage screen, expand each of the discovered hosts to ensure that the storage devices were discovered properly. Check that the disk that you want to install RHCOS onto is correctly set with a Role of "***installation disk***". If not, use the drop-down selection to change the Role to the correct disk. In the screenshot, the installation disk is correctly set to a 120GB SSD. Click on "***Next***" to continue.





11 Disks

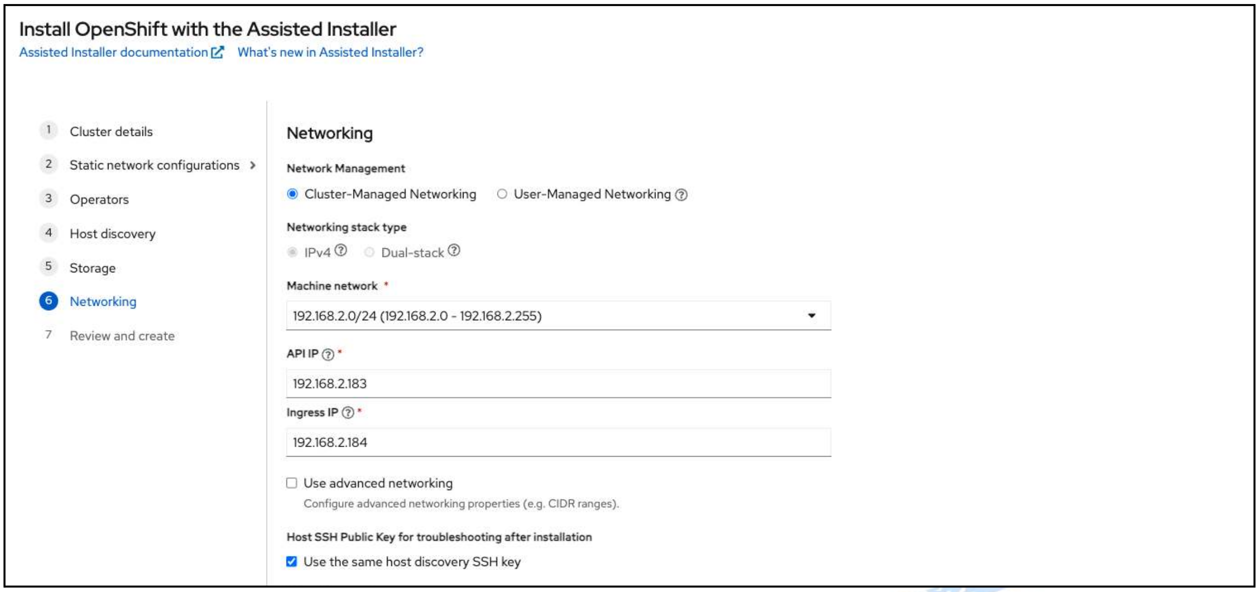
Name	Role	Limitat...	Format?	Drive t...	Size	Serial	Model	WWN
sda	None	1		SSD	0.00 B	20111102-00000002	vKVM-Mapped_vHDD	
sdb	None	1		SSD	0.00 B	20111102-00000002	vKVM-Mapped_vFDD	
sdc	None	1		SSD	0.00 B	20111102-00000002	CIMC-Mapped_vHDD	
sdd (bootable)	Installation disk			SSD	120.03 GB	PHWA638304H4120CGN	INTEL_SSDSC2BB12	0x55cd2e414d4af529
sde	None			SSD	400.09 GB	S18PNWAGB03349	MZ6ER400HAGL_003	0x5002538475b1fda0
sdf	None			SSD	400.09 GB	S18PNEAFC01271	MZ6ER400HAGL_003	0x5002538454c716e0
sdg	None			SSD	1.60 TB	PHDV6325005Y1P6EGN	INTEL_SSDSC2BB01	0x55cd2e414d453bd4
sdh	None			SSD	200.05 GB	S18NNEAF800120	MZ6ER200HAGM_003	0x500253845481bfd0
sdi	None			SSD	200.05 GB	S18NNEAFA00354	MZ6ER200HAGM_003	0x5002538454a497d0
sr0	None	2		ODD	1.07 GB	20111102-00000002	vKVM-Mapped_vDVD	
sr1 (bootable)	None	2		ODD	1.32 GB	20111102-00000002	CIMC-Mapped_vDVD	

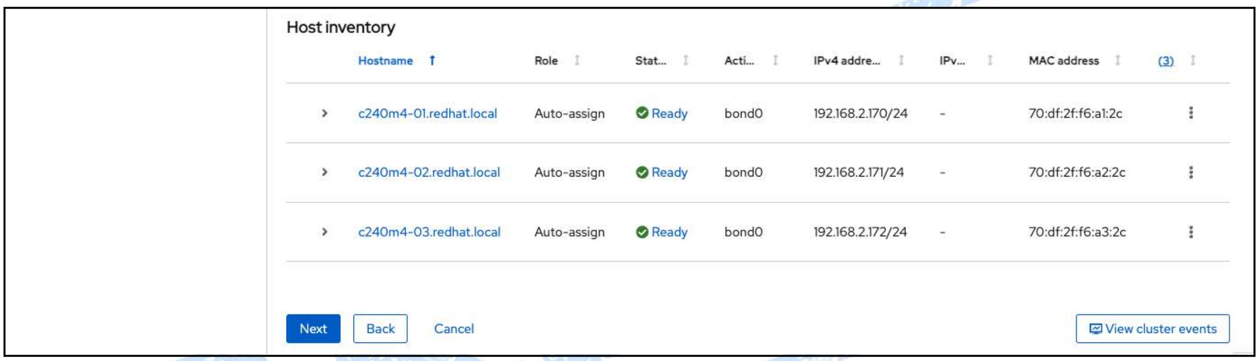
All bootable disks, except for read-only disks, will be formatted during installation. Make sure to back up any critical data before proceeding.

Next Back Cancel View cluster events

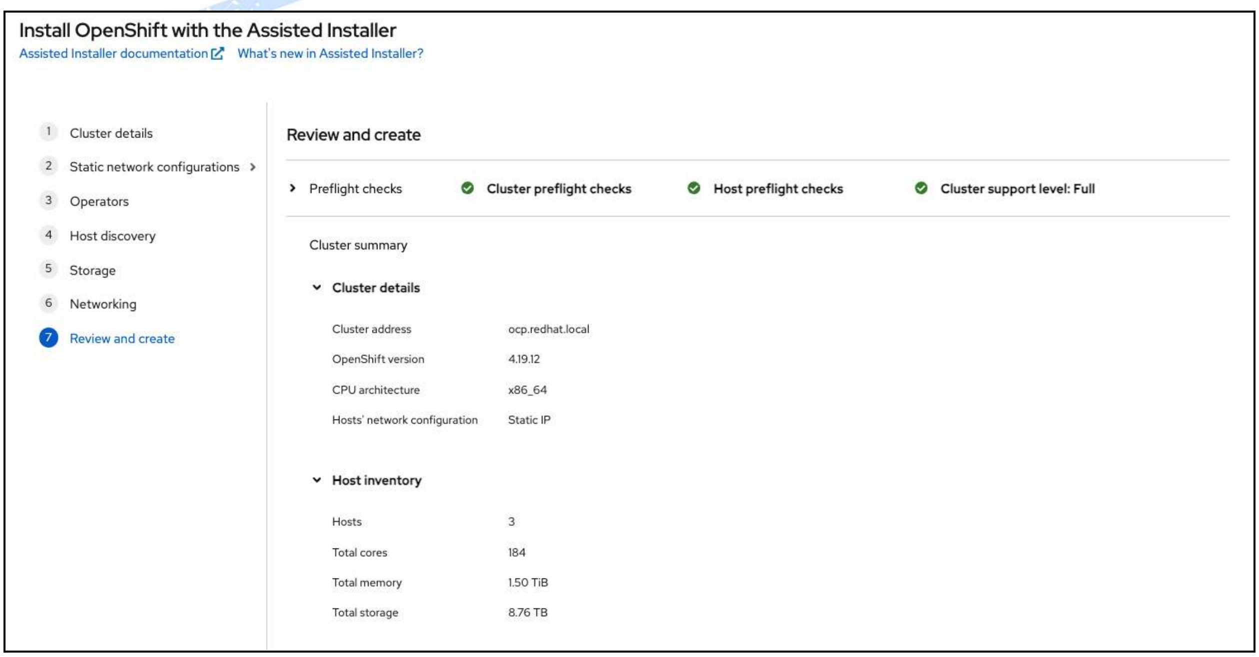
STEP 12. From the Networking screen, enter the following parameters, and click on ***Next*** to continue.

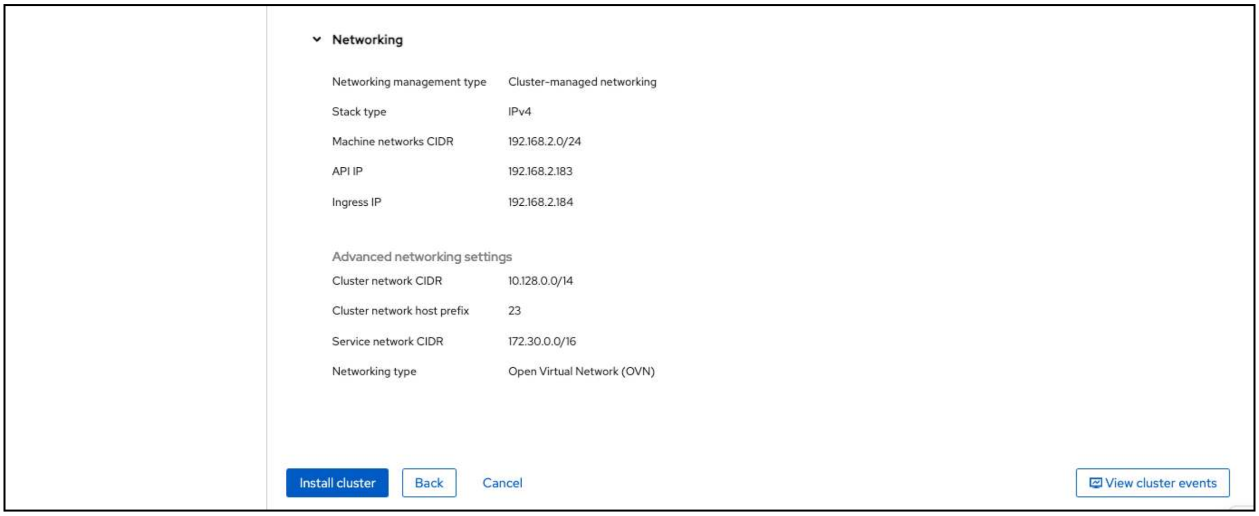
Parameter	Setting/Value
Network Management	Cluster-Managed Networking
Machine Network	192.168.2.0/24 (192.168.2.0 - 192.168.2.255)
API IP	192.168.2.183
Ingress IP	192.168.2.184
Use advanced networking	Not selected
Use the same host discovery SSH key	Selected



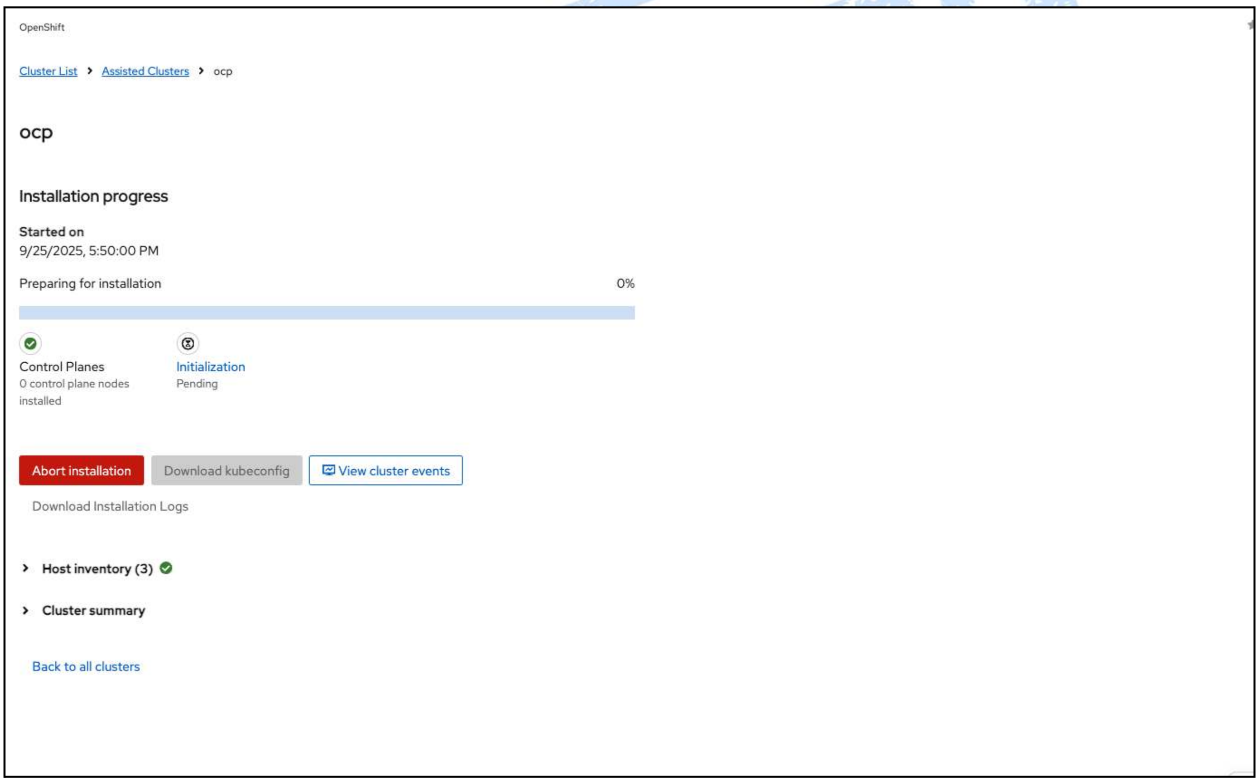


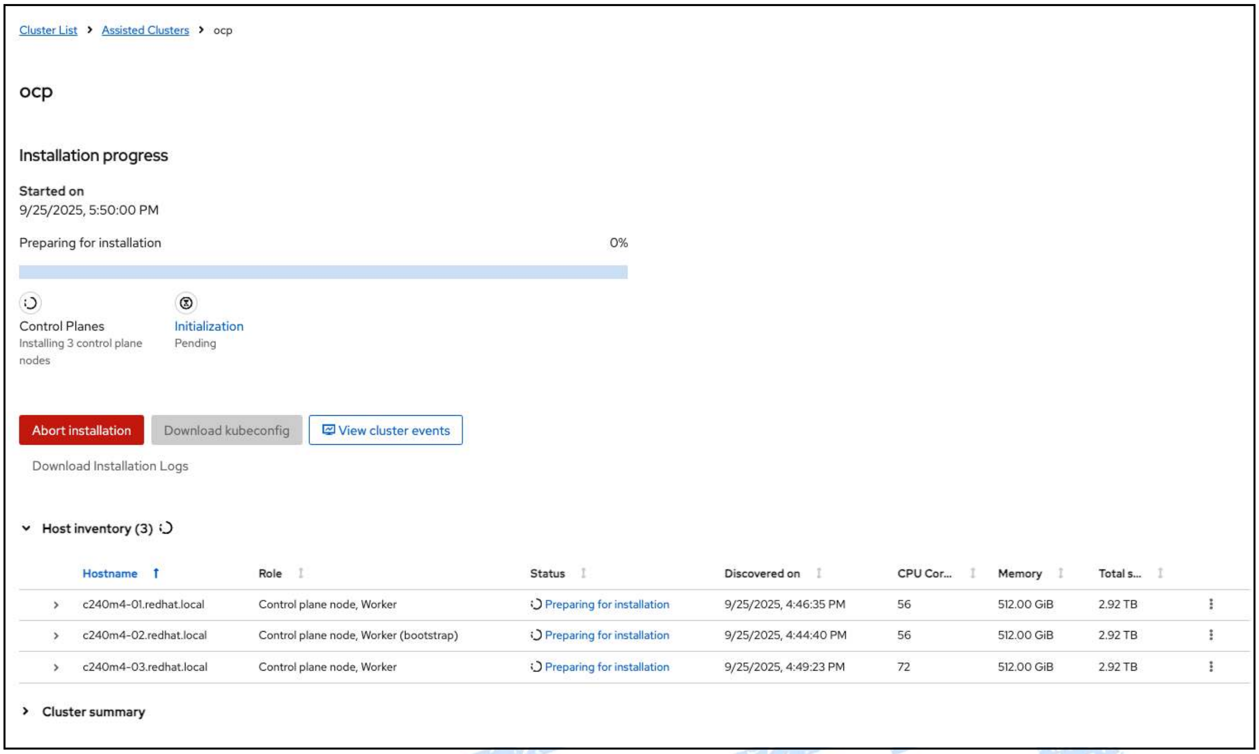
STEP 13. From the Review and create screen, verify the installation parameters and click on "***Install cluster***" to start the installation.

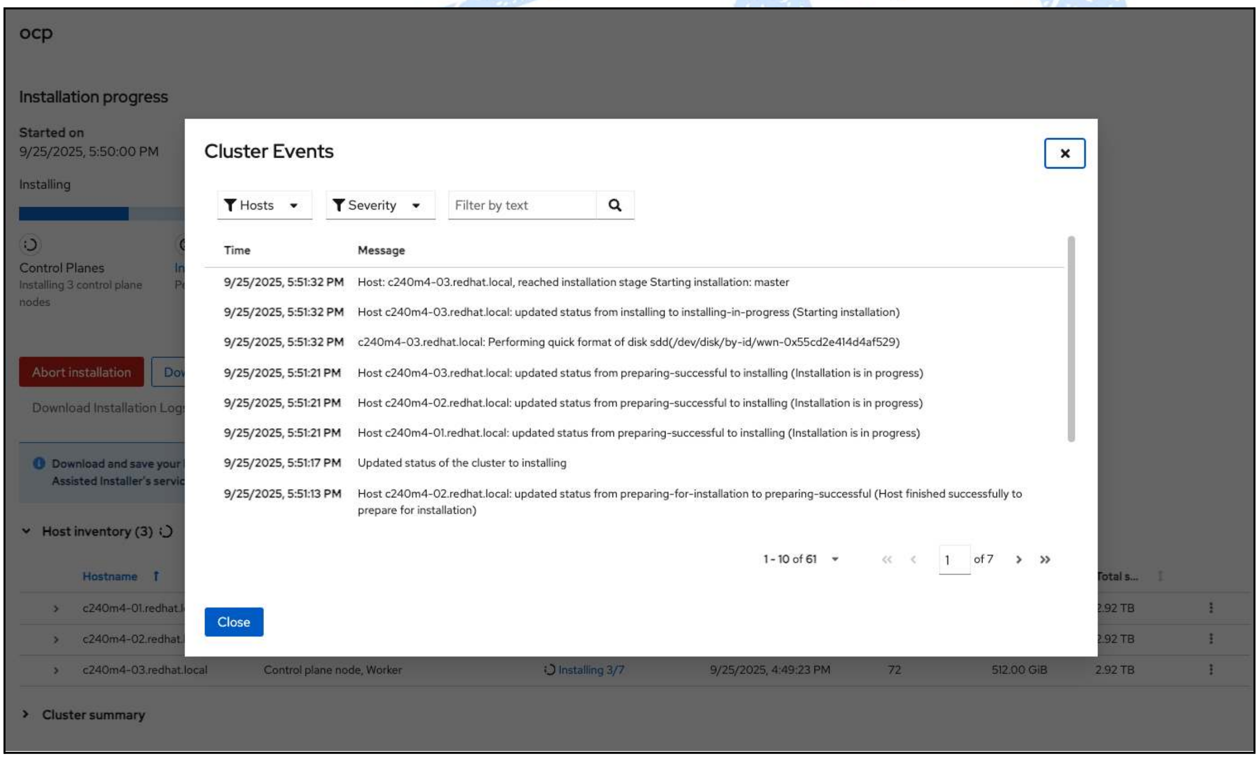




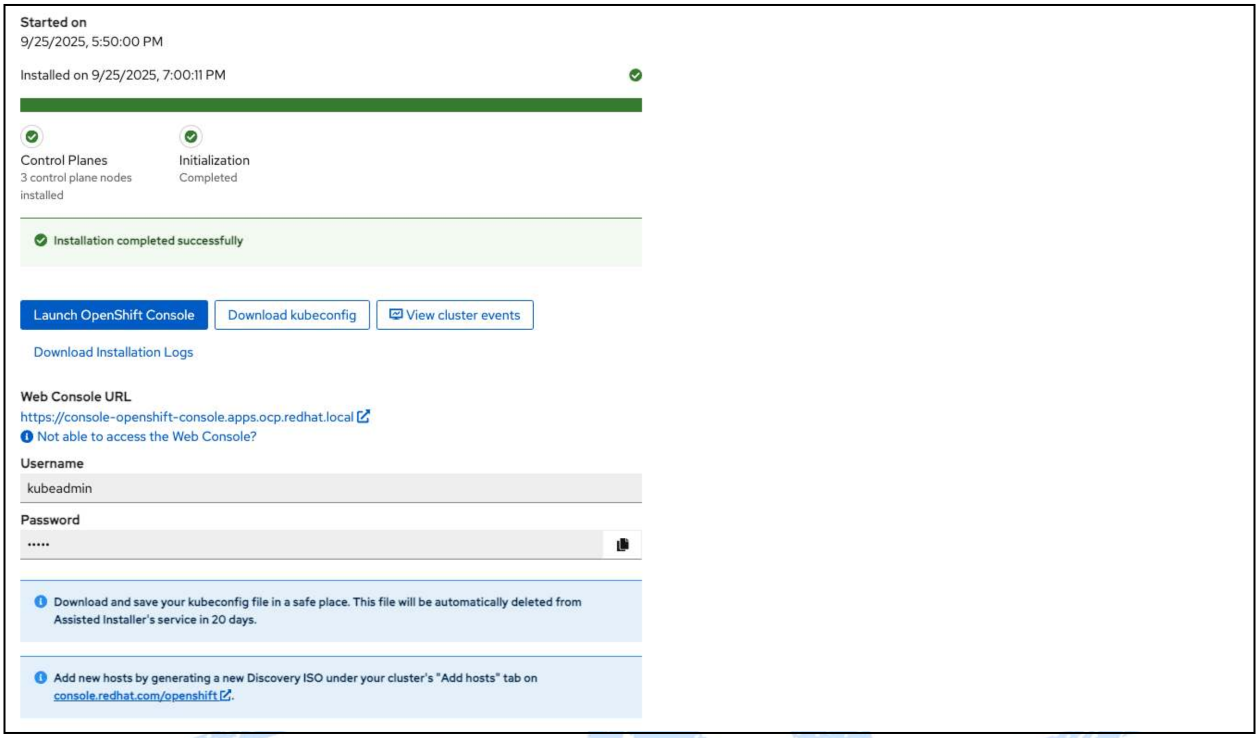
STEP 14. Monitor the progress of the installation. Select "View cluster events" and the "Status" field to the right of each server host.

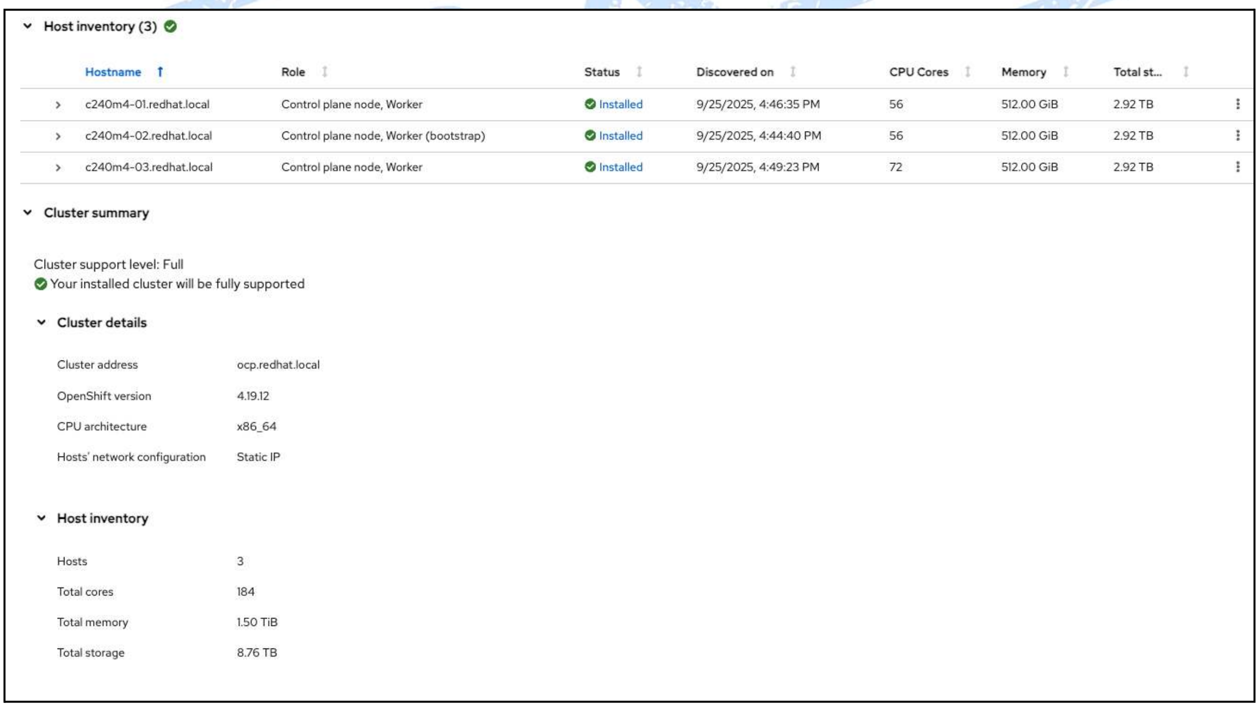


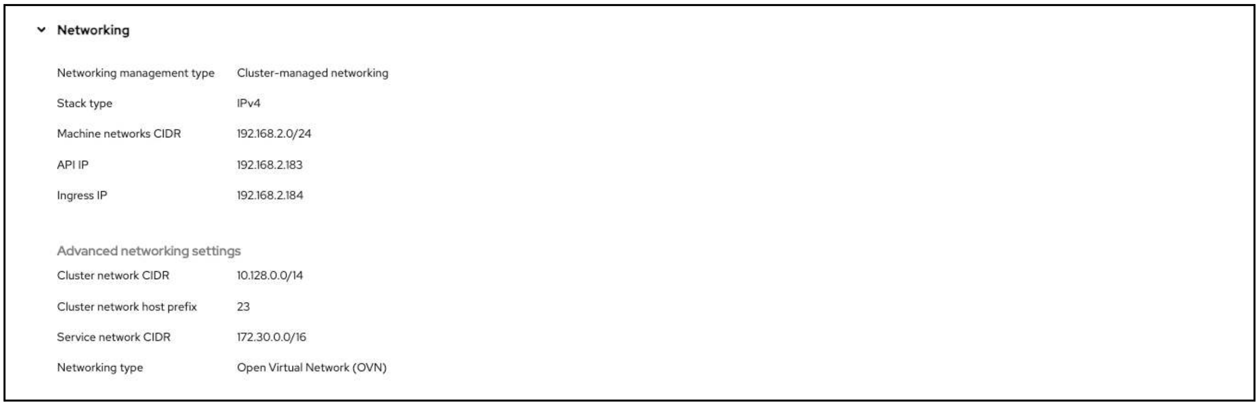

Cluster List > Assisted Clusters > ocp
ocp
Installation progress
Started on
9/25/2025, 5:50:00 PM
Preparing for installation
0%
Control Planes
Installing 3 control plane nodes
Initialization
Pending
Abort installation
Download kubeconfig
View cluster events
Download Installation Logs
Host inventory (3)
Hostname
Role
Status
Discovered on
CPU Cor...
Memory
Total s...
c240m4-01.redhat.local
Control plane node, Worker
Preparing for installation
9/25/2025, 4:46:35 PM
56
512.00 GiB
2.92 TB
c240m4-02.redhat.local
Control plane node, Worker (bootstrap)
Preparing for installation
9/25/2025, 4:44:40 PM
56
512.00 GiB
2.92 TB
c240m4-03.redhat.local
Control plane node, Worker
Preparing for installation
9/25/2025, 4:49:23 PM
72
512.00 GiB
2.92 TB
Cluster summary


ocp
Installation progress
Started on
9/25/2025, 5:50:00 PM
Installing
Control Planes
Installing 3 control plane nodes
Abort installation
Download Installation Logs
Host inventory (3)
Hostname
c240m4-01.redhat.l
c240m4-02.redhat.
c240m4-03.redhat.local
Control plane node, Worker
Installing 3/7
9/25/2025, 4:49:23 PM
72
512.00 GiB
2.92 TB
Cluster summary
Cluster Events
Hosts
Severity
Filter by text
Time
Message
9/25/2025, 5:51:32 PM
Host: c240m4-03.redhat.local, reached installation stage Starting installation: master
9/25/2025, 5:51:32 PM
Host c240m4-03.redhat.local: updated status from installing to installing-in-progress (Starting installation)
9/25/2025, 5:51:32 PM
c240m4-03.redhat.local: Performing quick format of disk sdd(/dev/disk/by-id/wwn-0x55cd2e414d4af529)
9/25/2025, 5:51:21 PM
Host c240m4-03.redhat.local: updated status from preparing-successful to installing (Installation is in progress)
9/25/2025, 5:51:21 PM
Host c240m4-02.redhat.local: updated status from preparing-successful to installing (Installation is in progress)
9/25/2025, 5:51:21 PM
Host c240m4-01.redhat.local: updated status from preparing-successful to installing (Installation is in progress)
9/25/2025, 5:51:17 PM
Updated status of the cluster to installing
9/25/2025, 5:51:13 PM
Host c240m4-02.redhat.local: updated status from preparing-for-installation to preparing-successful (Host finished successfully to prepare for installation)
1 - 10 of 61
1
of 7
Close

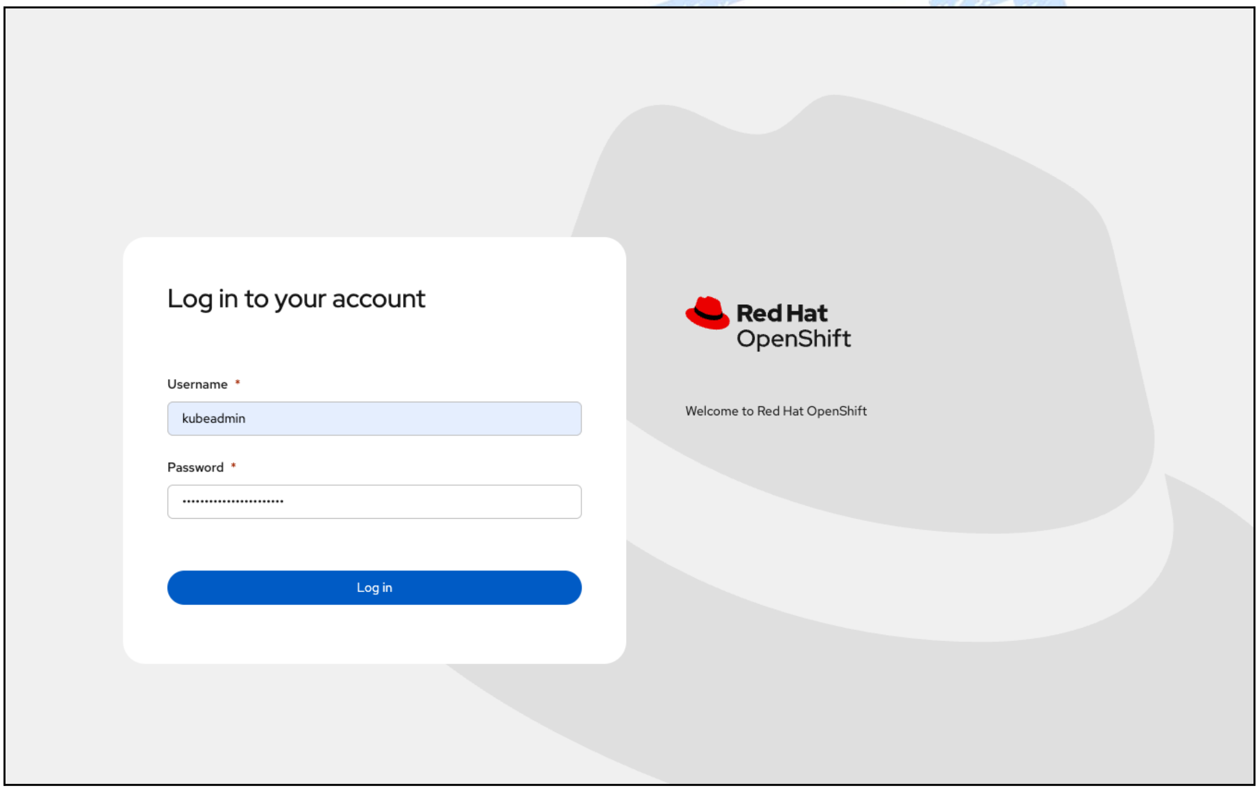
STEP 15. When the installation is complete, download the kubeconfig file, and record the password for the kubeadmin user.







STEP 16. Login to the OpenShift console, click on the Web Console URL: https://console-openshift-console.apps.ocp.redhat.local/
When prompted for the Username and Password, enter the kubeadmin user and the password recorded previously from the installation progress screen.



OpenShift Credentials

Web Console URL: https://console-openshift-console.apps.ocp.redhat.local/

https://console.redhat.com/openshift/assisted-installer/clusters/2e9aece0-66f8-42d5-92e0-6f802344fa1d

```
Username: [your_openshift_username] (select login with kube:admin)
Password: [your_openshift_password]

Username: admin (select login with ldap)
Password: redhat123
```

Vefify MTU setting on all nodes:

Run the following command to verify that the cluster installation set the MTU as expected:

```
[root@rhel8-bastion ~]# for i in $(oc get node -o jsonpath='{ .items[*].metadata.name  }'); do oc debug node/${i} -- chroot /host ip -d link |grep -1 maxmtu|grep -1 bond0; done
```

Sample output

```
To use host binaries, run `chroot /host`
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 allmulti 0 minmtu 0 maxmtu 0 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 524280 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
2: eno1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
    link/ether 70:df:2f:f6:a1:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
--
3: eno2: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
    link/ether 70:df:2f:f6:a1:2c brd ff:ff:ff:ff:ff:ff permaddr 70:df:2f:f6:a1:2d promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
    openvswitch numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 65536 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
10: bond0: <BROADCAST,MULTICAST,MASTER,UP,LOWER_UP> mtu 9000 qdisc noqueue master ovs-system state UP mode DEFAULT group default qlen 1000
    link/ether 70:df:2f:f6:a1:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 65535

Removing debug pod ...
Starting pod/c240m4-02redhatlocal-debug-4ds4j ...
To use host binaries, run `chroot /host`
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 allmulti 0 minmtu 0 maxmtu 0 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 524280 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
2: eno1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
    link/ether 70:df:2f:f6:a2:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
--
3: eno2: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
    link/ether 70:df:2f:f6:a2:2c brd ff:ff:ff:ff:ff:ff permaddr 70:df:2f:f6:a2:2d promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
    openvswitch numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 65536 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
10: bond0: <BROADCAST,MULTICAST,MASTER,UP,LOWER_UP> mtu 9000 qdisc noqueue master ovs-system state UP mode DEFAULT group default qlen 1000
    link/ether 70:df:2f:f6:a2:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 65535

Removing debug pod ...
Starting pod/c240m4-03redhatlocal-debug-hq9tl ...
To use host binaries, run `chroot /host`
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 allmulti 0 minmtu 0 maxmtu 0 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 524280 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
2: eno1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
```

```
    link/ether 70:df:2f:f6:a3:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
--
3: eno2: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
    link/ether 70:df:2f:f6:a3:2c brd ff:ff:ff:ff:ff:ff permaddr 70:df:2f:f6:a3:2d promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
```

MACHINECONFIG:

[6.10.3. Automatically allocating resources for nodes]

KubeletConfig

OpenShift Container Platform can automatically determine the optimal **system-reserved** CPU and memory resources for nodes associated with a specific machine config pool and update the nodes with those values when the nodes start. By default, the **system-reserved** CPU is **500m** and **system-reserved** memory is **1Gi**.

To automatically determine and allocate the **system-reserved** resources on nodes, create a **KubeletConfig** custom resource (CR) to set the **autoSizingReserved: true** parameter. A script on each node calculates the optimal values for the respective reserved resources based on the installed CPU and memory capacity on each node. The script takes into account that increased capacity requires a corresponding increase in the reserved resources.

Automatically determining the optimal **system-reserved** settings ensures that your cluster is running efficiently and prevents node failure due to resource starvation of system components, such as CRI-O and kubelet, without needing to manually calculate and update the values.

This feature is disabled by default.

Procedure
STEP 1. Create a custom resource (CR) for your configuration change:
Sample configuration for a resource allocation CR

```
[root@bastion ]# cat << EOF | oc apply -f -
apiVersion: machineconfiguration.openshift.io/v1
kind: KubeletConfig
metadata:
  name: dynamic-node-master
spec:
  machineConfigPoolSelector:
    matchLabels:
      pools.operator.machineconfiguration.openshift.io/master: ""
  autoSizingReserved: true
EOF
```

```
[root@bastion ]# cat << EOF | oc apply -f -
apiVersion: machineconfiguration.openshift.io/v1
kind: KubeletConfig
metadata:
  name: dynamic-node-worker
spec:
  machineConfigPoolSelector:
    matchLabels:
      pools.operator.machineconfiguration.openshift.io/worker: ""
  autoSizingReserved: true
EOF
```

The previous example enables automatic resource allocation on all worker nodes. OpenShift Container Platform drains the nodes, applies the kubelet config, and restarts the nodes.

WARNING: This will trigger a serial restart of nodes using this MachineConfigPool. Refer also to the documentation for [Modifying nodes]() to see how to modify the reservations with those values

Verification

STEP 1. View the /etc/node-sizing.env file on a node(s) you configured by entering the following command:

```
[root@bastion ]# for i in $(oc get node -o jsonpath='{
.items[*].metadata.name  }'); do oc debug node/${i} -- chroot /host cat
/etc/node-sizing.env; done
Temporary namespace openshift-debug-spjgv is created for debugging node...
Starting pod/c240m4-01redhatlocal-debug-x56qq ...
To use host binaries, run `chroot /host`
SYSTEM_RESERVED_MEMORY=17Gi
SYSTEM_RESERVED_CPU=0.91
SYSTEM_RESERVED_ES=1Gi
Removing debug pod ...
Temporary namespace openshift-debug-spjgv was removed.
Temporary namespace openshift-debug-8wxrm is created for debugging node...
Starting pod/c240m4-02redhatlocal-debug-6k858 ...
To use host binaries, run `chroot /host`
SYSTEM_RESERVED_MEMORY=17Gi
SYSTEM_RESERVED_CPU=0.91
SYSTEM_RESERVED_ES=1Gi
Removing debug pod ...
Temporary namespace openshift-debug-8wxrm was removed.
Temporary namespace openshift-debug-m7tpk is created for debugging node...
Starting pod/c240m4-03redhatlocal-debug-s65tx ...
To use host binaries, run `chroot /host`
SYSTEM_RESERVED_MEMORY=17Gi
SYSTEM_RESERVED_CPU=0.91
SYSTEM_RESERVED_ES=1Gi
```

```
Removing debug pod ...
Temporary namespace openshift-debug-m7tpk was removed.
```

Or run it on a single node:

```
[root@bastion ]# oc debug node/c240m4-03.redhat.local -- chroot /host cat
/etc/node-sizing.env
```

INSTALL/CONFIGURE OPERATORS

Install Local Storage Operator

2.1. Install Local Storage Operator

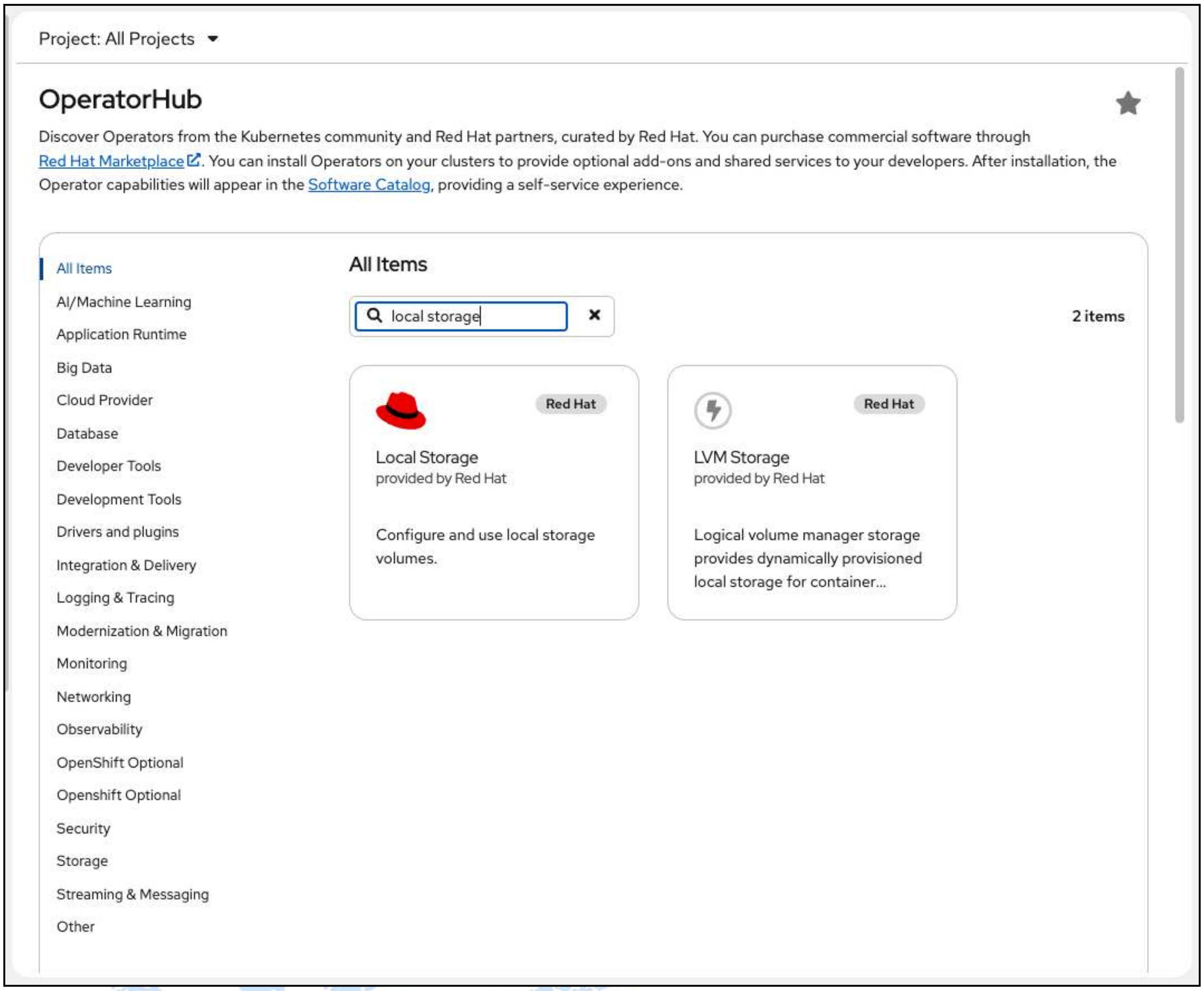
NOTE: If re-installing, ensure that the SSDs are empty, and wiped before configuring ODF.

Access each node using: **ssh core@c240m4-0x**, or **oc debug node/c240m4-0x, then run the following** commands:

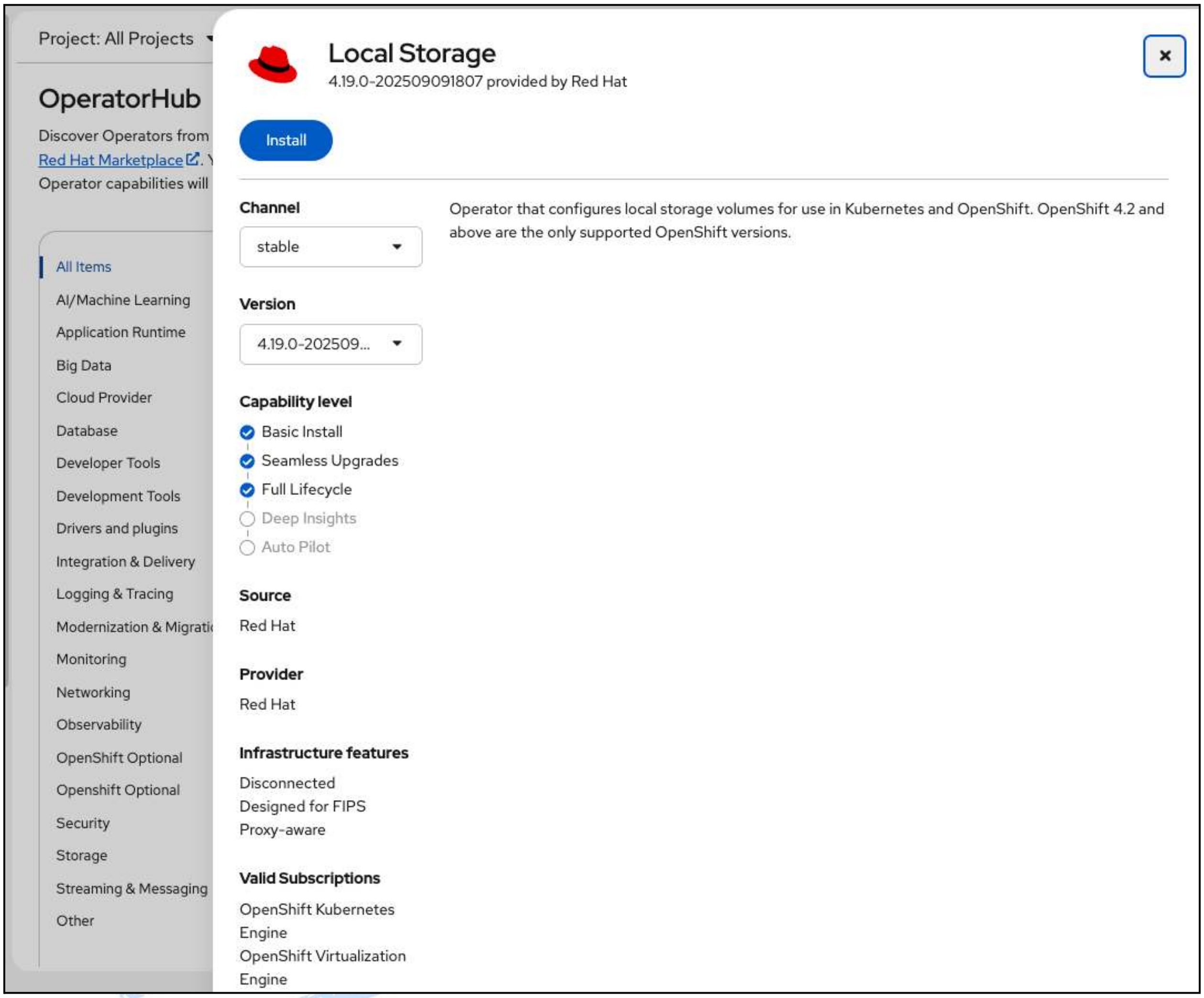
```
[root@bastion ]# ssh core@c240m4-01
[core@c240m4-01 ]# sudo sgdisk -Z /dev/sdX
[core@c240m4-01 ]# sudo wipefs -a /dev/sdX
```

Use the **`sudo lsblk`** command to identify the disks' device names, and use them in place of **/dev/sdX** above.

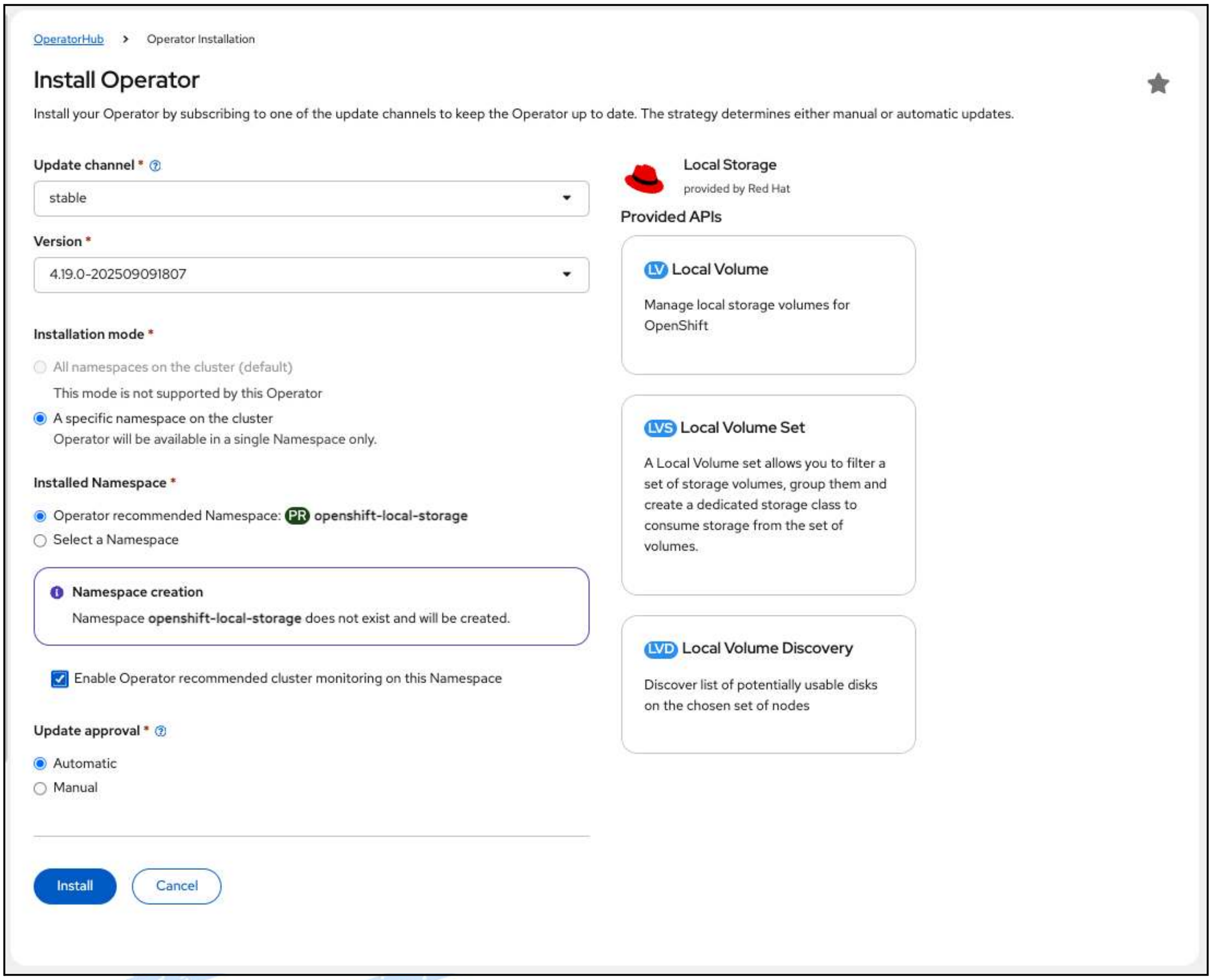
`STEP 1.` In OperatorHub, search on local storage, select the ***Local Storage*** Operator.


Project: All Projects
OperatorHub
Discover Operators from the Kubernetes community and Red Hat partners, curated by Red Hat. You can purchase commercial software through Red Hat Marketplace. You can install Operators on your clusters to provide optional add-ons and shared services to your developers. After installation, the Operator capabilities will appear in the Software Catalog, providing a self-service experience.
All Items
AI/Machine Learning
Application Runtime
Big Data
Cloud Provider
Database
Developer Tools
Development Tools
Drivers and plugins
Integration & Delivery
Logging & Tracing
Modernization & Migration
Monitoring
Networking
Observability
OpenShift Optional
Openshift Optional
Security
Storage
Streaming & Messaging
Other
All Items
local storage
2 items
Red Hat
Local Storage
provided by Red Hat
Configure and use local storage volumes.
Red Hat
LVM Storage
provided by Red Hat
Logical volume manager storage provides dynamically provisioned local storage for container...

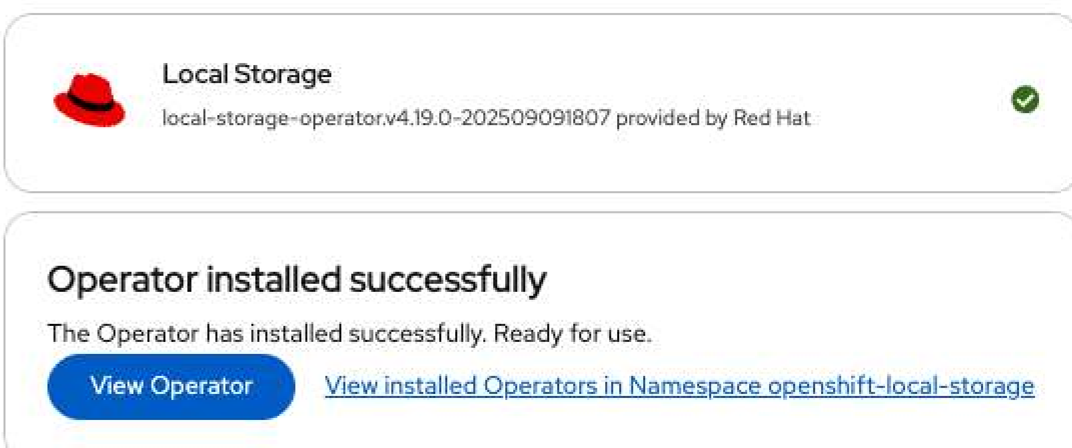
STEP 2. Click on ***Install***.



STEP 3. Click ***Install***.

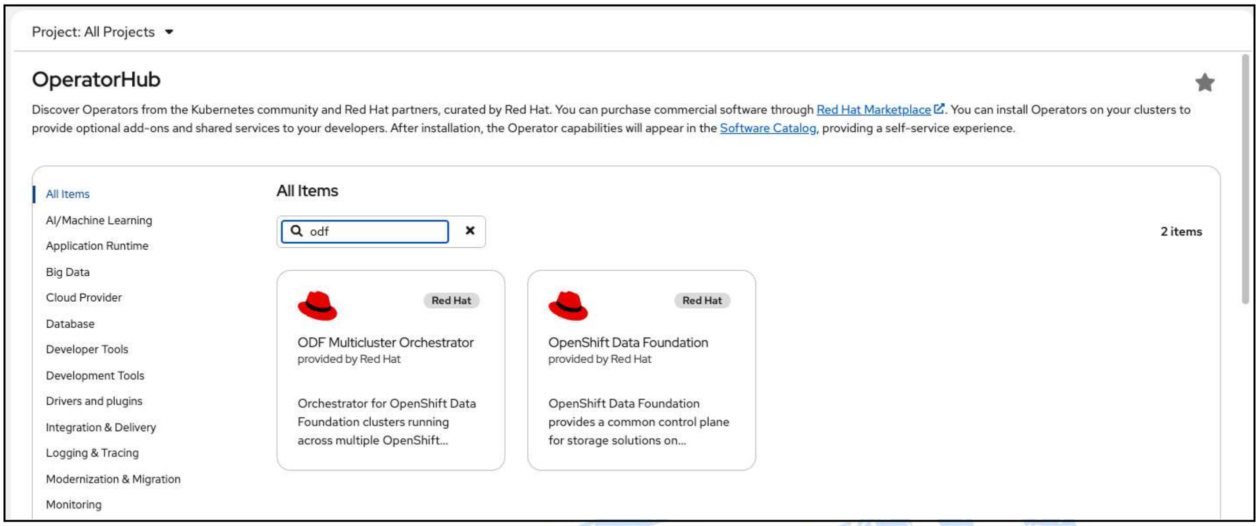


STEP 4. Ensure that the operator installs successfully. Then continue with the next step by installing ODF. The installation process for ODF will configure the Local Storage operator.

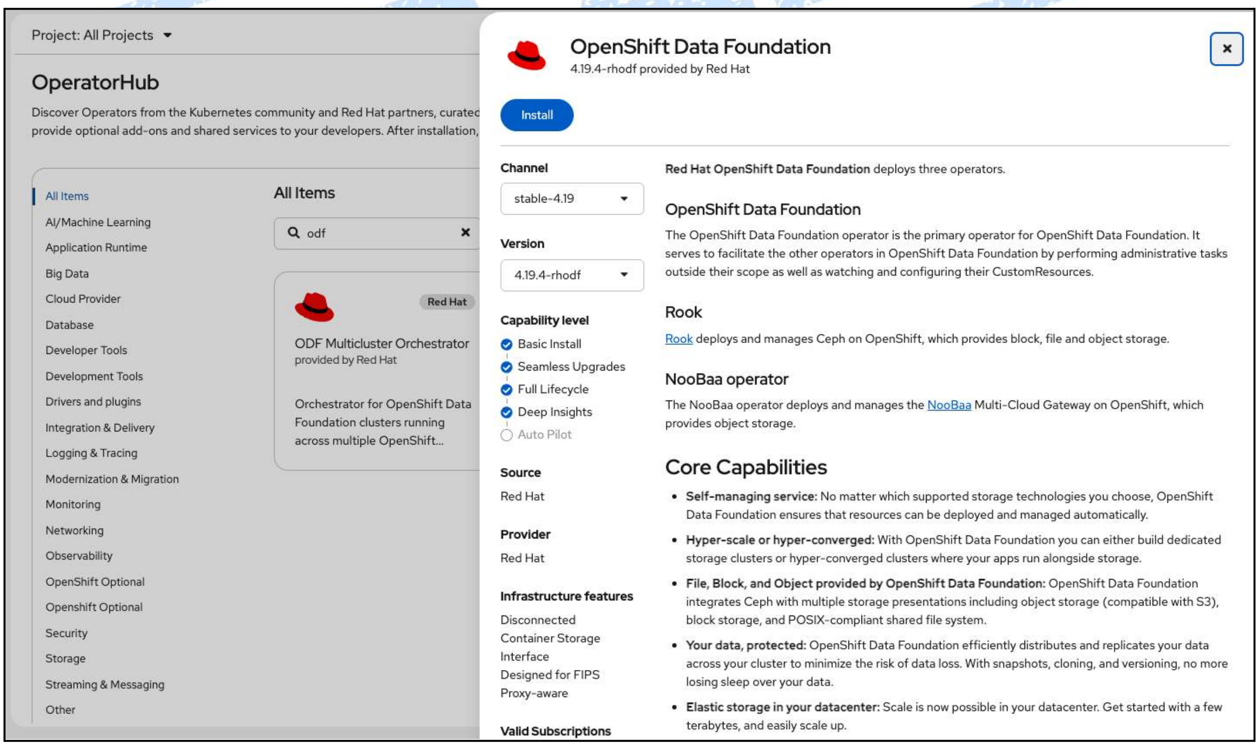


Install OpenShift Data Foundation (Internal mode)

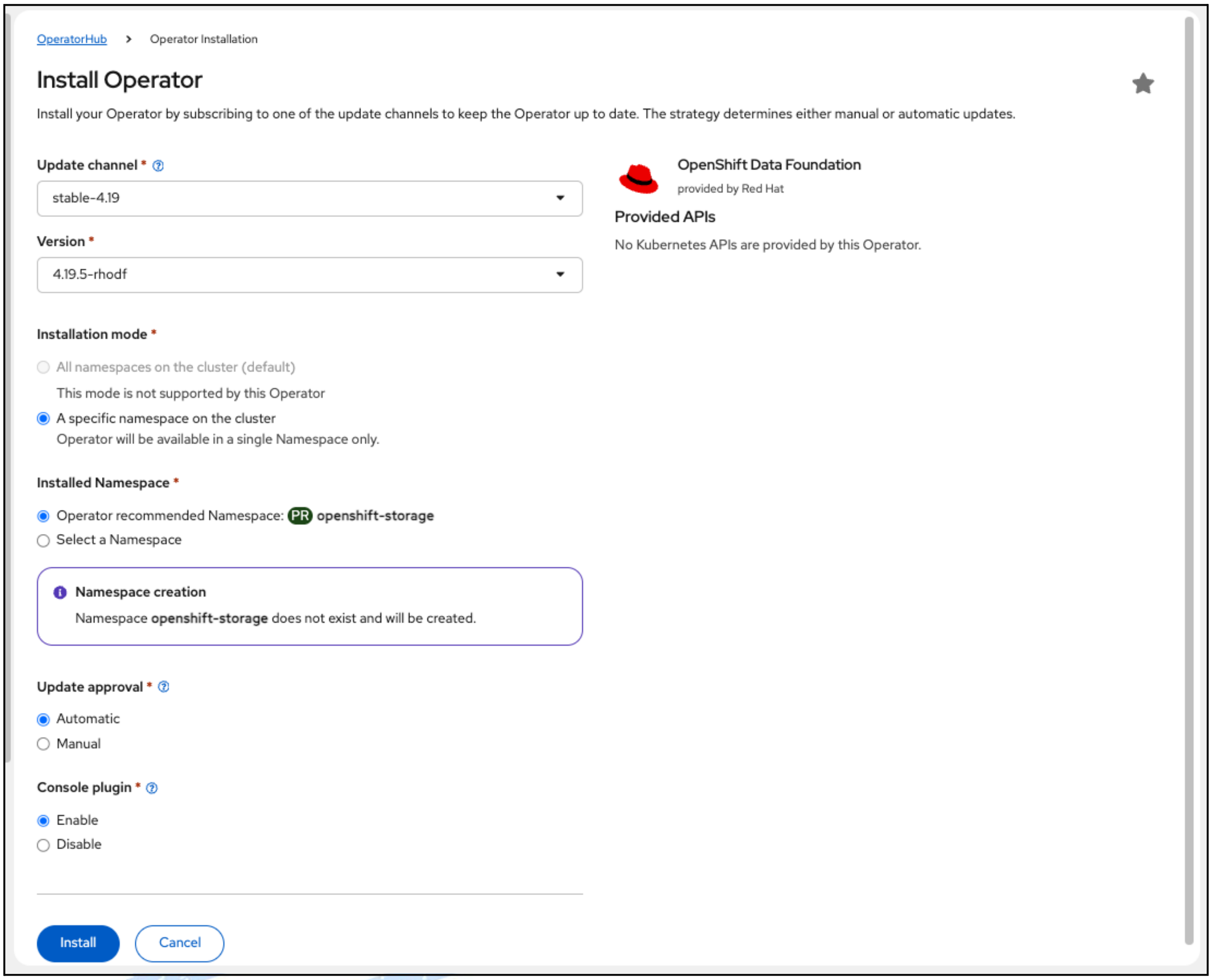
STEP 1. In OperatorHub, search on odf, select "***OpenShift Data Foundation***".



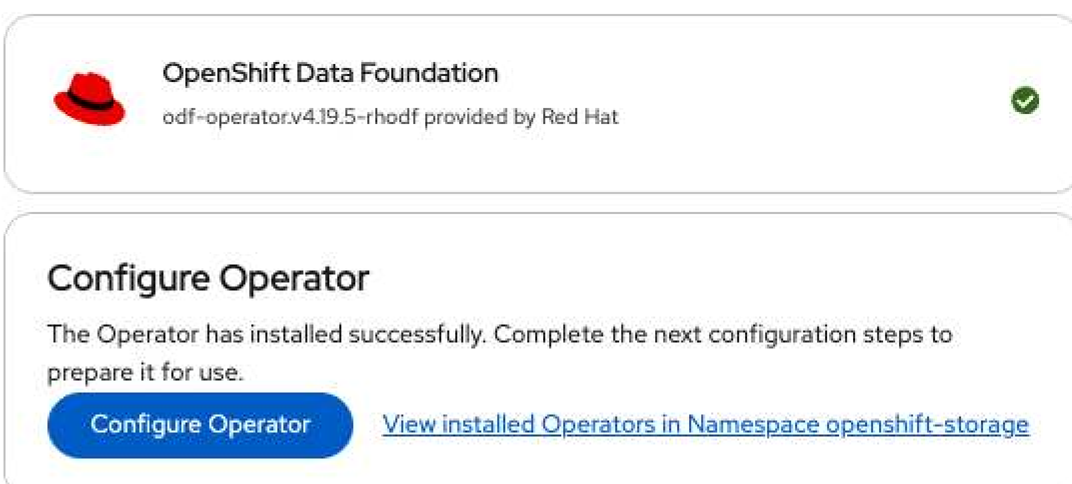
STEP 2. Click on "***Install***".



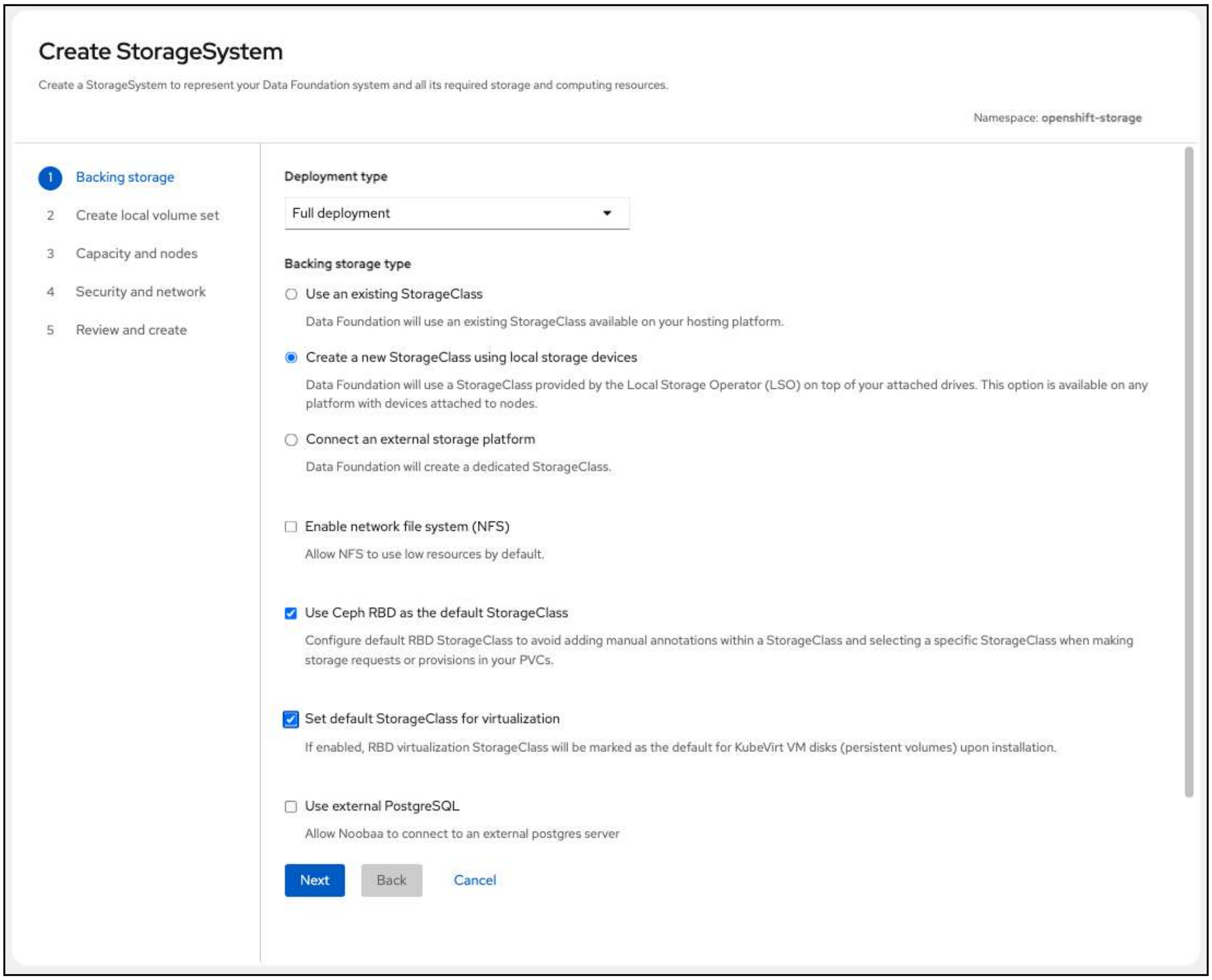
STEP 3. Click on "***Install***".



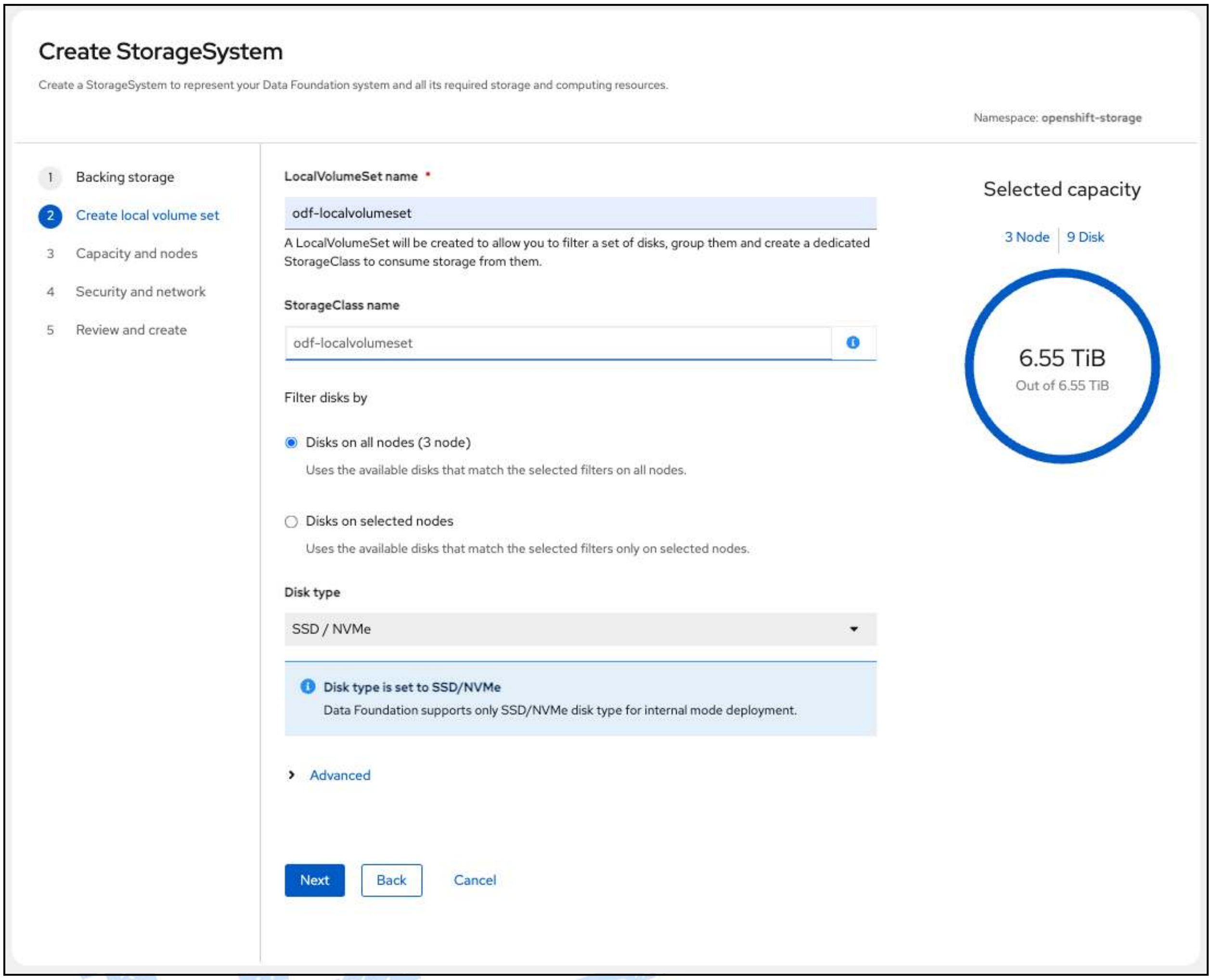
STEP 4. Ensure that the operator installs successfully. Then, click on "***Configure Operator***".



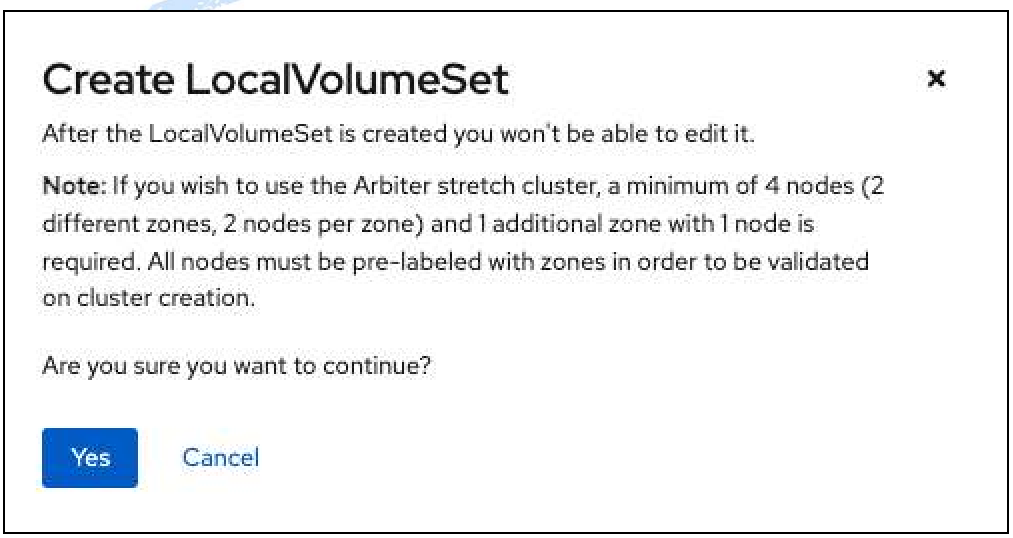
STEP 4. Select "***Create a new StorageClass using local storage devices***", "***Use Ceph RBD as the default StorageClass***", and "***Set default StorageClass for virtualization***" then select "***Next***".



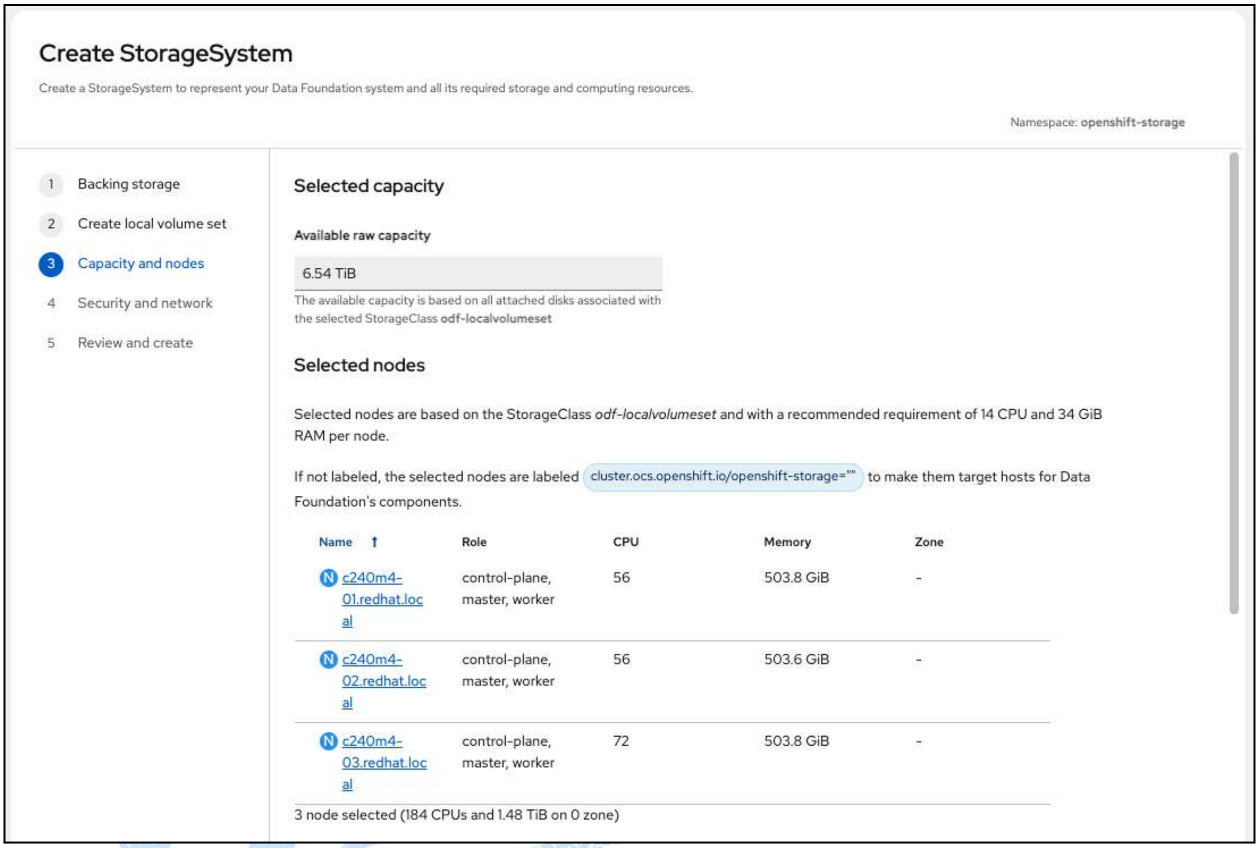
STEP 5. Enter a name for the LocalVolumeSet and StorageClass, and select the appropriate settings for the storage across your cluster.
In this example, a 3-node OpenShift cluster is being used, with each node having 3 SSDs installed, for a total of 9 Disks.
Then click on "***Next***".

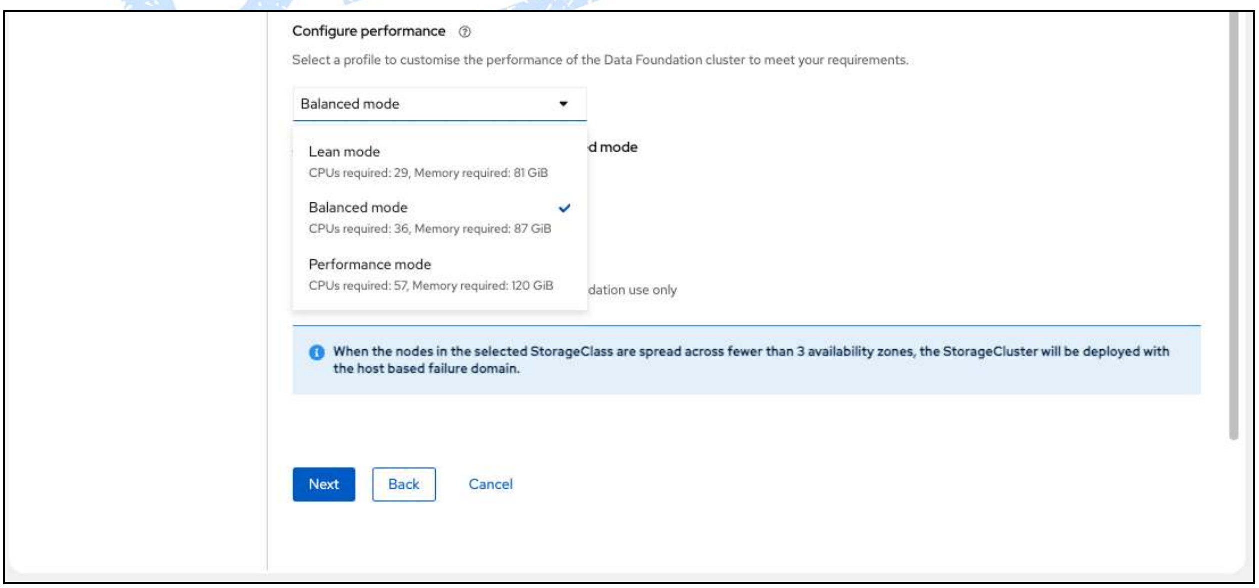


STEP 6. Select "***Yes***" to continue.

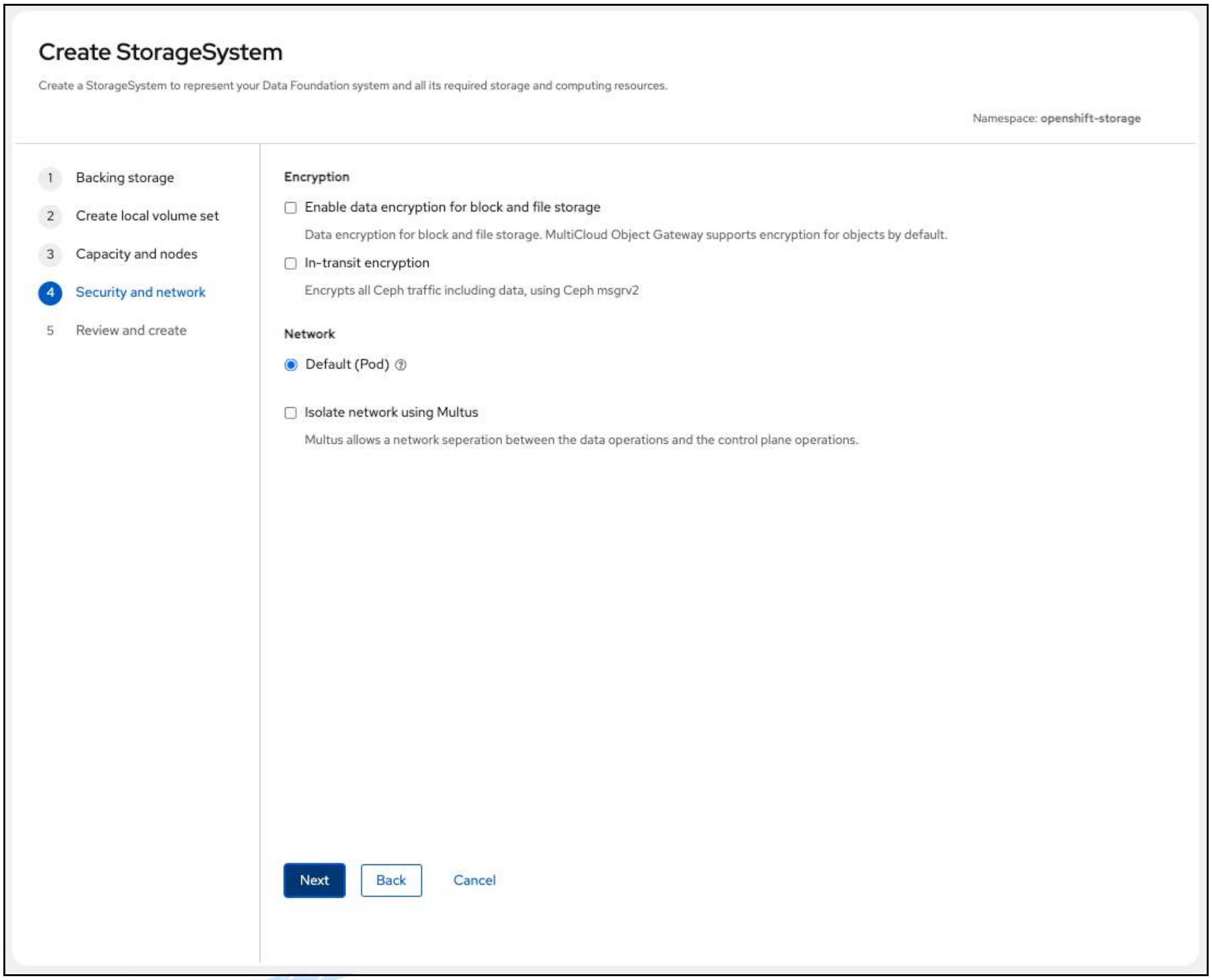


STEP 7. On the “Capacity and nodes” screen you will see confirmation of the selected capacity, and nodes. Select the amount of resources to assign to ODF; either ***Lean***, ***Balanced*** or ***Performance*** mode. For most cases, you would select the default “*Balanced*” mode setting. Select “***Next***”.

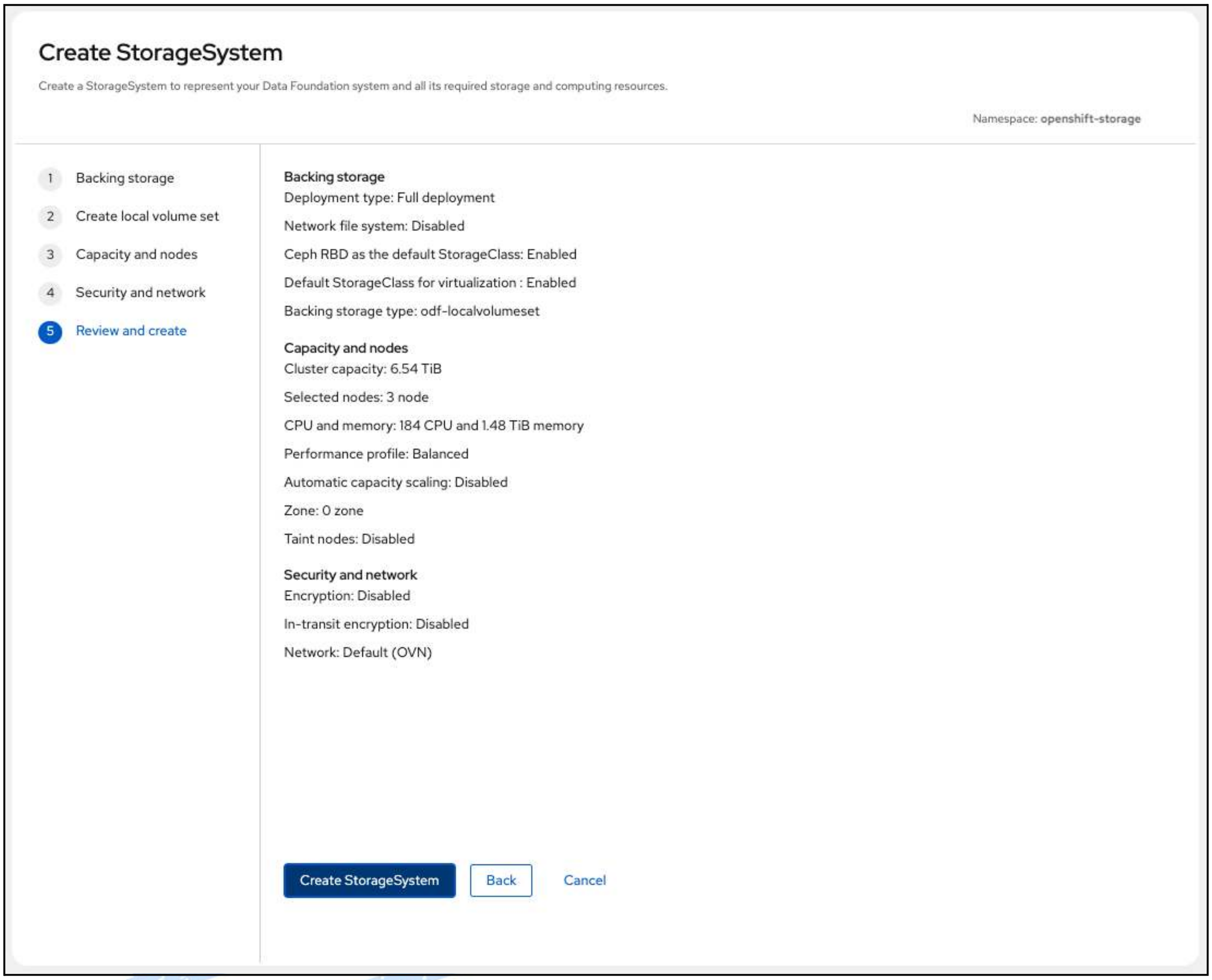




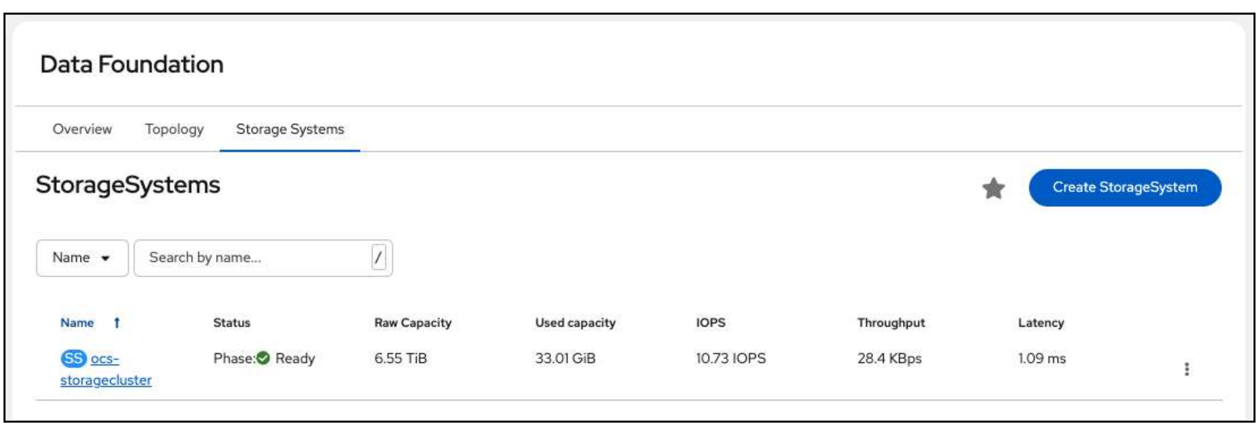
STEP 8. Optionally enable encryption or to specify advanced networking settings, then select "***Next***".



STEP 9. Click on "***Create StorageSystem***".



STEP 10. When complete, you will see a **Ready** status.



Create a Replica 2 Block Pool and StorageClass for ODF

Perform the following from the OpenShift console to create the Replica 2 Block Pool:

STEP 1. Click ***Storage*** > ***Data Foundation***.

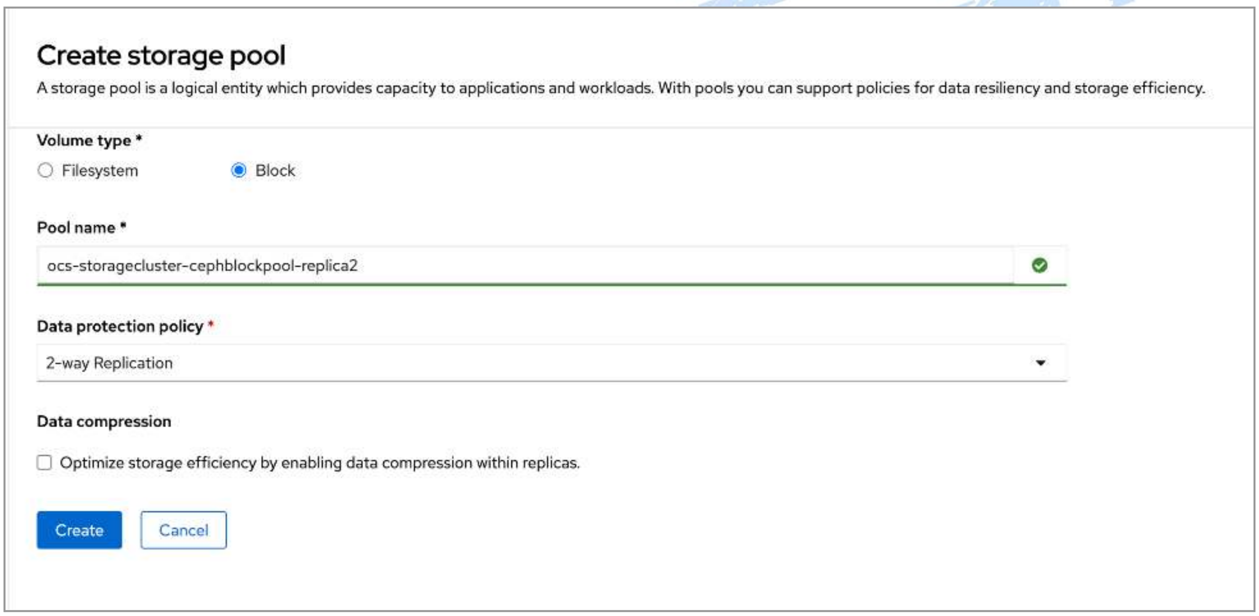
STEP 2. In the ***Storage systems*** tab, select the storage system and then click the ***Storage pools*** tab.

STEP 3. Click ***Create storage pool***.

STEP 4. Select ***Volume type*** as ***Block***.

STEP 5. Enter ***Pool name***, such as "***ocs-storagecluster-cephblockpool-replica2***".

STEP 6. Select Data protection policy as ***2-way Replication***.



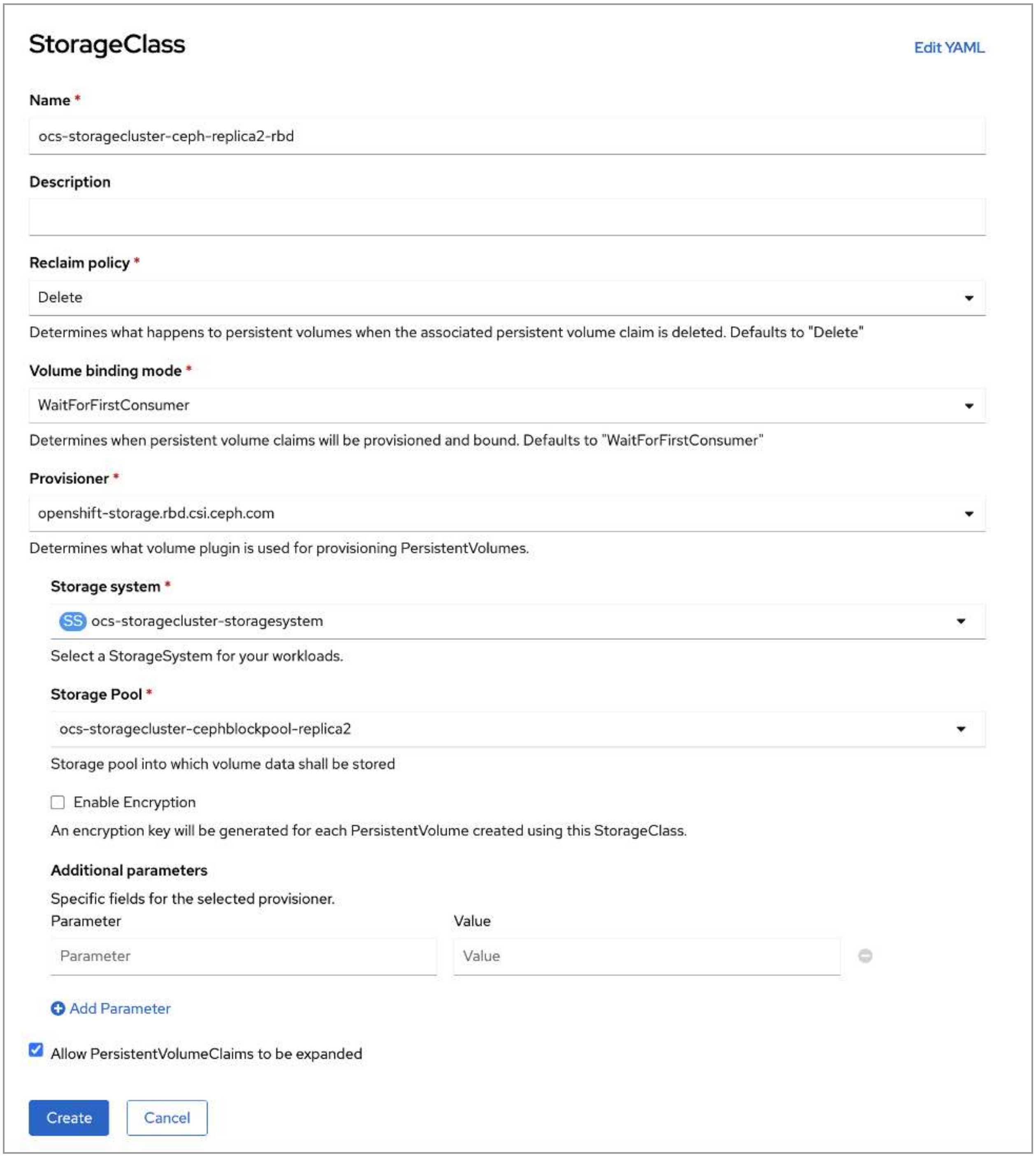
STEP 7. Click ***Create***.

After the Pool is created, perform the following from the OpenShift console to create the Replica 2 StorageClass:

STEP 1. Navigate to ***Storage*** > ***StorageClasses***.

STEP 2. Click ***Create Storage Class***.

STEP 3. Enter a ***name***, such as "***ocs-storagecluster-ceph-replica2-rbd***"; then select the ***openshift-storage.rbd,csi.ceph.com*** provisioner, and select the Storage Pool entered in the previous step, such as "***ocs-storagecluster-cephblockpool-replica2***".



STEP 4. Click ***Create***.

Configuring the Image Registry Operator to use CephFS storage with Red Hat OpenShift Data Foundation

Procedure

STEP 1. Create a PVC to use the cephfs storage class. For example:

```
[root@bastion ]# cat <<EOF | oc apply -f -
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: registry-storage-pvc
 namespace: openshift-image-registry
spec:
 accessModes:
 - ReadWriteMany
 resources:
   requests:
     storage: 100Gi
 storageClassName: ocs-storagecluster-cephfs
EOF
```

STEP 2. Configure the image registry to use the CephFS file system storage by entering the following command:

```
[root@bastion ]# oc patch config.image/cluster -p
'{"spec":{"managementState":"Managed","replicas":2,"storage":{"managementStat
e":"Unmanaged","pvc":{"claim":"registry-storage-pvc"}}}}' --type=merge
```

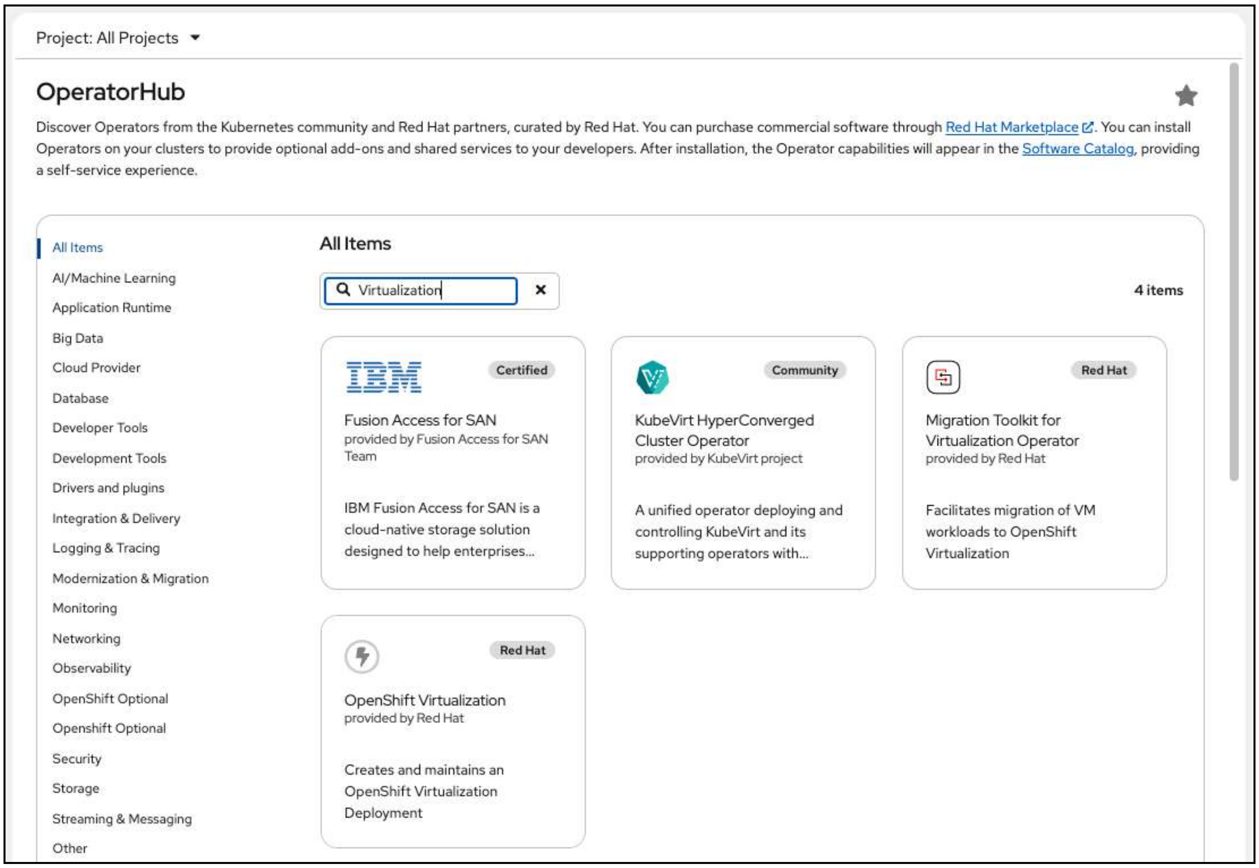
Example

```
apiVersion: imageregistry.operator.openshift.io/v1
kind: Config
metadata:
  name: cluster
spec:
  logLevel: Normal
  managementState: Managed
  replicas: 2
  storage:
    managementState: Unmanaged
    pvc:
      claim: registry-storage-pvc
```

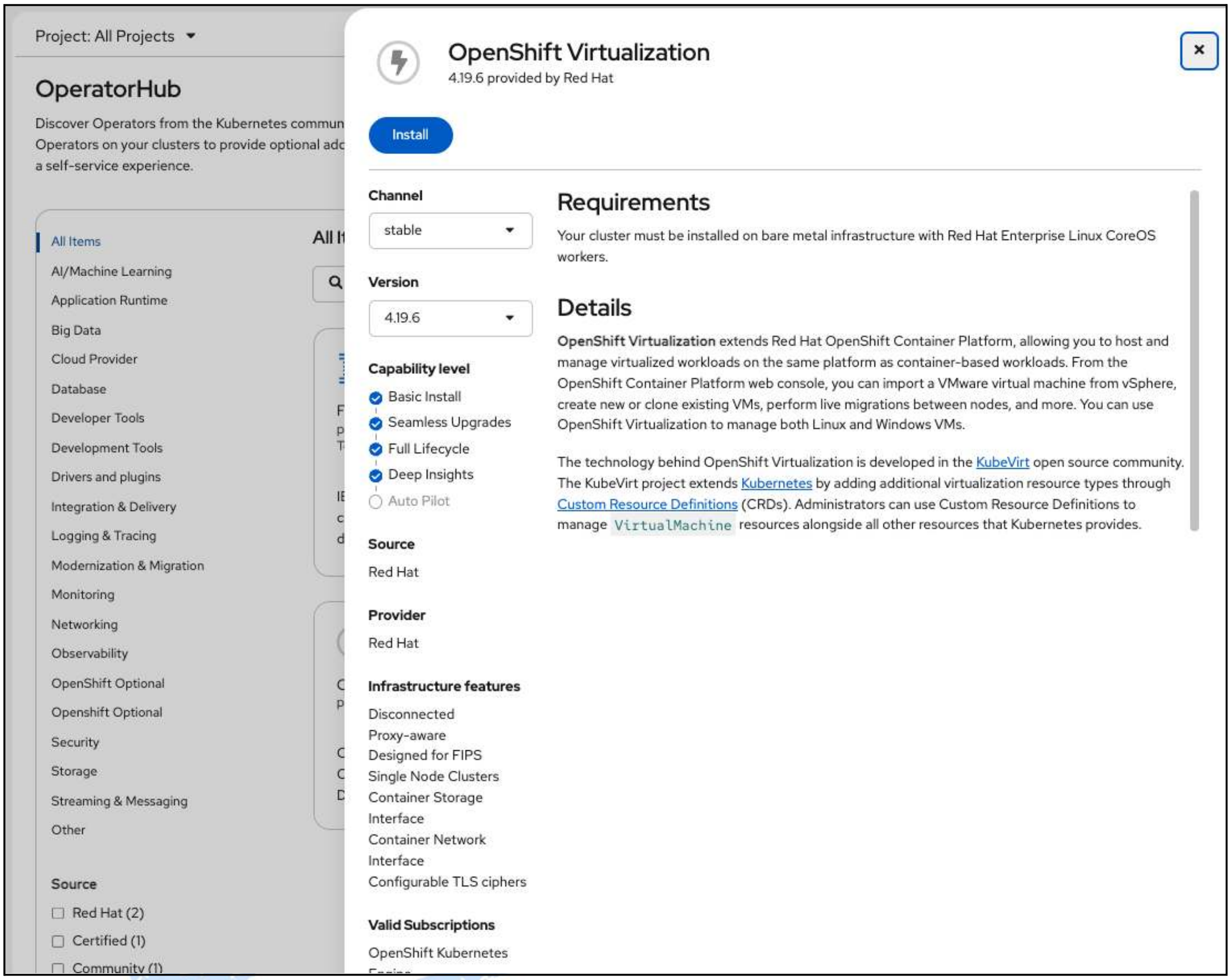
Also for reference: [Configuring Image Registry to use OpenShift Data Foundation]

Install OpenShift Virtualization Operator

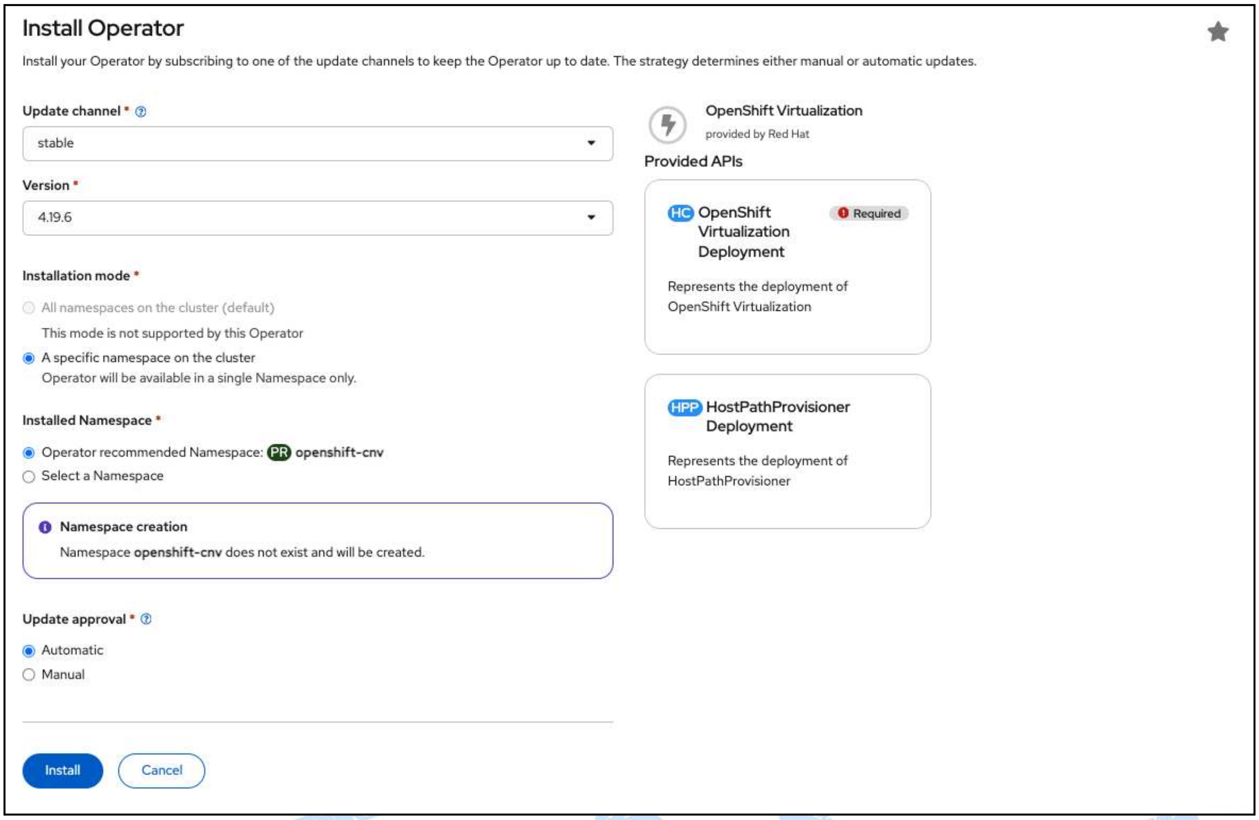
STEP 1. Search on ***Virtualization***, select ***OpenShift Virtualization***. Click on "***Install***".



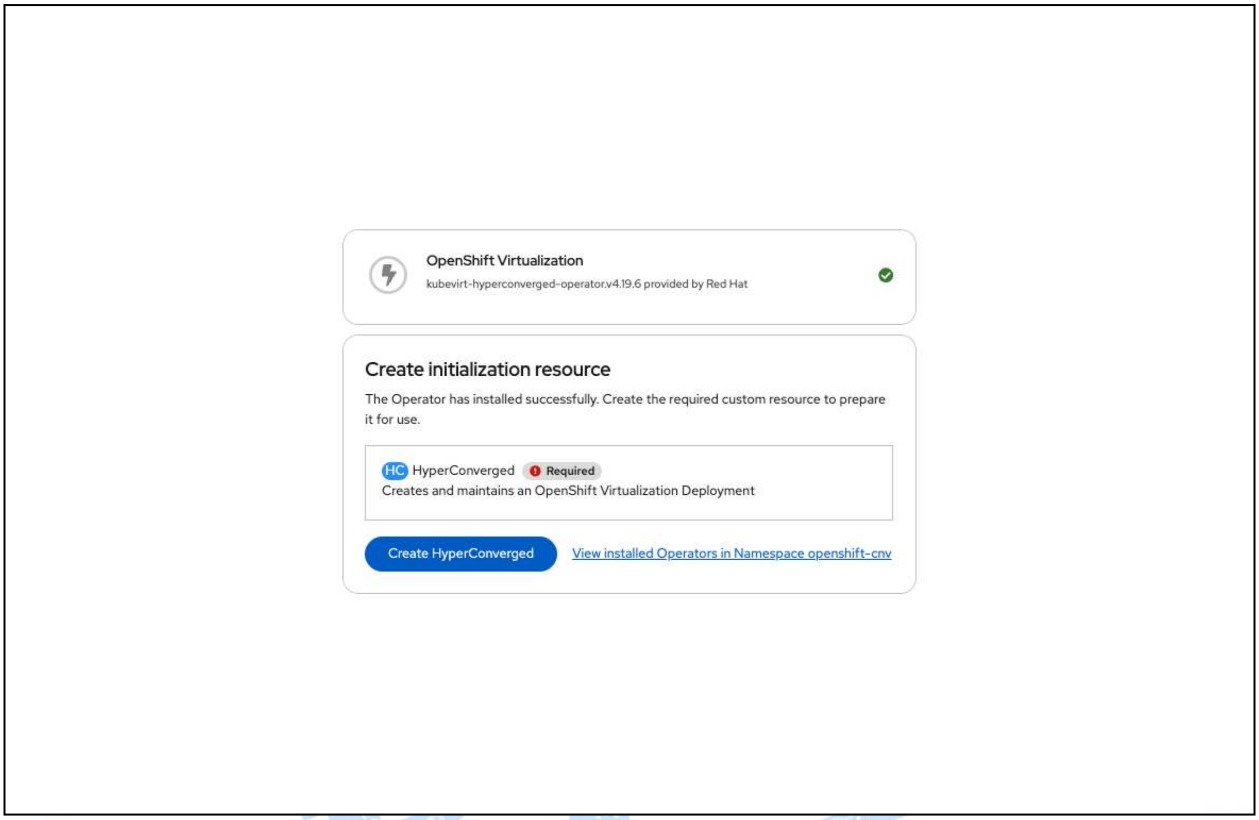
STEP 2. Click on "***Install***".



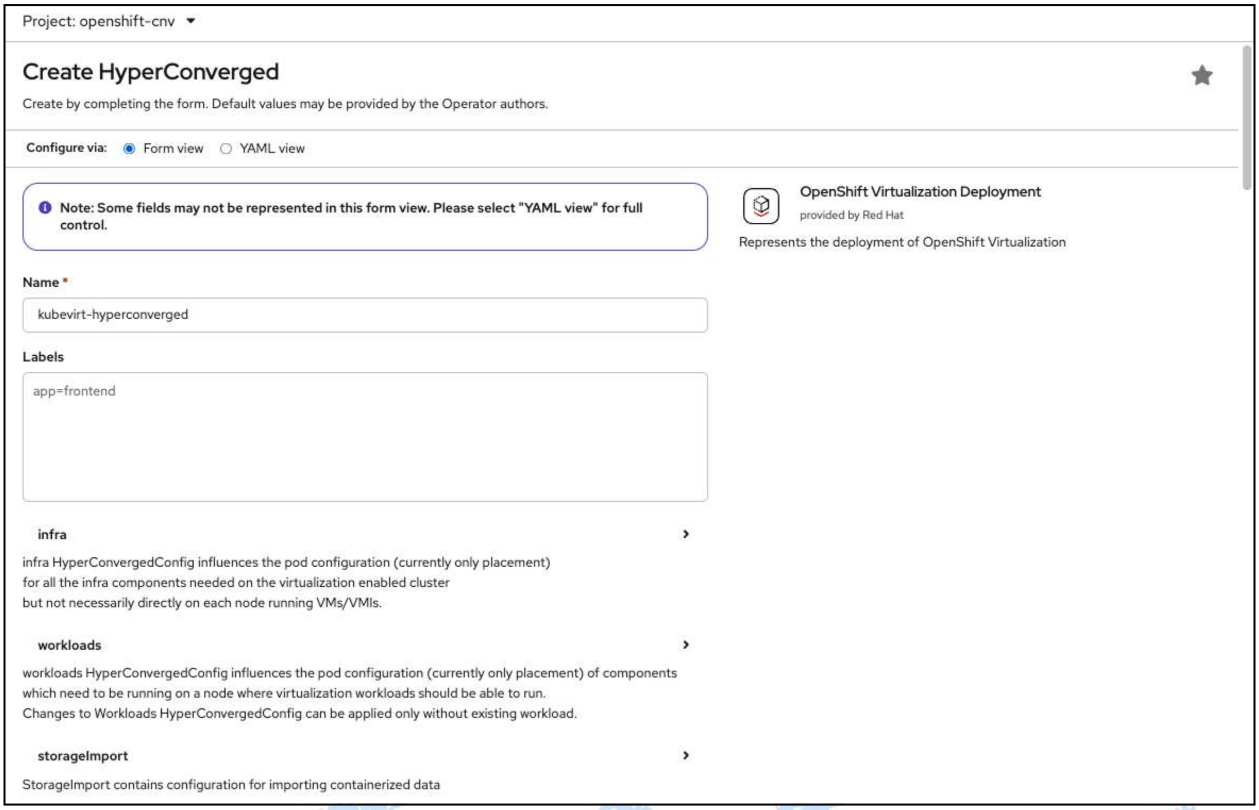
`STEP 2.` `Select "Install".`

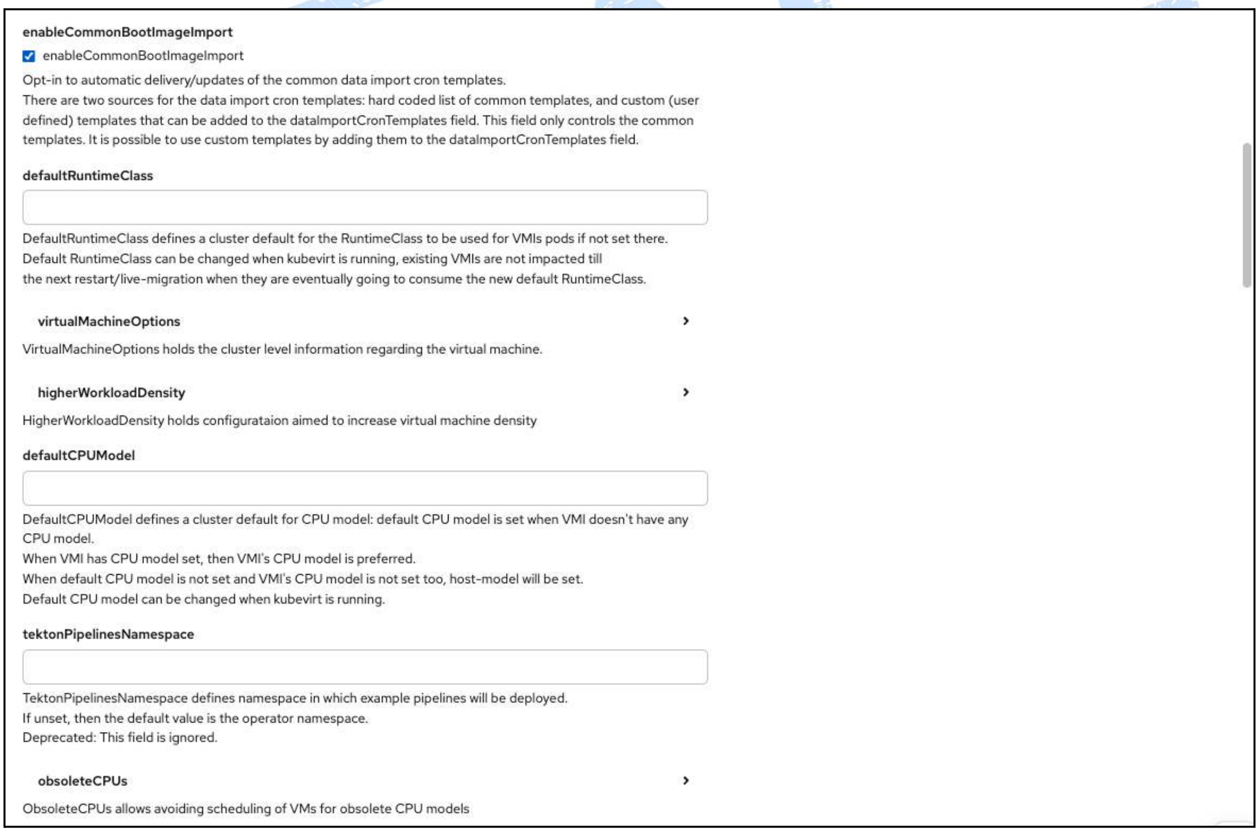

Install Operator
Install your Operator by subscribing to one of the update channels to keep the Operator up to date. The strategy determines either manual or automatic updates.
Update channel *
stable
Version *
4.19.6
Installation mode *
All namespaces on the cluster (default)
This mode is not supported by this Operator
A specific namespace on the cluster
Operator will be available in a single Namespace only.
Installed Namespace *
Operator recommended Namespace: PR openshift-cnv
Select a Namespace
Namespace creation
Namespace openshift-cnv does not exist and will be created.
Update approval *
Automatic
Manual
Install
Cancel
OpenShift Virtualization
provided by Red Hat
Provided APIs
HC OpenShift Virtualization Deployment
Required
Represents the deployment of OpenShift Virtualization
HPP HostPathProvisioner Deployment
Represents the deployment of HostPathProvisioner

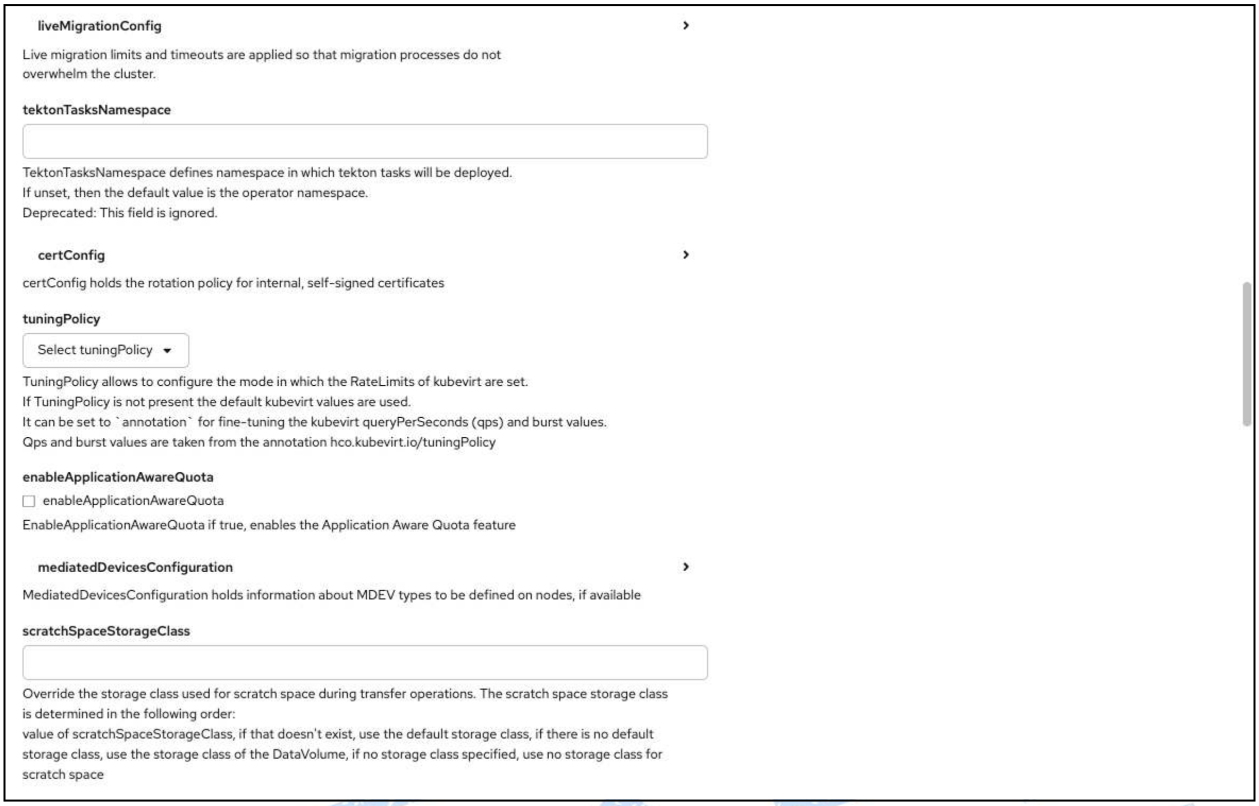
STEP 3. Once the operator installation is completed, select ***"Create HyperConverged"*** to create a deployment of OpenShift Virtualization.

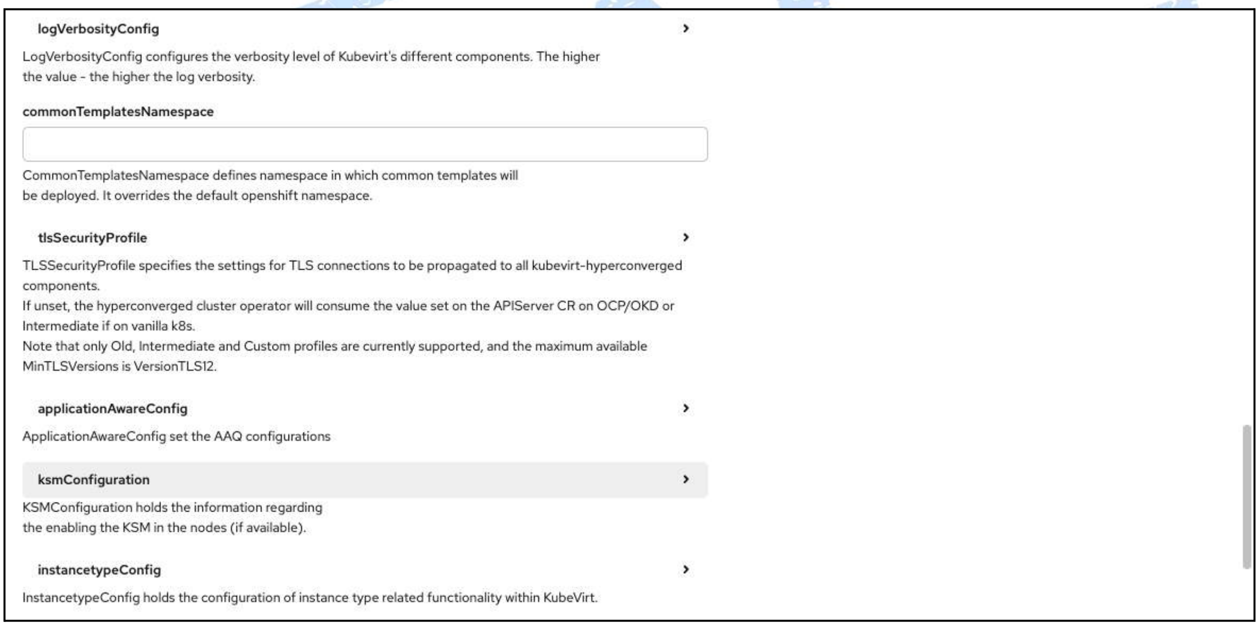


STEP 4. Create a deployment using the default settings; scroll down to the bottom of the page, and click on ***"Create"***.


Project: openshift-cnv
Create HyperConverged
Create by completing the form. Default values may be provided by the Operator authors.
Configure via: Form view YAML view
Note: Some fields may not be represented in this form view. Please select "YAML view" for full control.
OpenShift Virtualization Deployment
provided by Red Hat
Represents the deployment of OpenShift Virtualization
Name *
kubevirt-hyperconverged
Labels
app=frontend
infra
infra HyperConvergedConfig influences the pod configuration (currently only placement)
for all the infra components needed on the virtualization enabled cluster
but not necessarily directly on each node running VMs/VMIs.
workloads
workloads HyperConvergedConfig influences the pod configuration (currently only placement) of components
which need to be running on a node where virtualization workloads should be able to run.
Changes to Workloads HyperConvergedConfig can be applied only without existing workload.
storageImport
StorageImport contains configuration for importing containerized data


enableCommonBootImageImport
enableCommonBootImageImport
Opt-in to automatic delivery/updates of the common data import cron templates.
There are two sources for the data import cron templates: hard coded list of common templates, and custom (user defined) templates that can be added to the dataImportCronTemplates field. This field only controls the common templates. It is possible to use custom templates by adding them to the dataImportCronTemplates field.
defaultRuntimeClass
DefaultRuntimeClass defines a cluster default for the RuntimeClass to be used for VMIs pods if not set there.
Default RuntimeClass can be changed when kubevirt is running, existing VMIs are not impacted till
the next restart/live-migration when they are eventually going to consume the new default RuntimeClass.
virtualMachineOptions
VirtualMachineOptions holds the cluster level information regarding the virtual machine.
higherWorkloadDensity
HigherWorkloadDensity holds configurataion aimed to increase virtual machine density
defaultCPUModel
DefaultCPUModel defines a cluster default for CPU model: default CPU model is set when VMI doesn't have any CPU model.
When VMI has CPU model set, then VMI's CPU model is preferred.
When default CPU model is not set and VMI's CPU model is not set too, host-model will be set.
Default CPU model can be changed when kubevirt is running.
tektonPipelinesNamespace
TektonPipelinesNamespace defines namespace in which example pipelines will be deployed.
If unset, then the default value is the operator namespace.
Deprecated: This field is ignored.
obsoleteCPUs
ObsoleteCPUs allows avoiding scheduling of VMs for obsolete CPU models


liveMigrationConfig
Live migration limits and timeouts are applied so that migration processes do not overwhelm the cluster.
tektonTasksNamespace
TektonTasksNamespace defines namespace in which tekton tasks will be deployed.
If unset, then the default value is the operator namespace.
Deprecated: This field is ignored.
certConfig
certConfig holds the rotation policy for internal, self-signed certificates
tuningPolicy
Select tuningPolicy
TuningPolicy allows to configure the mode in which the RateLimits of kubevirt are set.
If TuningPolicy is not present the default kubevirt values are used.
It can be set to `annotation` for fine-tuning the kubevirt queryPerSeconds (qps) and burst values.
Qps and burst values are taken from the annotation hco.kubevirt.io/tuningPolicy
enableApplicationAwareQuota
enableApplicationAwareQuota
EnableApplicationAwareQuota if true, enables the Application Aware Quota feature
mediatedDevicesConfiguration
MediatedDevicesConfiguration holds information about MDEV types to be defined on nodes, if available
scratchSpaceStorageClass
Override the storage class used for scratch space during transfer operations. The scratch space storage class is determined in the following order:
value of scratchSpaceStorageClass, if that doesn't exist, use the default storage class, if there is no default storage class, use the storage class of the DataVolume, if no storage class specified, use no storage class for scratch space


logVerbosityConfig
LogVerbosityConfig configures the verbosity level of Kubevirt's different components. The higher the value - the higher the log verbosity.
commonTemplatesNamespace
CommonTemplatesNamespace defines namespace in which common templates will be deployed. It overrides the default openshift namespace.
tlsSecurityProfile
TLSSecurityProfile specifies the settings for TLS connections to be propagated to all kubevirt-hyperconverged components.
If unset, the hyperconverged cluster operator will consume the value set on the APIServer CR on OCP/OKD or Intermediate if on vanilla k8s.
Note that only Old, Intermediate and Custom profiles are currently supported, and the maximum available MinTLSVersions is VersionTLS12.
applicationAwareConfig
ApplicationAwareConfig set the AAQ configurations
ksmConfiguration
KSMConfiguration holds the information regarding the enabling the KSM in the nodes (if available).
instancetypeConfig
InstancetypeConfig holds the configuration of instance type related functionality within KubeVirt.

evictionStrategy

Select evictionStrategy

EvictionStrategy defines at the cluster level if the VirtualMachineInstance should be
migrated instead of shut-off in case of a node drain. If the VirtualMachineInstance specific
field is set it overrides the cluster level one.
Allowed values:
- `None` no eviction strategy at cluster level.
- `LiveMigrate` migrate the VM on eviction; a not live migratable VM with no specific strategy will block the drain of the node util manually evicted.
- `LiveMigrateIfPossible` migrate the VM on eviction if live migration is possible, otherwise directly evict.
- `External` block the drain, track eviction and notify an external controller.

Defaults to LiveMigrate with multiple worker nodes, None on single worker clusters.

featureGates

featureGates is a map of feature gate flags. Setting a flag to `true` will enable
the feature. Setting `false` or removing the feature gate, disables the feature.

localStorageClassName

Deprecated: LocalStorageClassName the name of the local storage class.

workloadUpdateStrategy

WorkloadUpdateStrategy defines at the cluster level how to handle automated workload updates

deployVmConsoleProxy

☐ deployVmConsoleProxy

deploy VM console proxy resources in SSP operator

uninstallStrategy

BlockUninstallIfWorkloadsExist

UninstallStrategy defines how to proceed on uninstall when workloads (VirtualMachines, DataVolumes) still exist.
BlockUninstallIfWorkloadsExist will prevent the CR from being removed when workloads still exist.
BlockUninstallIfWorkloadsExist is the safest choice to protect your workloads from accidental data loss, so it's strongly advised.
RemoveWorkloads will cause all the workloads to be cascading deleted on uninstallation.
WARNING: please notice that RemoveWorkloads will cause your workloads to be deleted as soon as this CR will be, even accidentally, deleted.
Please correctly consider the implications of this option before setting it.
BlockUninstallIfWorkloadsExist is the default behaviour.

vddkInitImage

VDDK Init Image eventually used to import VMs from external providers

Deprecated: please use the Migration Toolkit for Virtualization

vmStateStorageClass

VMStateStorageClass is the name of the storage class to use for the PVCs created to preserve VM state, like TPM.

kubeSecondaryDNSNameServerIP

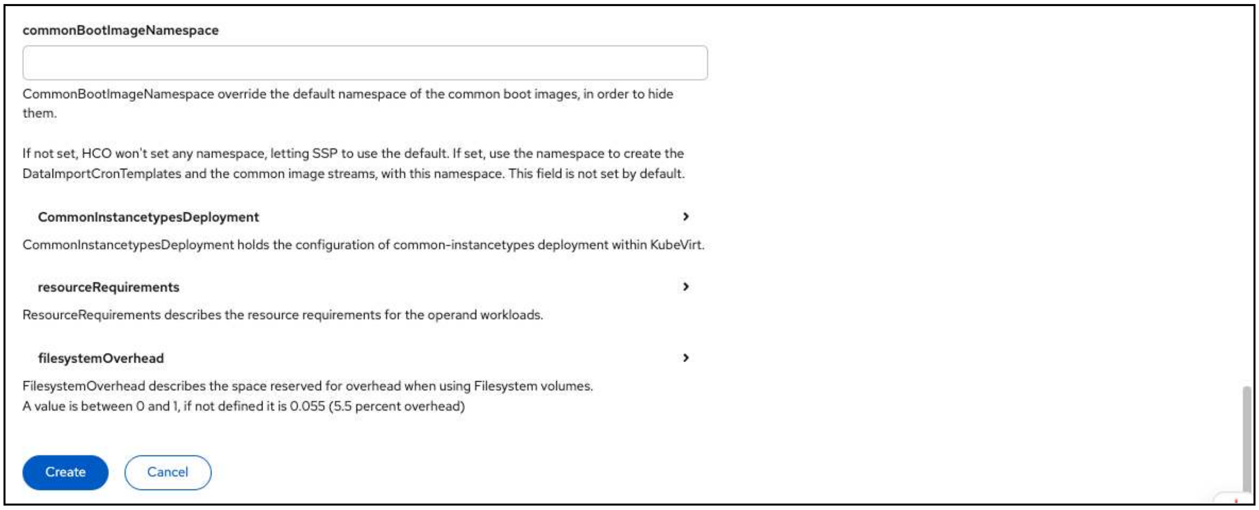
KubeSecondaryDNSNameServerIP defines name server IP used by KubeSecondaryDNS

dataImportCronTemplates

DataImportCronTemplates holds list of data import cron templates (golden images)

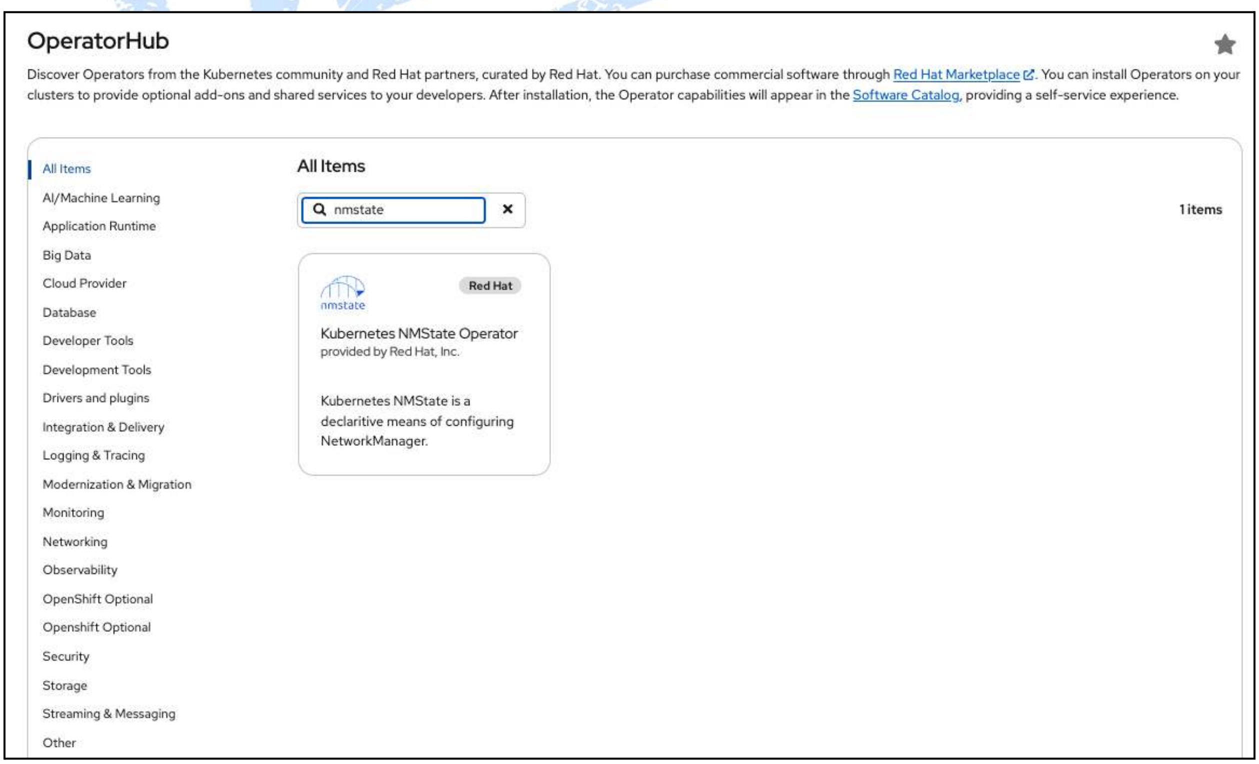
permittedHostDevices

PermittedHostDevices holds information about devices allowed for passthrough

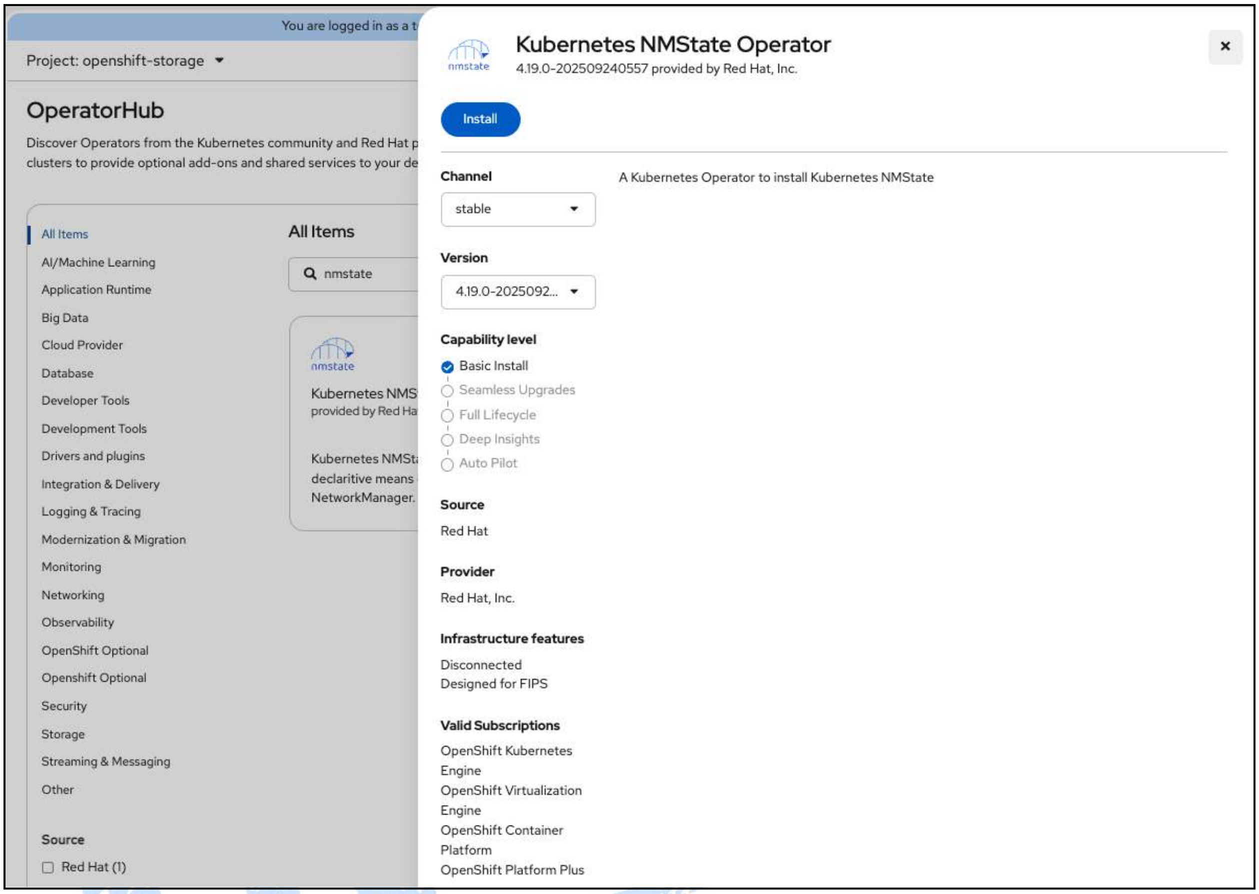


Install NMState Operator

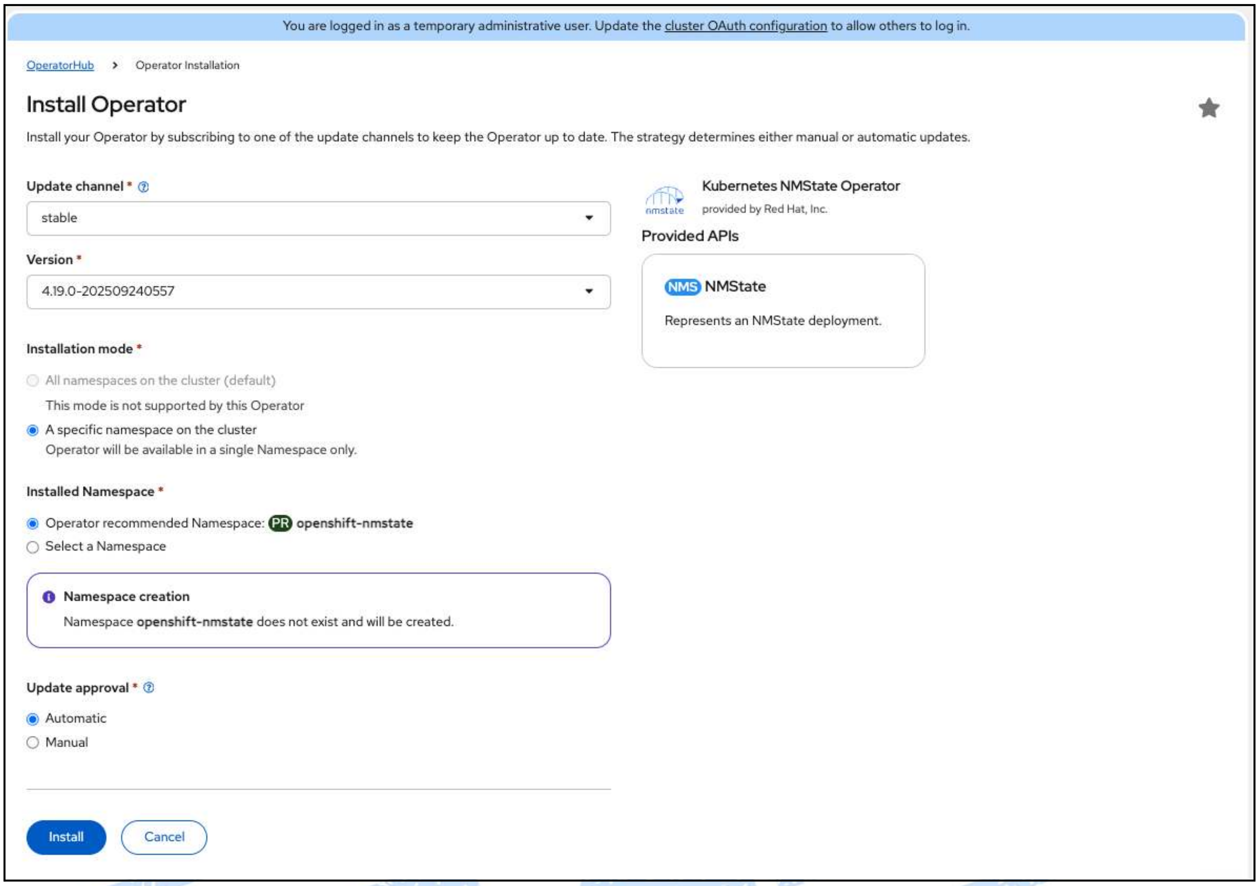
STEP 1. In OperatorHub, search on ***NMState***, select the ***Kubernetes NMState Operator***.



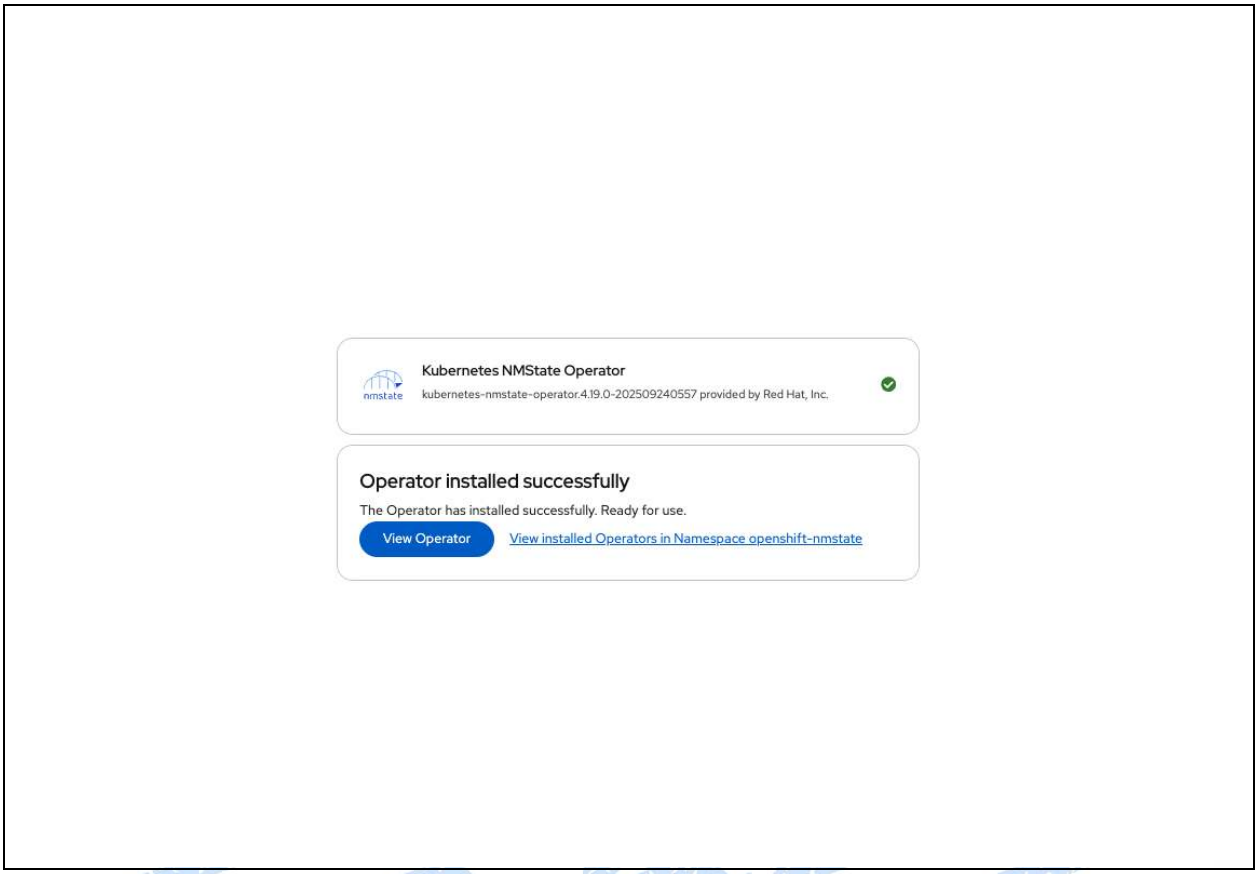
STEP 2. Click on "***Install***".



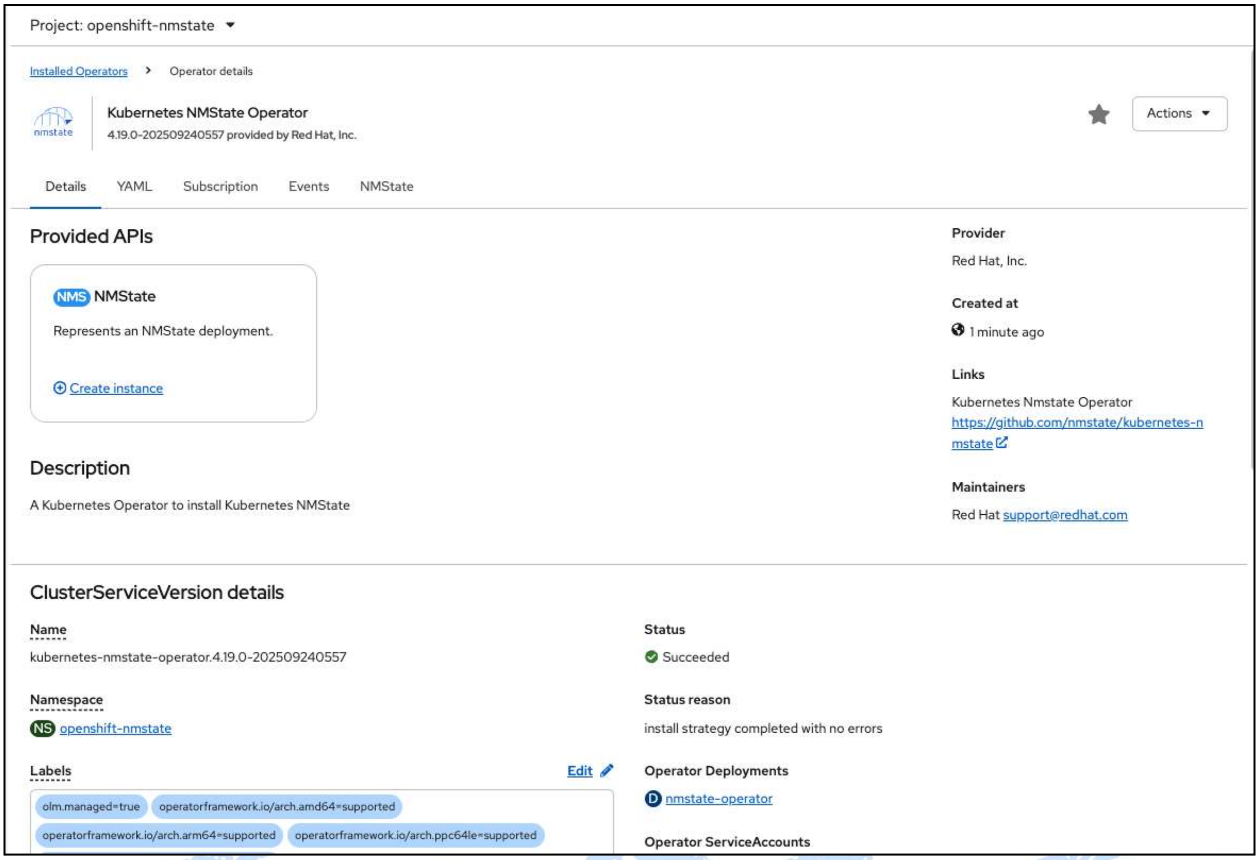
STEP 3. Select "***Install***".



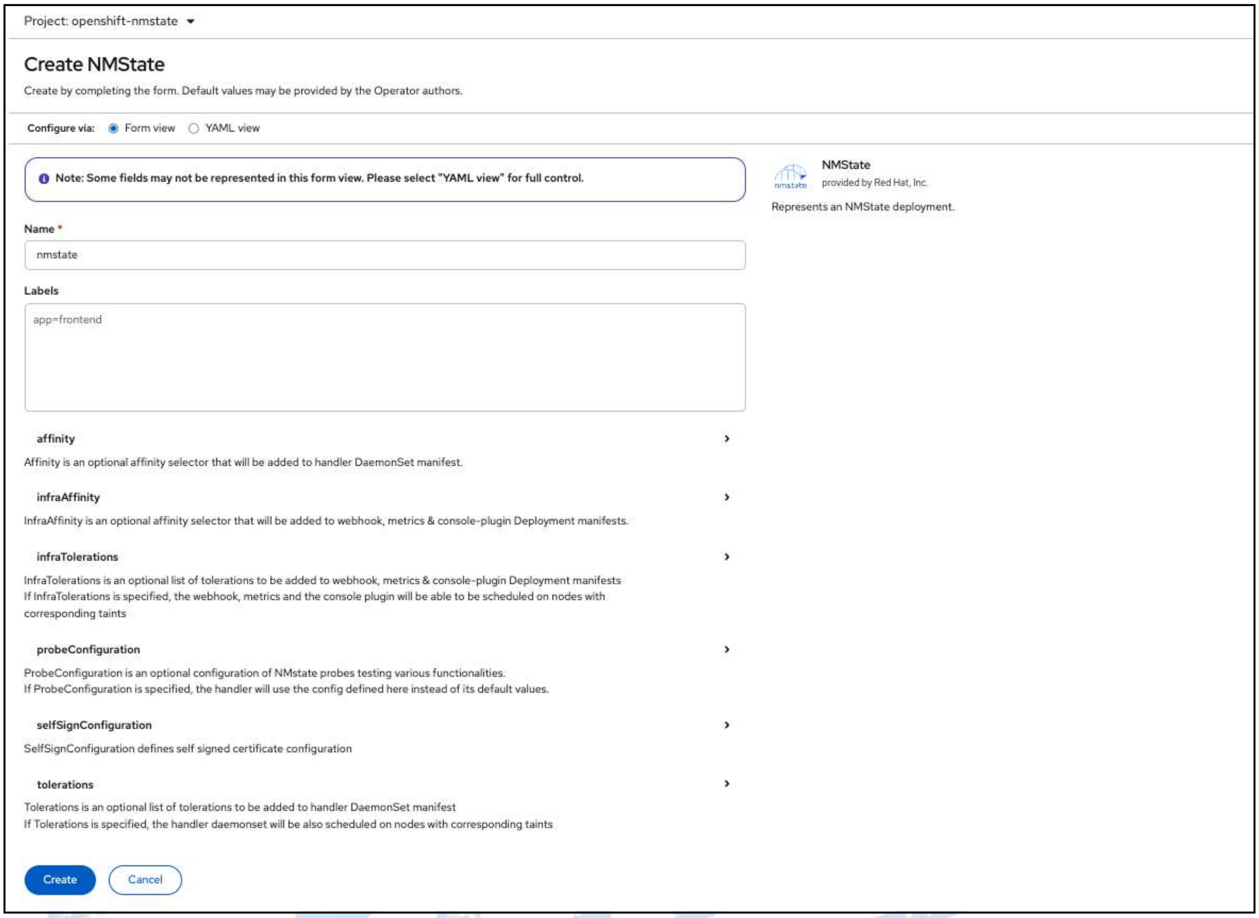
STEP 4. Once the operator installation is completed, select "***View Operator***".



STEP 5. Select ***"Create instance"***.



STEP 6. Select ***"Create"***.



Create Secondary Network

OpenShift Container Platform 4.18: Virtualization – Networking – [10.9 Connecting a virtual machine to an OVN-Kubernetes secondary network]

Create the following **NodeNetworkConfigurationPolicy** and **NetworkAttachmentDefinition.**

`STEP 1.` From the OpenShift console, navigate to **Networking > NodeNetworkConfigurationPolicy**, then click on "***Create NodeNetworkConfigurationPolicy***". Click on "***Edit YAML***", and then copy/paste the following YAML into the editor window. When complete, click "***Create***".

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: localnet-network-mapping
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ''
  desiredState:
    ovn:
```

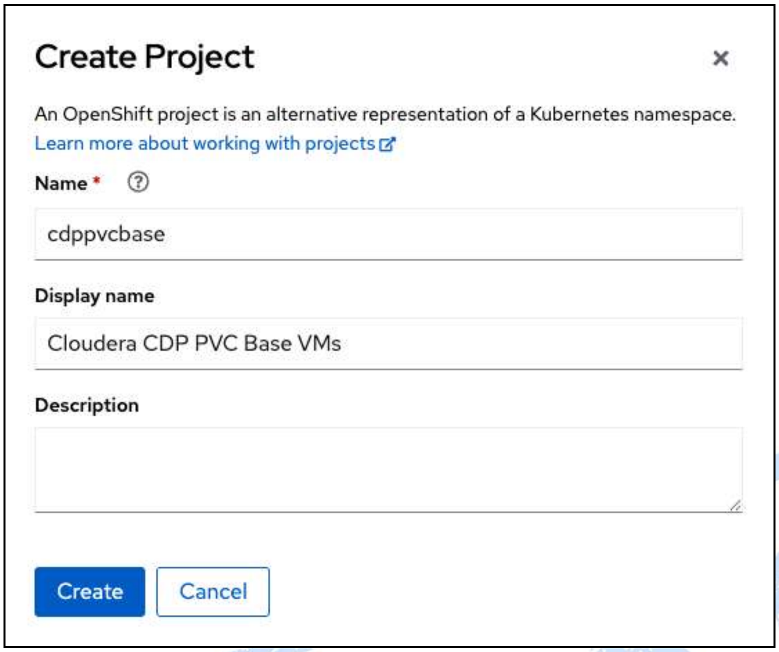
```
    bridge-mappings:
    - localnet: localnet-network
      bridge: br-ex
      state: present
```

STEP 2. From the OpenShift console, navigate to **Networking > NetworkAttachmentDefinition**, then click on "***Create NetworkAttachmentDefinition***". Click on "***YAML view***", and then copy/paste the following YAML into the editor window. When complete, click "***Create***".

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: localnet-network
  namespace: default
spec:
  config: |2
    {
            "cniVersion": "0.3.1",
            "name": "localnet-network",
            "type": "ovn-k8s-cni-overlay",
            "topology": "localnet",
            "netAttachDefName": "default/localnet-network"
    }
```

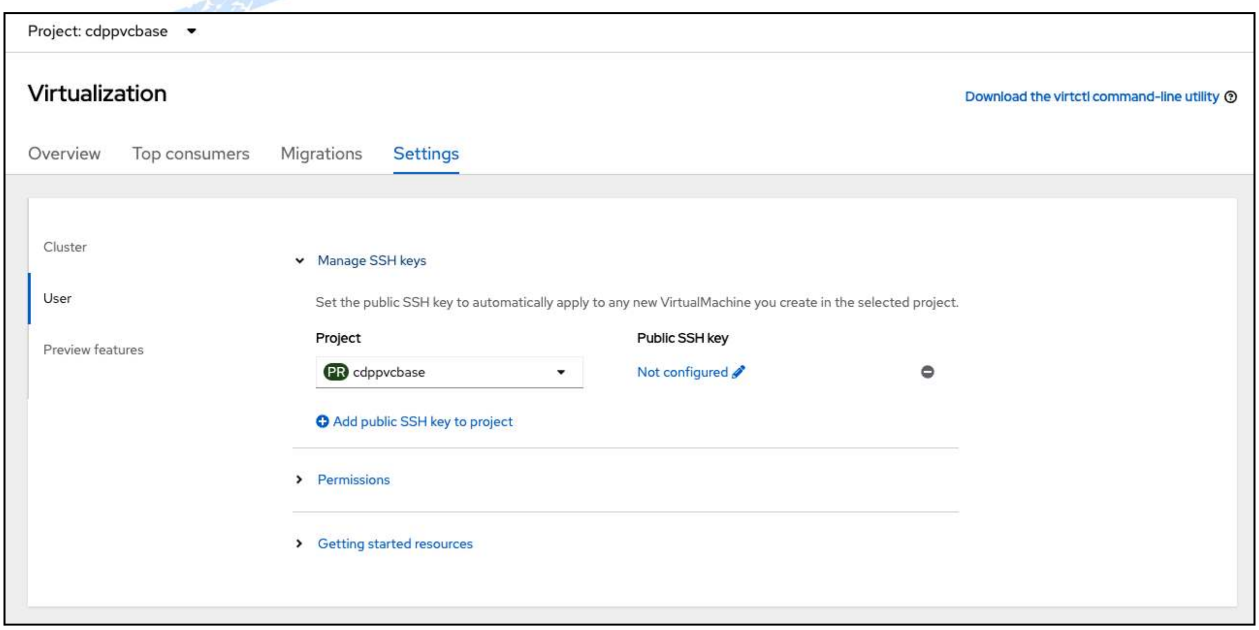

Create Project for CDP PVC Base VMs

STEP 1. Navigate to **Home** > **Projects** from the OpenShift Console. Select the Create Project button to create the ***cdppvcbase*** project:

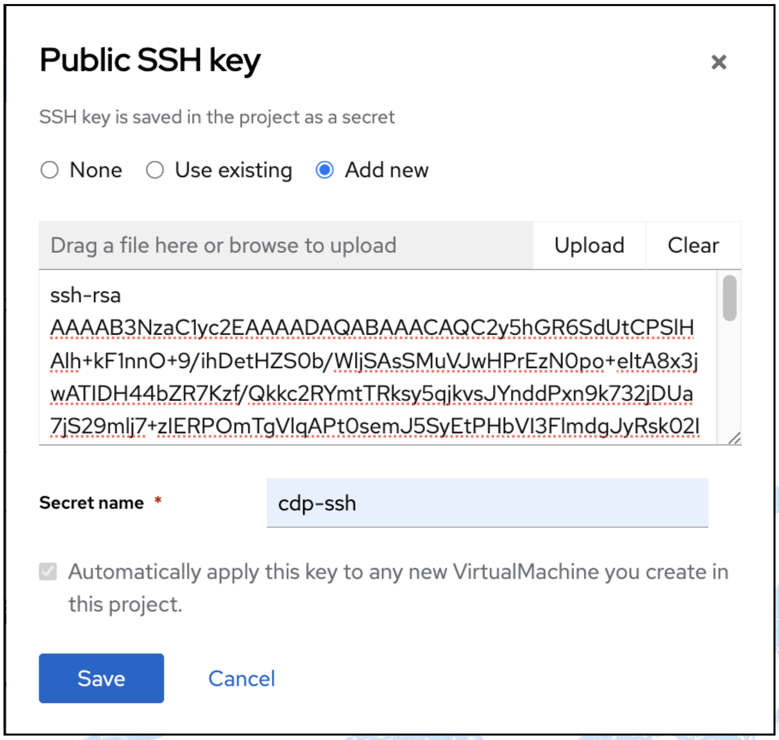


Create SSH Keys in Project

STEP 1. Navigate to **Virtualization** > **Overview** > **Settings** > **User** > **Manage SSH Keys**. Select the project ***cdppvcbase***, then click on ***Not configured*** to add the public SSH key to the project.

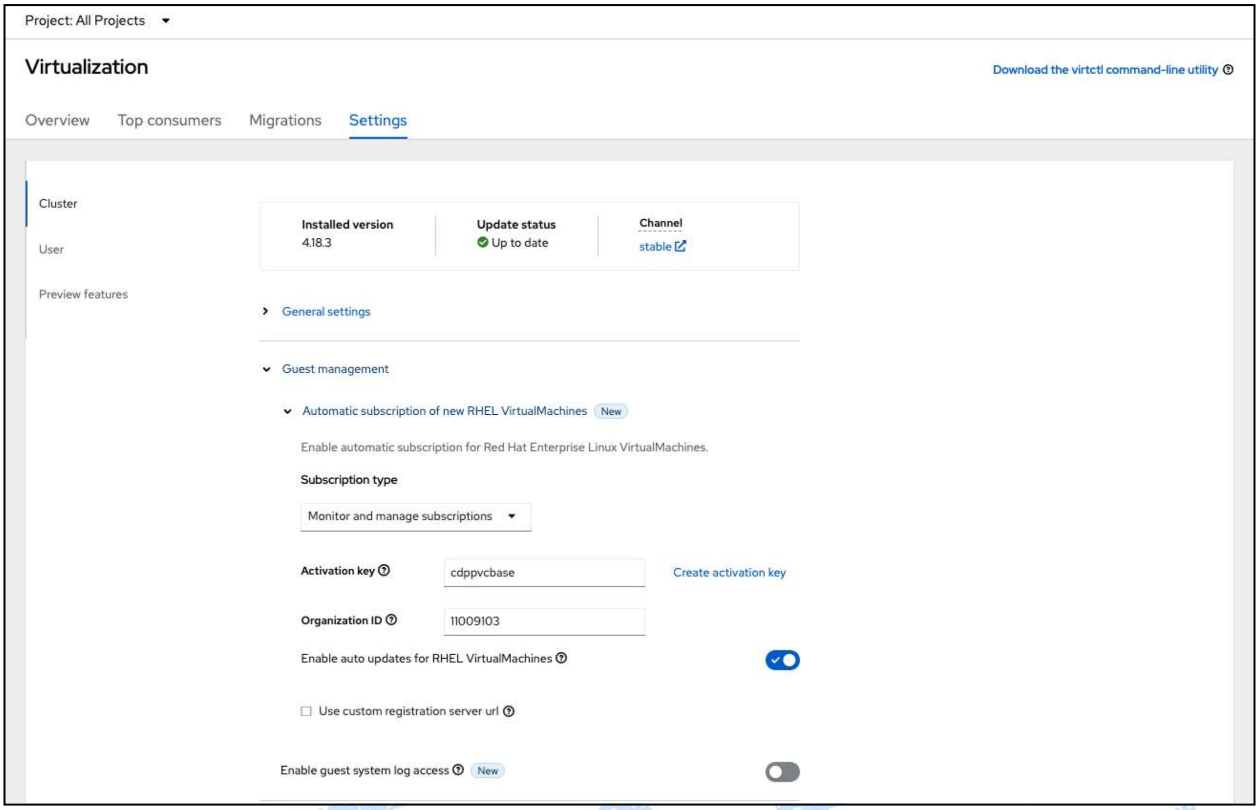


STEP 2. Select ***Add new***, and paste the contents of the public key into the page, provide a ***secret name***, and click on ***Save***.



Auto Register RHEL VMs

STEP 1. From the OpenShift console, navigate to ***Virtualization > Overview > Settings > Cluster > Guest Management > Automatic subscription of new RHEL VirtualMachines***. Select ***Monitor and manage subscriptions***. Enter the ***Activation key***, ***Organization ID*** and select ***Enable auto updates for RHEL VirtualMachines***.



For more information, refer to [Blog: Subscribing RHEL VMs in OpenShift Virtualization]()

To create an activation key for your Red Hat Account and Org, you can follow the link provided at:
https://console.redhat.com/settings/connector/activation-keys

This registration only happens during VM creation via the CloudInit script.

NOTE: In case the VM does not register with Red Hat Insights, you can use the following command to manually register the VM: rhc connect

Example:

```
# rhc connect --activation-key [your_activation_key] --organization [your_org_#]
```

This can also be performed by using the ansible command, in case it isn’t applied during VM creation.

```
ansible all -m shell -a "subscription-manager register --org=[your_org_#] --activationkey=[your_activation_key]"
```


Create VMs on OpenShift Virtualization

The following table is provided as a reference for the VMs that will be created for CDP Base:

Hostname	CPU	RAM	Disk	IP Address
cldr-mngr.<domain_name>	32	64 GiB	600 GiB	10.1.49.100/23
cldr-utility.<domain_name>	32	64 GiB	600 GiB	10.1.49.101/23
pvcbase-master[01-03].<domain_name>	32	64 GiB	500 GiB	10.1.49.102-104/23
pvcbase-worker[01-96].<domain_name>	32	64 GiB	500 GiB	10.1.49.105-199/23

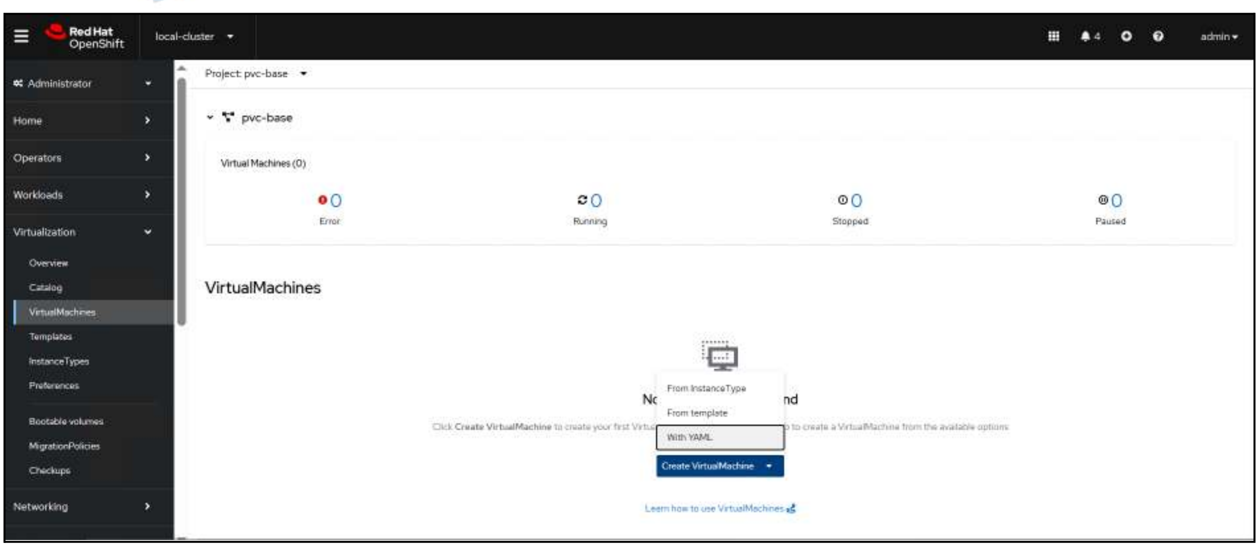
<domain_name> for the RH scale lab will be: cdp.rdu2.scalelab.redhat.com

The IPA server was created in a server (or a VM) running outside from OpenShift Virtualization:

ipaserver.<domain_name>	16	32 GiB	250 GiB	10.1.49.1/23

STEP 1. We will quickly create the VirtualMachines using the YAML definition provided below.

From the OpenShift console, navigate to ***Virtualization*** > ***VirtualMachines***. Make sure that the ***"cdppvcbase"*** Project is shown in the upper left part of the screen. Click on the ***"Create VirtualMachine"*** button and select ***"With YAML"***.

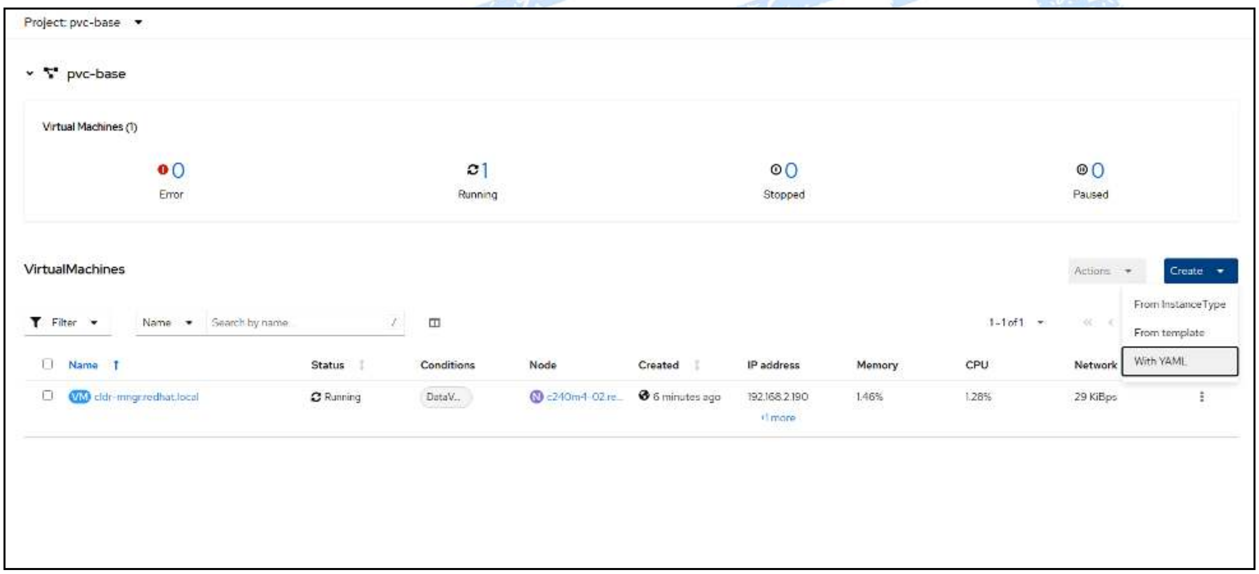


STEP 2. From the Create VirtualMachine screen, copy/paste the YAML code provided below into the editor window replacing the code that is initially shown.

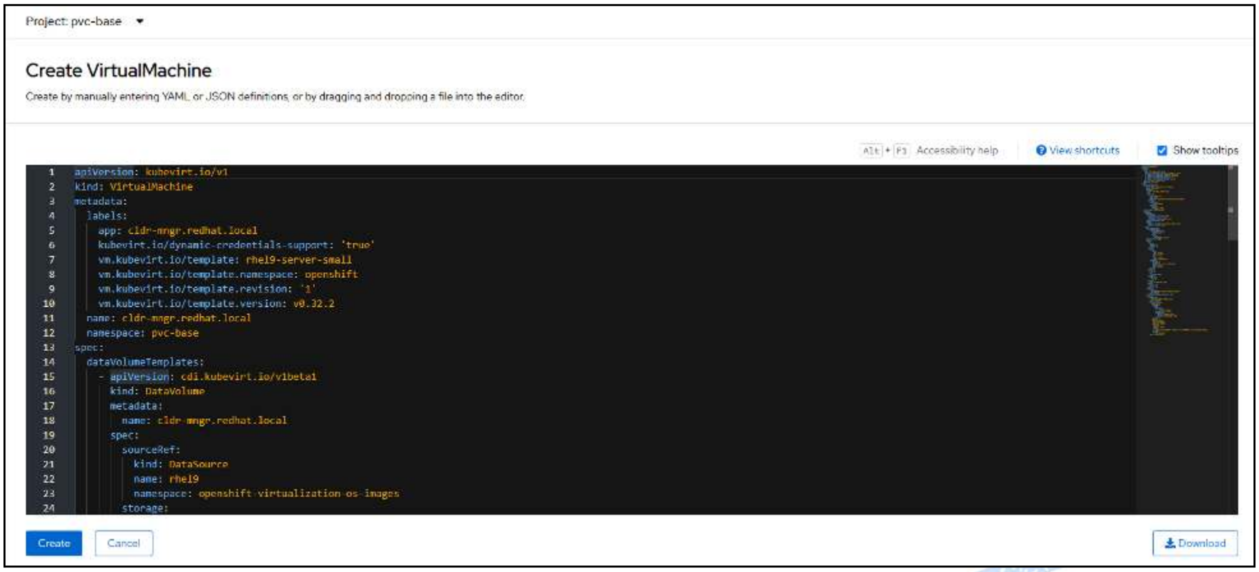
STEP 3. To create the VM for **cldr-mngr**, no changes are needed to the provided YAML code. Click on "Create".

STEP 4. Next, we will create the other VMs; using the VirtualMachine YAML code for the remaining VMs that is provided at the bottom of this section; after step 8.

STEP 5. Navigate back to ***Virtualization*** > ***VirtualMachines***, and click on the "***Create***" button on the right side of the screen and then select "***With YAML***".

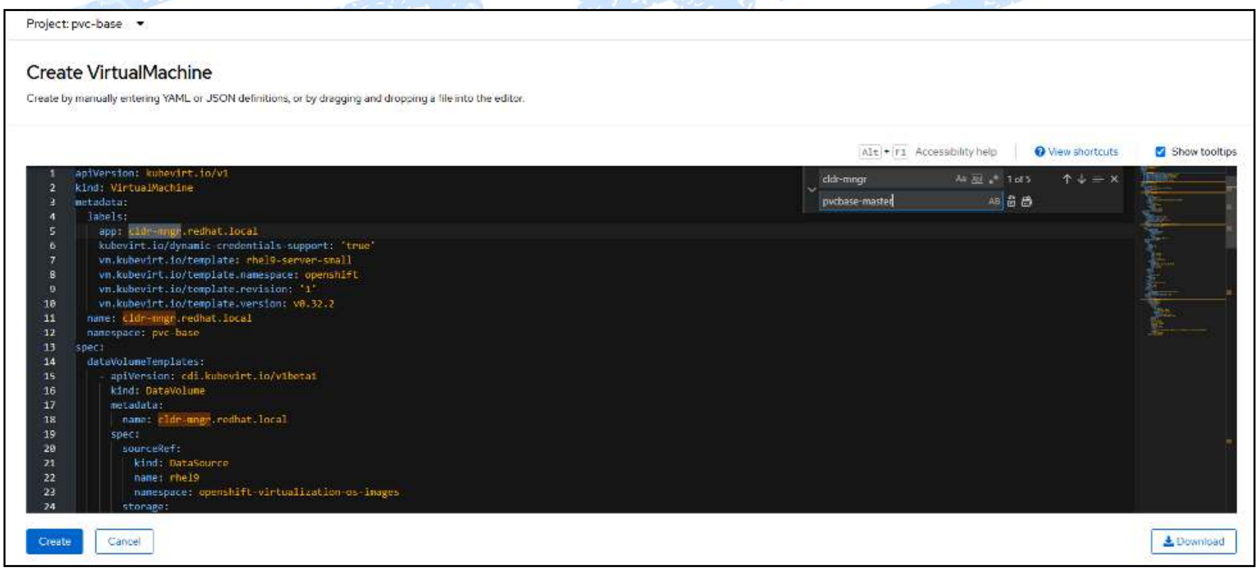


STEP 6. Again, copy/paste the provided YAML code into the editor window, replacing the code that is initially shown.



NOTE: Use the VirtualMachine YAML code for each of the remaining VMs that is provided at the bottom of this section; after step 8.

STEP 7. When any additional/optional changes have been made to the configuration parameters, click on "Create".



STEP 8. Repeat the last three steps until all of the VMs have been created.

The following is the cloud-init section that is included in each of the VM YAML definitions provided.

cloud-init section of the VirtualMachine YAML

```
      - cloudInitNoCloud:
           networkData: |
             ethernets:
               eth0:
                 addresses:
                   - 10.1.49.100/23       # Set IP v4 address
                 gateway4: 10.1.49.254     # Set default gateway
                 nameservers:
                   search: [cdp.rdu2.scalelab.redhat.com]  # Set DNS search domain
                   addresses: [10.1.49.1]  # Set DNS server address
             version: 2
           userData: |-
             #cloud-config
             user: cloud-user              # Create user account
             password: r3dh4t!             # Set password for user account
             chpasswd: { expire: False }   # Disable password expiration
             ssh_pwauth: True              # Enable password login for root over ssh
             hostname: cldr-mngr.cdp.rdu2.scalelab.redhat.com    # set hostname for VM
             packages:
               - dnf-automatic
             runcmd:
               - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
               - subscription-manager release --set=9.5          # Set RHEL minor version
               - systemctl enable --now dnf-automatic-install.timer
               - |
                 #!/bin/bash
                 # Add entries for sysctl
                 echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
                 echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf
                 echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
                 echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
                 echo "vm.swappiness=1" >> /etc/sysctl.conf
                 # Enable sysctl
                 sysctl -p
```

The VirtualMachine YAML code for all 7 of the VMs are provided below. Refer to the section OPTIONAL Method to create VMs for the process to create the VM configuration and its associated YAML code.

cldr-mngr VirtualMachine YAML

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  annotations:
    kubemacpool.io/transaction-timestamp: '2025-10-27T20:30:50.413104701Z'
    kubevirt.io/latest-observed-api-version: v1
    kubevirt.io/storage-observed-api-version: v1
    vm.kubevirt.io/validations: |
      [
```

```
      {
        "name": "minimal-required-memory",
        "path": "jsonpath::.spec.domain.memory.guest",
        "rule": "integer",
        "message": "This VM requires more memory.",
        "min": 1610612736
      }
    ]
resourceVersion: '8837376'
name: cldr-mngr
uid: 38f278e3-7b56-4f65-8586-c4e16adbc40e
creationTimestamp: '2025-10-27T20:30:50Z'
generation: 1
managedFields:
  - apiVersion: kubevirt.io/v1
    fieldsType: FieldsV1
    fieldsV1:
      'f:metadata':
        'f:annotations':
          .: {}
          'f:kubemacpool.io/transaction-timestamp': {}
          'f:vm.kubevirt.io/validations': {}
        'f:labels':
          .: {}
          'f:app': {}
          'f:kubevirt.io/dynamic-credentials-support': {}
          'f:vm.kubevirt.io/template': {}
          'f:vm.kubevirt.io/template.namespace': {}
          'f:vm.kubevirt.io/template.revision': {}
          'f:vm.kubevirt.io/template.version': {}
      'f:spec':
        .: {}
        'f:dataVolumeTemplates': {}
        'f:runStrategy': {}
        'f:template':
          .: {}
          'f:metadata':
            .: {}
            'f:annotations':
              .: {}
              'f:vm.kubevirt.io/flavor': {}
              'f:vm.kubevirt.io/os': {}
              'f:vm.kubevirt.io/workload': {}
            'f:creationTimestamp': {}
            'f:labels':
              .: {}
              'f:affinity_group_1': {}
              'f:kubevirt.io/domain': {}
              'f:kubevirt.io/size': {}
          'f:spec':
            'f:accessCredentials': {}
```

```
          'f:volumes': {}
          'f:hostname': {}
          'f:architecture': {}
          .: {}
          'f:terminationGracePeriodSeconds': {}
          'f:domain':
            .: {}
            'f:cpu':
              .: {}
              'f:cores': {}
              'f:sockets': {}
              'f:threads': {}
            'f:devices':
              .: {}
              'f:disks': {}
              'f:interfaces': {}
              'f:networkInterfaceMultiqueue': {}
              'f:rng': {}
            'f:features':
              .: {}
              'f:acpi': {}
              'f:smm':
                .: {}
                'f:enabled': {}
            'f:firmware':
              .: {}
              'f:bootloader':
                .: {}
                'f:efi': {}
            'f:machine':
              .: {}
              'f:type': {}
            'f:memory':
              .: {}
              'f:guest': {}
            'f:resources': {}
          'f:networks': {}
          'f:affinity':
            .: {}
            'f:podAntiAffinity':
              .: {}
              'f:requiredDuringSchedulingIgnoredDuringExecution': {}
  manager: OpenAPI-Generator
  operation: Update
  time: '2025-10-27T20:30:50Z'
- apiVersion: kubevirt.io/v1
  fieldsType: FieldsV1
  fieldsV1:
    'f:metadata':
      'f:annotations':
        'f:kubevirt.io/latest-observed-api-version': {}
```



```
          'f:kubevirt.io/storage-observed-api-version': {}
        'f:finalizers':
          .: {}
          'v:"kubevirt.io/virtualMachineControllerFinalize"': {}
    manager: virt-controller
    operation: Update
    time: '2025-10-27T20:30:50Z'
  - apiVersion: kubevirt.io/v1
    fieldsType: FieldsV1
    fieldsV1:
      'f:status':
        'f:printableStatus': {}
        'f:runStrategy': {}
        'f:conditions': {}
        .: {}
        'f:ready': {}
        'f:volumeSnapshotStatuses': {}
        'f:observedGeneration': {}
        'f:created': {}
        'f:desiredGeneration': {}
    manager: virt-controller
    operation: Update
    subresource: status
    time: '2025-10-27T20:31:39Z'
  namespace: cdppvcbase
  finalizers:
    - kubevirt.io/virtualMachineControllerFinalize
  labels:
    app: cldr-mngr
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-large
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.34.0
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        creationTimestamp: null
        name: cldr-mngr-dv
      spec:
        source:
          pvc:
            name: rhel9.5
            namespace: cdppvcbase
        storage:
          resources:
            requests:
              storage: 600Gi
          storageClassName: ocs-storagecluster-ceph-replica2-rbd
```



```
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: large
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        affinity_group_1: ''
        kubevirt.io/domain: cldr-mngr
        kubevirt.io/size: large
    spec:
      accessCredentials:
        - sshPublicKey:
            propagationMethod:
              noCloud: {}
            source:
              secret:
                secretName: cdp-ssh
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            - labelSelector:
                matchExpressions:
                  - key: affinity_group_1
                    operator: Exists
              topologyKey: kubernetes.io/hostname
      architecture: amd64
      domain:
        cpu:
          cores: 1
          sockets: 8
          threads: 1
        devices:
          disks:
            - disk:
                bus: virtio
              name: rootdisk
            - disk:
                bus: virtio
              name: cloudinitdisk
          interfaces:
            - bridge: {}
#              macAddress: '02:00:e3:00:02:3f'
              model: virtio
              name: default
              state: up
          networkInterfaceMultiqueue: true
          rng: {}
        features:
```

```
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.6.0
        memory:
          guest: 32Gi
        resources: {}
      hostname: cldr-mngr
      networks:
        - multus:
            networkName: default/localnet-network
          name: default
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: cldr-mngr-dv
          name: rootdisk
        - cloudInitNoCloud:
            networkData: |
              ethernets:
                eth0:
                  addresses:
                    - 10.1.49.100/23
                  gateway4: 10.1.49.254
                  nameservers:
                    search: [cdp.rdu2.scalelab.redhat.com]
                    addresses: [10.1.49.1]
              version: 2
            userData: |-
              #cloud-config
              user: cloud-user
              password: r3dh4t!
              chpasswd: { expire: False }
              ssh_pwauth: True
              hostname: cldr-mngr.cdp.rdu2.scalelab.redhat.com
              packages:
                - dnf-automatic
              runcmd:
                - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
                - subscription-manager release --set=9.5
                - systemctl enable --now dnf-automatic-install.timer
                - |
                  #!/bin/bash
                  # Add entries for sysctl
                  echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
                  echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf
```

```
                  echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
                  echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
                  echo "vm.swappiness=1" >> /etc/sysctl.conf
                  # Enable sysctl
                  sysctl -p
          name: cloudinitdisk
```

cldr-utility VirtualMachine YAML


```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  annotations:
    kubemacpool.io/transaction-timestamp: '2025-10-27T20:21:32.838262833Z'
    kubevirt.io/latest-observed-api-version: v1
    kubevirt.io/storage-observed-api-version: v1
    vm.kubevirt.io/validations: |
      [
        {
          "name": "minimal-required-memory",
          "path": "jsonpath::.spec.domain.memory.guest",
          "rule": "integer",
          "message": "This VM requires more memory.",
          "min": 1610612736
        }
      ]
  resourceVersion: '8817926'
  name: cldr-utility
  uid: 610bbc39-d416-4c0c-8d53-a06cb55c459f
  creationTimestamp: '2025-10-27T20:21:32Z'
  generation: 1
  managedFields:
    - apiVersion: kubevirt.io/v1
      fieldsType: FieldsV1
      fieldsV1:
        'f:metadata':
          'f:annotations':
            .: {}
            'f:kubemacpool.io/transaction-timestamp': {}
            'f:vm.kubevirt.io/validations': {}
          'f:labels':
            .: {}
            'f:app': {}
            'f:kubevirt.io/dynamic-credentials-support': {}
            'f:vm.kubevirt.io/template': {}
            'f:vm.kubevirt.io/template.namespace': {}
            'f:vm.kubevirt.io/template.revision': {}
            'f:vm.kubevirt.io/template.version': {}
```

```
        'f:spec':
          .: {}
          'f:dataVolumeTemplates': {}
          'f:runStrategy': {}
          'f:template':
            .: {}
            'f:metadata':
              .: {}
              'f:annotations':
                .: {}
                'f:vm.kubevirt.io/flavor': {}
                'f:vm.kubevirt.io/os': {}
                'f:vm.kubevirt.io/workload': {}
              'f:creationTimestamp': {}
              'f:labels':
                .: {}
                'f:affinity_group_1': {}
                'f:kubevirt.io/domain': {}
                'f:kubevirt.io/size': {}
            'f:spec':
              'f:accessCredentials': {}
              'f:volumes': {}
              'f:hostname': {}
              'f:architecture': {}
              .: {}
              'f:terminationGracePeriodSeconds': {}
              'f:domain':
                .: {}
                'f:cpu':
                  .: {}
                  'f:cores': {}
                  'f:sockets': {}
                  'f:threads': {}
                'f:devices':
                  .: {}
                  'f:disks': {}
                  'f:interfaces': {}
                  'f:networkInterfaceMultiqueue': {}
                  'f:rng': {}
                'f:features':
                  .: {}
                  'f:acpi': {}
                  'f:smm':
                    .: {}
                    'f:enabled': {}
                'f:firmware':
                  .: {}
                  'f:bootloader':
                    .: {}
                    'f:efi': {}
                'f:machine':
```

```
                .: {}
                'f:type': {}
              'f:memory':
                .: {}
                'f:guest': {}
              'f:resources': {}
            'f:networks': {}
            'f:affinity':
              .: {}
              'f:podAntiAffinity':
                .: {}
                'f:requiredDuringSchedulingIgnoredDuringExecution': {}
    manager: OpenAPI-Generator
    operation: Update
    time: '2025-10-27T20:21:32Z'
  - apiVersion: kubevirt.io/v1
    fieldsType: FieldsV1
    fieldsV1:
      'f:metadata':
        'f:annotations':
          'f:kubevirt.io/latest-observed-api-version': {}
          'f:kubevirt.io/storage-observed-api-version': {}
        'f:finalizers':
          .: {}
          'v:"kubevirt.io/virtualMachineControllerFinalize"': {}
    manager: virt-controller
    operation: Update
    time: '2025-10-27T20:21:32Z'
  - apiVersion: kubevirt.io/v1
    fieldsType: FieldsV1
    fieldsV1:
      'f:status':
        'f:printableStatus': {}
        'f:runStrategy': {}
        'f:conditions': {}
        .: {}
        'f:ready': {}
        'f:volumeSnapshotStatuses': {}
        'f:observedGeneration': {}
        'f:created': {}
        'f:desiredGeneration': {}
    manager: virt-controller
    operation: Update
    subresource: status
    time: '2025-10-27T20:22:14Z'
namespace: cdppvcbase
finalizers:
  - kubevirt.io/virtualMachineControllerFinalize
labels:
  app: cldr-utility
  kubevirt.io/dynamic-credentials-support: 'true'
```

```yaml
    vm.kubevirt.io/template: rhel9-server-large
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.34.0
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        creationTimestamp: null
        name: cldr-utility-dv
      spec:
        source:
          pvc:
            name: rhel9.5
            namespace: cdppvcbase
        storage:
          resources:
            requests:
              storage: 600Gi
          storageClassName: ocs-storagecluster-ceph-replica2-rbd
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: large
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        affinity_group_1: ''
        kubevirt.io/domain: cldr-utility
        kubevirt.io/size: large
    spec:
      accessCredentials:
        - sshPublicKey:
            propagationMethod:
              noCloud: {}
            source:
              secret:
                secretName: cdp-ssh
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            - labelSelector:
                matchExpressions:
                  - key: affinity_group_1
                    operator: Exists
              topologyKey: kubernetes.io/hostname
      architecture: amd64
      domain:
```

```
        cpu:
          cores: 1
          sockets: 8
          threads: 1
        devices:
          disks:
            - disk:
                bus: virtio
              name: rootdisk
            - disk:
                bus: virtio
              name: cloudinitdisk
          interfaces:
            - bridge: {}
#                macAddress: '02:00:e3:00:02:29'
              model: virtio
              name: default
              state: up
          networkInterfaceMultiqueue: true
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.6.0
        memory:
          guest: 32Gi
        resources: {}
      hostname: cldr-utility
      networks:
        - multus:
            networkName: default/localnet-network
          name: default
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: cldr-utility-dv
          name: rootdisk
        - cloudInitNoCloud:
            networkData: |
              ethernets:
                eth0:
                  addresses:
                    - 10.1.49.101/23
                  gateway4: 10.1.49.254
                  nameservers:
                    search: [cdp.rdu2.scalelab.redhat.com]
```

```
                    addresses: [10.1.49.1]
              version: 2
            userData: |-
              #cloud-config
              user: cloud-user
              password: r3dh4t!
              chpasswd: { expire: False }
              ssh_pwauth: True
              hostname: cldr-utility.cdp.rdu2.scalelab.redhat.com
              packages:
                - dnf-automatic
              runcmd:
                - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
                - subscription-manager release --set=9.5
                - systemctl enable --now dnf-automatic-install.timer
                - |
                  #!/bin/bash
                  # Add entries for sysctl
                  echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
                  echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf
                  echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
                  echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
                  echo "vm.swappiness=1" >> /etc/sysctl.conf
                  # Enable sysctl
                  sysctl -p
          name: cloudinitdisk
```

pvcbase-masterXX VirtualMachine YAML

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: pvcbase-master01
  generation: 1
  namespace: cdppvcbase
  finalizers:
    - kubevirt.io/virtualMachineControllerFinalize
  labels:
    app: pvcbase-master01
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-large
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.34.0
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
```

```
      creationTimestamp: null
      name: pvcbase-master01-dv
    spec:
      source:
        pvc:
          name: rhel9.5
          namespace: cdppvcbase
      storage:
        resources:
          requests:
            storage: 500Gi
        storageClassName: ocs-storagecluster-ceph-replica2-rbd
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: large
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      affinity_group_1: ''
      kubevirt.io/domain: pvcbase-master01
      kubevirt.io/size: large
  spec:
    accessCredentials:
      - sshPublicKey:
          propagationMethod:
            noCloud: {}
          source:
            secret:
              secretName: cdp-ssh
    affinity:
      podAntiAffinity:
        requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
                - key: affinity_group_1
                  operator: Exists
            topologyKey: kubernetes.io/hostname
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 8
        threads: 1
      devices:
        disks:
          - disk:
              bus: virtio
            name: rootdisk
```

```
          - disk:
              bus: virtio
            name: cloudinitdisk
        interfaces:
          - bridge: {}
            macAddress: '02:00:e3:00:02:a1'
            model: virtio
            name: default
            state: up
        networkInterfaceMultiqueue: true
        rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
      firmware:
        bootloader:
          efi: {}
      machine:
        type: pc-q35-rhel9.6.0
      memory:
        guest: 32Gi
      resources: {}
    hostname: pvcbase-master01
    networks:
      - multus:
          networkName: default/localnet-network
        name: default
    terminationGracePeriodSeconds: 180
    volumes:
      - dataVolume:
          name: pvcbase-master01-dv
        name: rootdisk
      - cloudInitNoCloud:
          networkData: |
            ethernets:
              eth0:
                addresses:
                  - 10.1.49.102/23
                gateway4: 10.1.49.254
                nameservers:
                  search: [cdp.rdu2.scalelab.redhat.com]
                  addresses: [10.1.49.1]
            version: 2
          userData: |-
            #cloud-config
            user: cloud-user
            password: r3dh4t!
            chpasswd: { expire: False }
            ssh_pwauth: True
            hostname: pvcbase-master01.cdp.rdu2.scalelab.redhat.com
```

```
            packages:
              - dnf-automatic
            runcmd:
              - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
              - subscription-manager release --set=9.5
              - systemctl enable --now dnf-automatic-install.timer
              - |
                #!/bin/bash
                # Add entries for sysctl
                echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
                echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf
                echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
                echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
                echo "vm.swappiness=1" >> /etc/sysctl.conf
                # Enable sysctl
                sysctl -p
          name: cloudinitdisk
```

pvcbase-workerXX VirtualMachine YAML

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: pvcbase-worker01
  labels:
    app: pvcbase-worker01
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-large
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.34.0
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        creationTimestamp: null
        name: pvcbase-worker01-dv
      spec:
        source:
          pvc:
            name: rhel9.5
            namespace: cdppvcbase
        storage:
          resources:
            requests:
              storage: 500Gi
          storageClassName: ocs-storagecluster-ceph-replica2-rbd
```



```yaml
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: large
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      affinity_group_1: ''
      kubevirt.io/domain: pvcbase-worker01
      kubevirt.io/size: large
  spec:
    accessCredentials:
      - sshPublicKey:
          propagationMethod:
            noCloud: {}
          source:
            secret:
              secretName: cdp-ssh
    affinity:
      podAntiAffinity:
        requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
                - key: affinity_group_1
                  operator: Exists
            topologyKey: kubernetes.io/hostname
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 8
        threads: 1
      devices:
        disks:
          - disk:
              bus: virtio
            name: rootdisk
          - disk:
              bus: virtio
            name: cloudinitdisk
        interfaces:
          - bridge: {}
            macAddress: '02:00:e3:00:02:42'
            model: virtio
            name: default
            state: up
        networkInterfaceMultiqueue: true
        rng: {}
      features:
```

```
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.6.0
        memory:
          guest: 32Gi
        resources: {}
      hostname: pvcbase-worker01
      networks:
        - multus:
            networkName: default/localnet-network
          name: default
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcbase-worker01-dv
          name: rootdisk
        - cloudInitNoCloud:
            networkData: |
              ethernets:
                eth0:
                  addresses:
                    - 10.1.49.105/23
                  gateway4: 10.1.49.254
                  nameservers:
                    search: [cdp.rdu2.scalelab.redhat.com]
                    addresses: [10.1.49.1]
              version: 2
            userData: |-
              #cloud-config
              user: cloud-user
              password: r3dh4t!
              chpasswd: { expire: False }
              ssh_pwauth: True
              hostname: pvcbase-worker01.cdp.rdu2.scalelab.redhat.com
              packages:
                - dnf-automatic
              runcmd:
                - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
                - subscription-manager release --set=9.5
                - systemctl enable --now dnf-automatic-install.timer
                - |
                  #!/bin/bash
                  # Add entries for sysctl
                  echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
                  echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf
```

```
                echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
                echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
                echo "vm.swappiness=1" >> /etc/sysctl.conf
                # Enable sysctl
                sysctl -p
        name: cloudinitdisk
```

OPTIONAL Method to create VMs

If needed, please refer to the optional method to create VMs from the VM Template Catalog:
[OPTIONAL Method to create VMs]
This method can be used to create the VirtualMachine YAML definitions.

CONFIGURE THE LDAP IDENTITY PROVIDER WITH OPENSHIFT:

`##### If needed, you can access the IPA Admin console:`
`https://idm.redhat.local/ipa/ui/`

`From the WebUI for IPAServer Administration, enter the same admin credentials used for CLI authentication: (i.e.` *admin/redhat123*`)` [root@idm ~]# **vi /etc/openldap/ldap.conf**

```
# File modified by ipa-client-install
. . .
# When no CA certificates are specified the Shared System Certificates
# are in use. In order to have these available along with the ones specified
# by TLS_CACERTDIR one has to include them explicitly:
#TLS_CACERT /etc/pki/tls/cert.pem

# System-wide Crypto Policies provide up to date cipher suite which should
# be used unless one needs a finer grinded selection of ciphers. Hence, the
# PROFILE=SYSTEM value represents the default behavior which is in place
# when no explicit setting is used. (see openssl-ciphers(1) for more info)
#TLS_CIPHER_SUITE PROFILE=SYSTEM

# Turning this off breaks GSSAPI used with krb5 when rdns = false
SASL_NOCANON      on

URI ldaps://ipaserver.cdp.rdu2.scalelab.redhat.com
BASE dc=cdp,dc=rdu2,dc=scalelab,dc=redhat,dc=com
SASL_MECH GSSAPI
```

If any changes are made to the IPA/IdM configuration, use the following command to restart the IPA server:
[root@idm ~]# **ipactl restart**

STEP 1. Create a secret with the Identity Management (IdM) admin user password. This secret will be used for the LDAP identity provider:

[root@bastion]# **oc create secret generic ldap-secret -n openshift-config \
--from-literal=bindPassword=redhat123**

STEP 2. TLS communication needs certificate authority validation, in this case, the CA is obtained from IdM.
The certificate is stored on the ipaserver in the file: **/etc/ipa/ca.crt**

Even when IdM is configured not to create/deploy a Certificate Authority, but instead use certificates provided by an external CA, the installer still stores the CA certificate in the file: **/etc/ipa/ca.crt**

Use the following command to download the Certificate Authority's root certificate to a file named ca.crt.

[root@bastion]# **scp
root@ipaserver.cdp.rdu2.scalelab.redhat.com:/etc/ipa/ca.crt /root/ca.crt**

```
or
[root@bastion ]# wget -c -nv
http://ipaserver.cdp.rdu2.scalelab.redhat.com/ipa/config/ca.crt
```

STEP 3. Create a configuration map containing the IdM Certificate Authority's root certificate to make OpenShift trust the IdM certificates.
The ConfigMap key must be named ca.crt.

```
[root@bastion ]# oc create configmap ldap-ca-configmap -n openshift-config \
--from-file=ca.crt=/root/ca.crt
```

STEP 4. Create a file to modify the OpenShift OAuth configuration by adding the LDAP identity provider and the other elements required for proper configuration.

```
[root@bastion ]# cat > ldap-cr.yaml << EOF
```

```
apiVersion: config.openshift.io/v1
kind: OAuth
metadata:
  name: cluster
spec:
  identityProviders:
    - name: ldap
      mappingMethod: claim
      type: LDAP
      ldap:
        attributes:
          email: []
          id:
            - dn
          name:
            - cn
          preferredUsername:
            - uid
        bindDN:
'uid=admin,cn=users,cn=accounts,dc=cdp,dc=rdu2,dc=scalelab,dc=redhat,dc=com'
        bindPassword:
          name: ldap-secret
        ca:
          name: ldap-ca-configmap
        insecure: false
        url:
'ldaps://ipaserver.cdp.rdu2.scalelab.redhat.com/cn=users,cn=accounts,dc=cdp,dc=rdu2,d
c=scalelab,dc=redhat,dc=com?uid'
```



```
EOF
```

STEP 5. Apply the custom resource and wait until the pods in the openshift-authentication namespace are restarted so that the identity provider is active:

```
[root@bastion ]# oc apply -f ldap-cr.yaml
```

STEP 6. View the status, as the pods in the openshift-authentication namespace redeploy. This can take ten minutes or more before the pods are ready.

```
[root@bastion ]# oc get clusteroperator/authentication
```

```
[root@bastion ]# oc get pods -l app=oauth-openshift -n
openshift-authentication
```

If there are any issues with the IdM integration, check the pod status and logs of the deployment.apps/oauth-openshift deployment in the openshift-authentication namespace.

```
[root@bastion ]# oc logs deployment.apps/oauth-openshift -n
openshift-authentication
```

STEP 7. Add the cluster-admin role to the IdM admin user that needs cluster administration privileges.

```
[root@bastion ]# oc adm policy add-cluster-role-to-user cluster-admin admin
Warning: User 'admin' not found
clusterrole.rbac.authorization.k8s.io/cluster-admin added: "admin"
```

STEP 8. Verify that you are able to log in with the IdM admin user.

```
[root@bastion ]# oc login -u admin -p redhat123
https://api.vlan601.rdu2.scalelab.redhat.com/:6443
```

```
[root@bastion ]# oc whoami
```

If necessary, the kubeadmin account is still available to use to login.

```
[root@bastion ]# oc login -u [your_openshift_username] -p
[your_openshift_password] https://api.vlan601.rdu2.scalelab.redhat.com:6443
```

Create OpenShift cluster using the Assisted Installer

The Assisted Installer is used to create an OpenShift cluster for CDP Data Services on bare-metal server nodes.

The following table is a reference of the bare-metal servers that will be used for the OpenShift cluster for CDP Data Services:

Hostname	CPU	RAM	Disk	MAC Address	IP Address
f36-h17-000-r640 (ocp master)	40	384 GiB	446 GiB	bc:97:e1:78:dd:91	10.1.49.216/23
f36-h18-000-r640 (ocp master)	40	384 GiB	446 GiB	bc:97:e1:78:d7:91	10.1.49.217/23
f36-h19-000-r640 (ocp master)	40	384 GiB	446 GiB	bc:97:e1:69:d2:41	10.1.49.218/23
d29-h11-000-r750 (ocp worker)	56	512 GiB	446 GiB	00:62:0b:1c:07:61	10.1.49.219/23
d29-h13-000-r750 (ocp worker)	56	512 GiB	446 GiB	00:62:0b:1d:d3:31	10.1.49.220/23
d30-h05-000-r750 (ocp worker)	56	512 GiB	446 GiB	00:62:0b:1d:9a:a1	10.1.49.221/23
d30-h07-000-r750 (ocp worker)	56	512 GiB	446 GiB	00:62:0b:1d:d1:a1	10.1.49.222/23

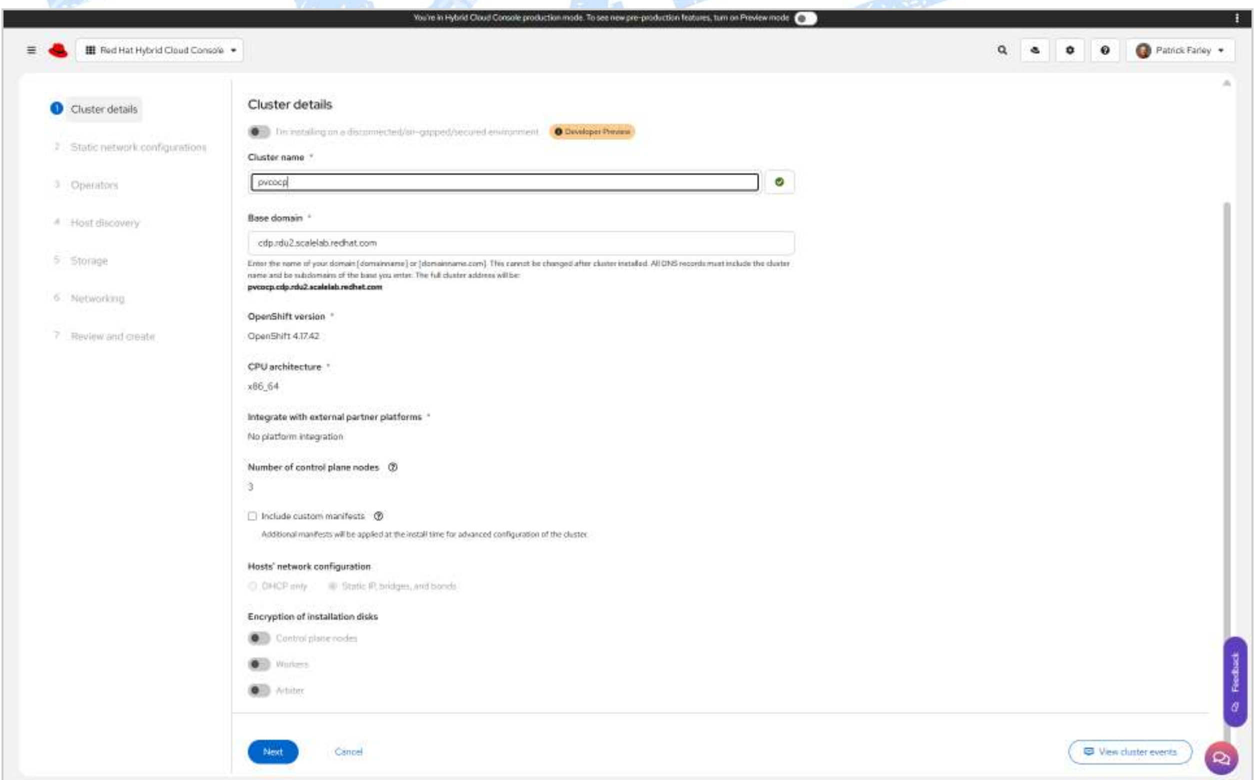
<domain_name> for the RH scale lab will be: cdp.rdu2.scalelab.redhat.com

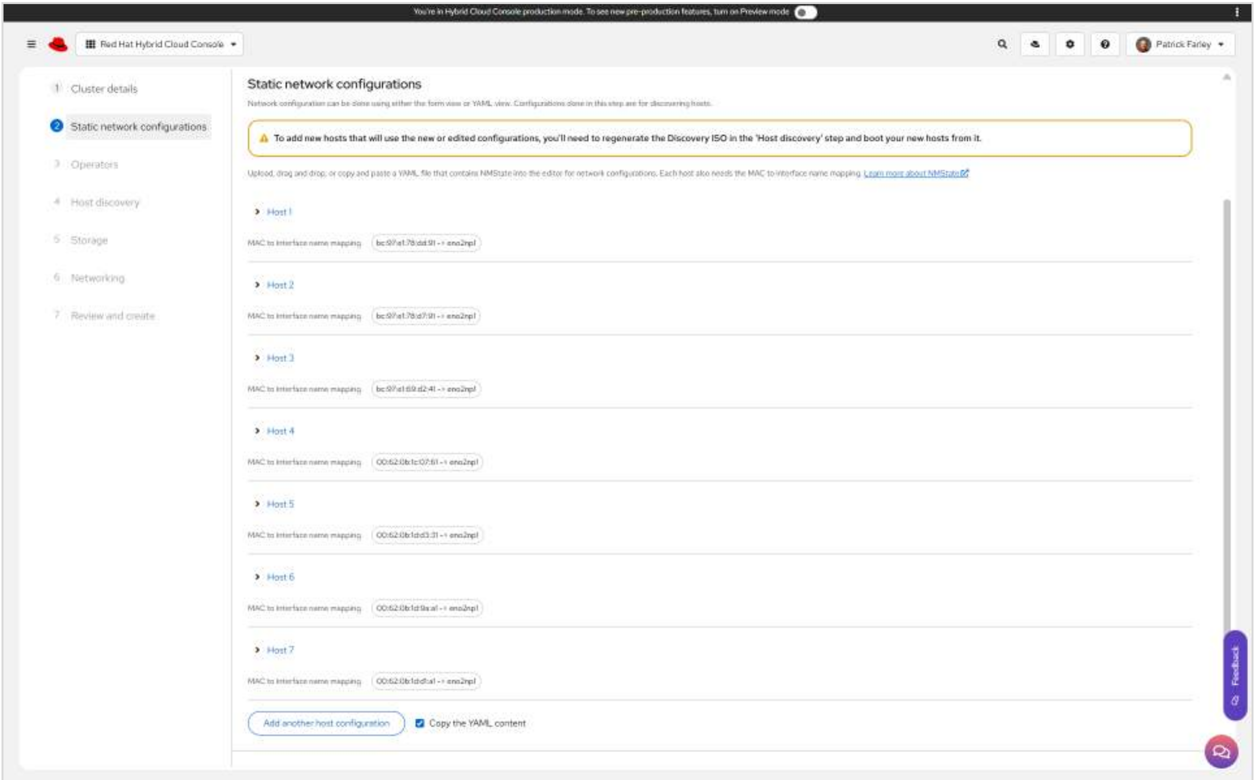
Installation parameters:

Parameter	Setting/Value
Cluster name	pvcocp
Base domain	cdp.rdu2.scalelab.redhat.com
OpenShift version	4.17.12
CPU architecture	x86_64
Integrate with external partner platforms	No platform integration
Number of control plane nodes	3 (highly available cluster)
Hosts’ network configuration	Static IP, bridges, and bonds
Configure via	YAML view

Networking stack type	IPv4
DNS	10.1.49.1
Machine network	10.1.48.0/23
Default gateway	10.1.49.254
api.pvcocp.cdp.rdu2.scalelab.redhat.com (API IP)	10.1.49.5
*.apps.pvcocp.cdp.rdu2.scalelab.redhat.com (Ingress IP)	10.1.49.6

STEP 1. Create a new OpenShift cluster using the OpenShift Assisted Installer from the Red Hat Hybrid Cloud Console:
https://console.redhat.com/openshift/assisted-installer/clusters/~new
Enter the cluster details using the installation parameters provided, then click ***Next***.





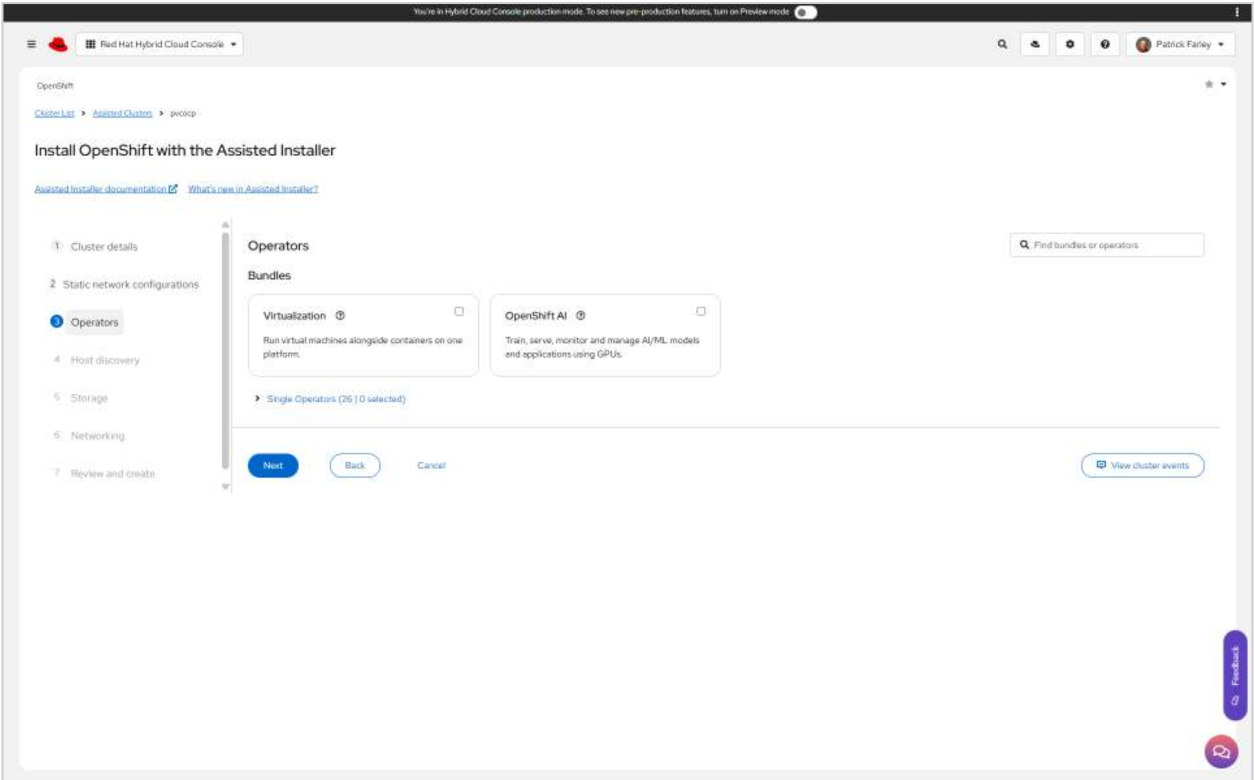
R640

```
interfaces:
- ipv4:
    address:
    - ip: 10.1.49.216  # Edit this IP address for each server
      prefix-length: 23
    dhcp: false
    enabled: true
  name: eno2np1
  state: up
  type: ethernet
  mtu: 9000
- ipv4:
    dhcp: false
    enabled: false
  name: eno1np0
  state: down
  type: ethernet
routes:
  config:
  - destination: 0.0.0.0/0
    next-hop-address: 10.1.49.254
    next-hop-interface: eno2np1
    table-id: 254
dns-resolver:
```

```
  config:
    server:
    - 10.1.49.1
    search:
    - cdp.rdu2.scalelab.redhat.com
```

R750

```
interfaces:
- ipv4:
    address:
    - ip: 10.1.49.216  # Edit this IP address for each server
      prefix-length: 23
    dhcp: false
    enabled: true
  name: eno12409np1
  state: up
  type: ethernet
  mtu: 9000
- ipv4:
    dhcp: false
    enabled: false
  name: eno12399np0
  state: down
  type: ethernet
routes:
  config:
  - destination: 0.0.0.0/0
    next-hop-address: 10.1.49.254
    next-hop-interface: eno2np1
    table-id: 254
dns-resolver:
  config:
    server:
    - 10.1.49.1
    search:
    - cdp.rdu2.scalelab.redhat.com
```



DISCOVERY ISO IMAGE

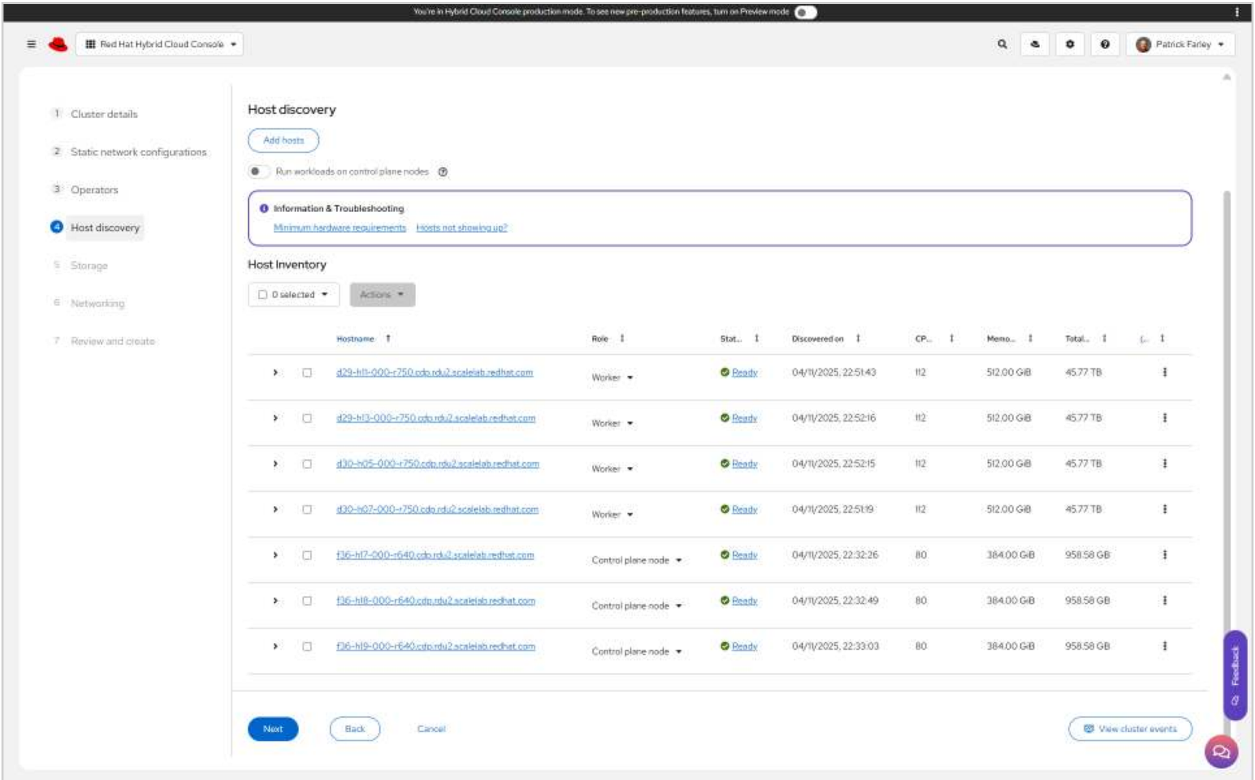
```
dnf install -y httpd

wget -O discovery_image_pvcocp.iso
'https://api.openshift.com/api/assisted-images/bytoken/eyJhbGciOiJIUzI1NiI
sInR5cCI6IkpXVCJ9.eyJleHAiOjE3NjIzNjM2MDAsInN1YiI6IjY5YTEyMzk2LTAyMTQtNDkx
Zi1hMWM4LWJjODUzZjlmYzY0NiJ9.3s-YXD5-BpeWK6MfWt3x9mR1wVaw7pPmJOkWWisv6Cc/4
.17/x86_64/full.iso'

cp discovery_image_pvcocp.iso /var/www/html

Verify that the discovery image is available at:
http://10.1.49.1/discovery_image_pvcocp.iso
```

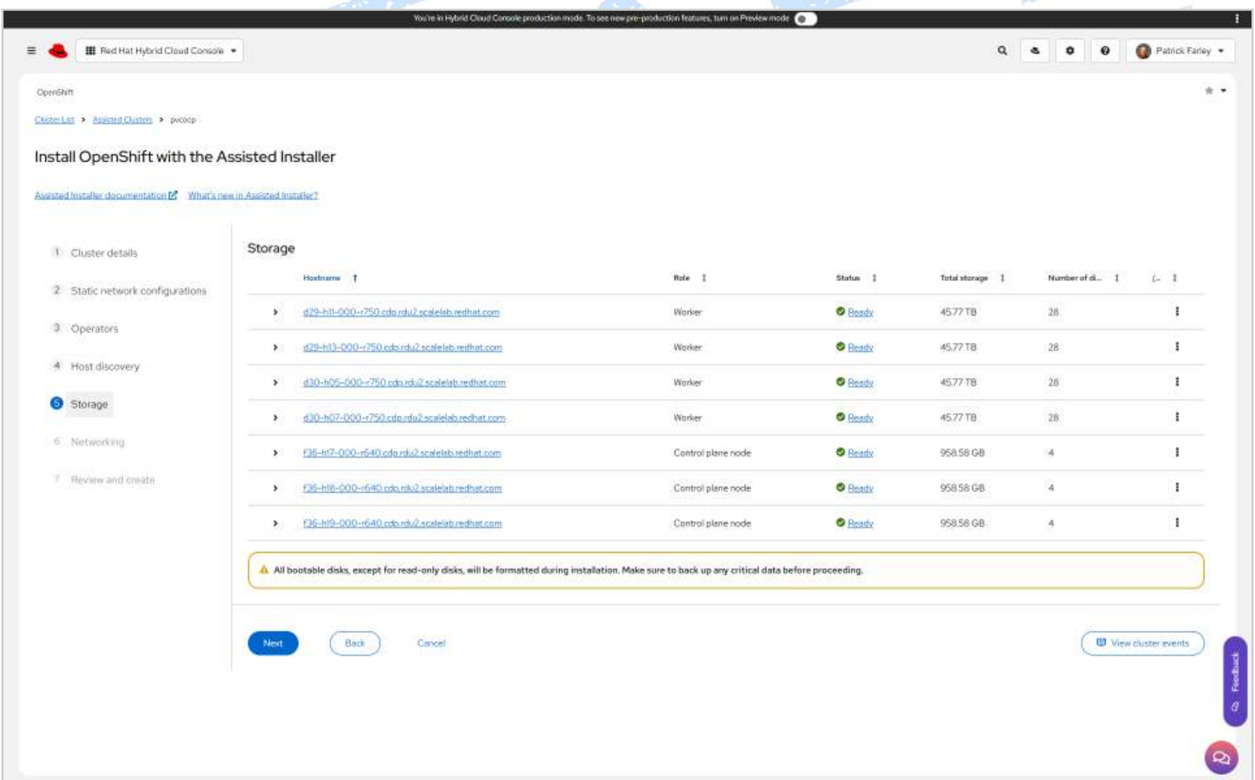
BADFISH

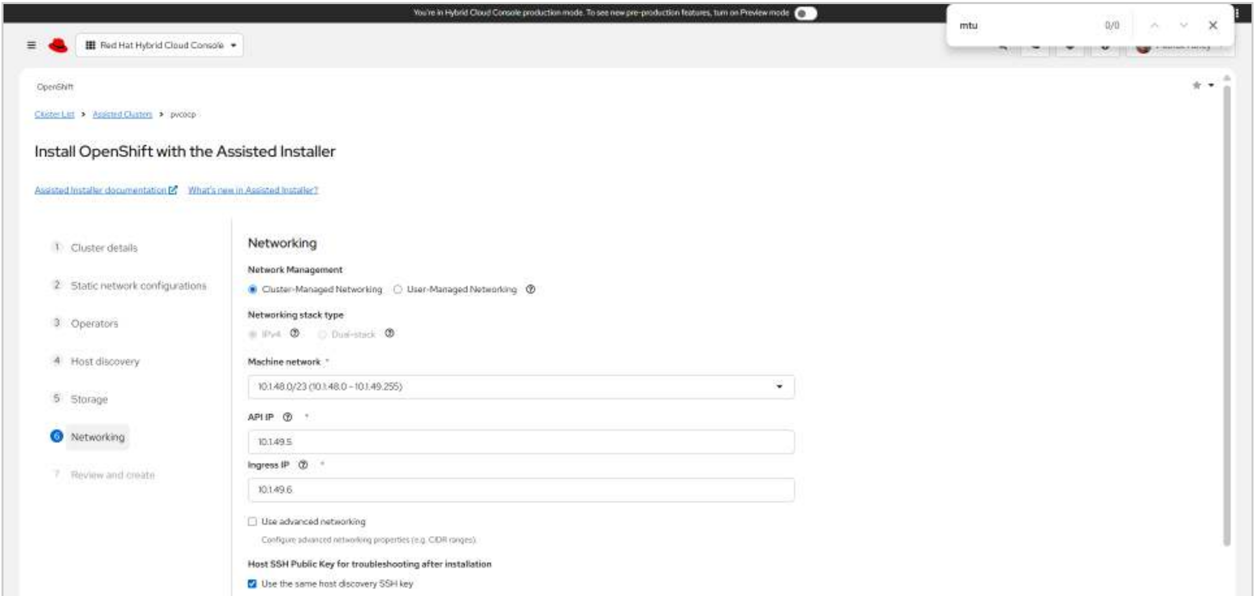
```
Configure the servers to boot from the Discovery image:

export USER=quads
export PASS=rdu2@4618
```

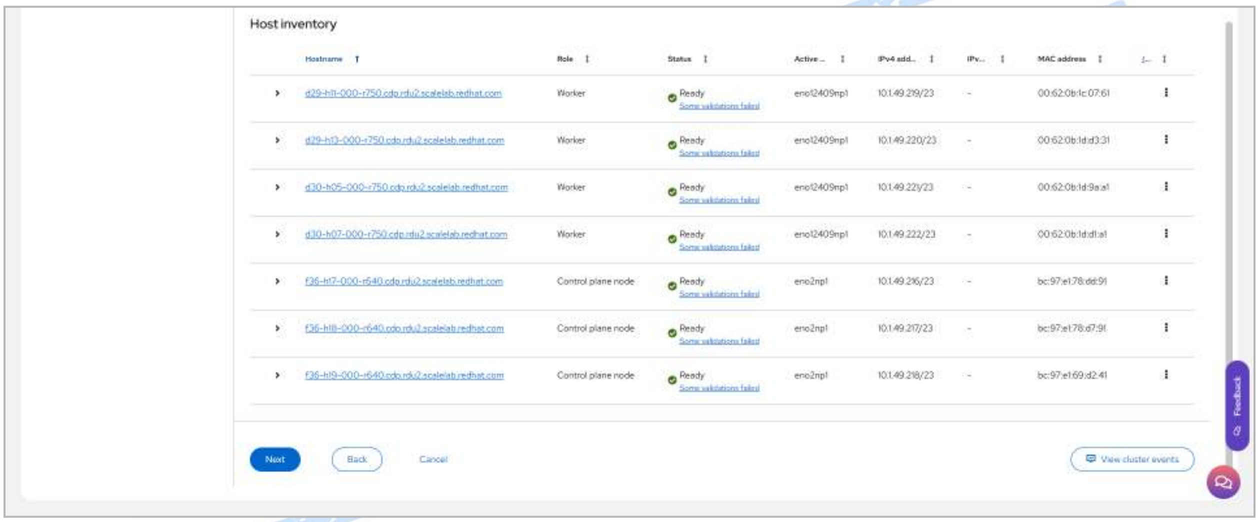
```
dnf install -y podman


vi masterlist
10.6.61.148
10.6.61.152
10.6.61.153


for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-off;done

for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --unmount-virtual-media;done

for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --mount-virtual-media
http://10.1.49.1/discovery_image_pvcocp.iso;done

for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --boot-to-virtual-media;done

for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-on;done

for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-state;done

Get Current Boot mode setting
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --get-bios-attribute --attribute BootMode;done

Set UEFI mode
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --set-bios-attribute --attribute BootMode --value
Uefi;done

Set BIOS mode
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --set-bios-attribute --attribute BootMode --value
Bios;done


vi workerlist
10.1.37.21
```

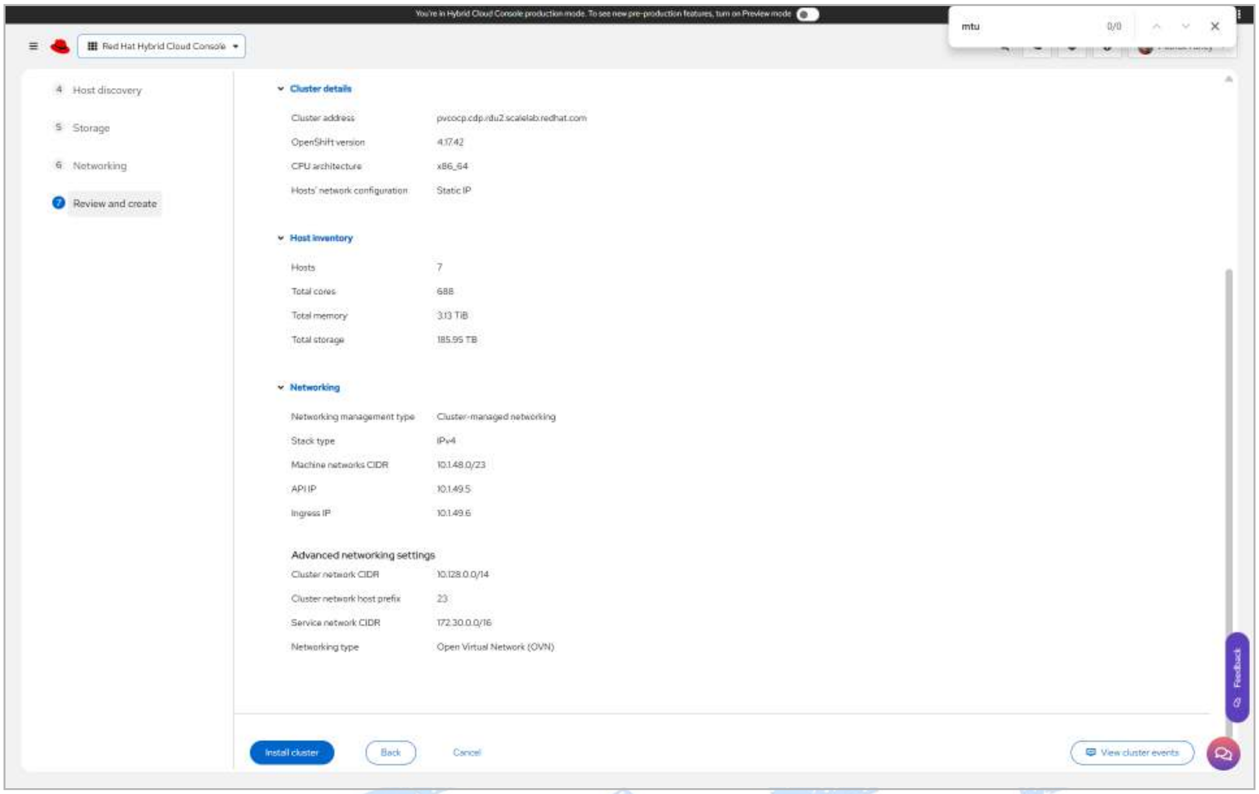
```
10.1.64.196
10.1.65.226
10.1.65.96
10.1.65.139
10.1.64.17
10.1.65.3

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-off;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --unmount-virtual-media;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --mount-virtual-media
http://10.1.49.1/discovery_image_pvcocp.iso;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --boot-to-virtual-media;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-on;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-state;done

Get Current Boot mode setting
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --get-bios-attribute --attribute BootMode;done

Set UEFI mode
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --set-bios-attribute --attribute BootMode --value
Uefi;done

Set BIOS mode
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --set-bios-attribute --attribute BootMode --value
Bios;done
```

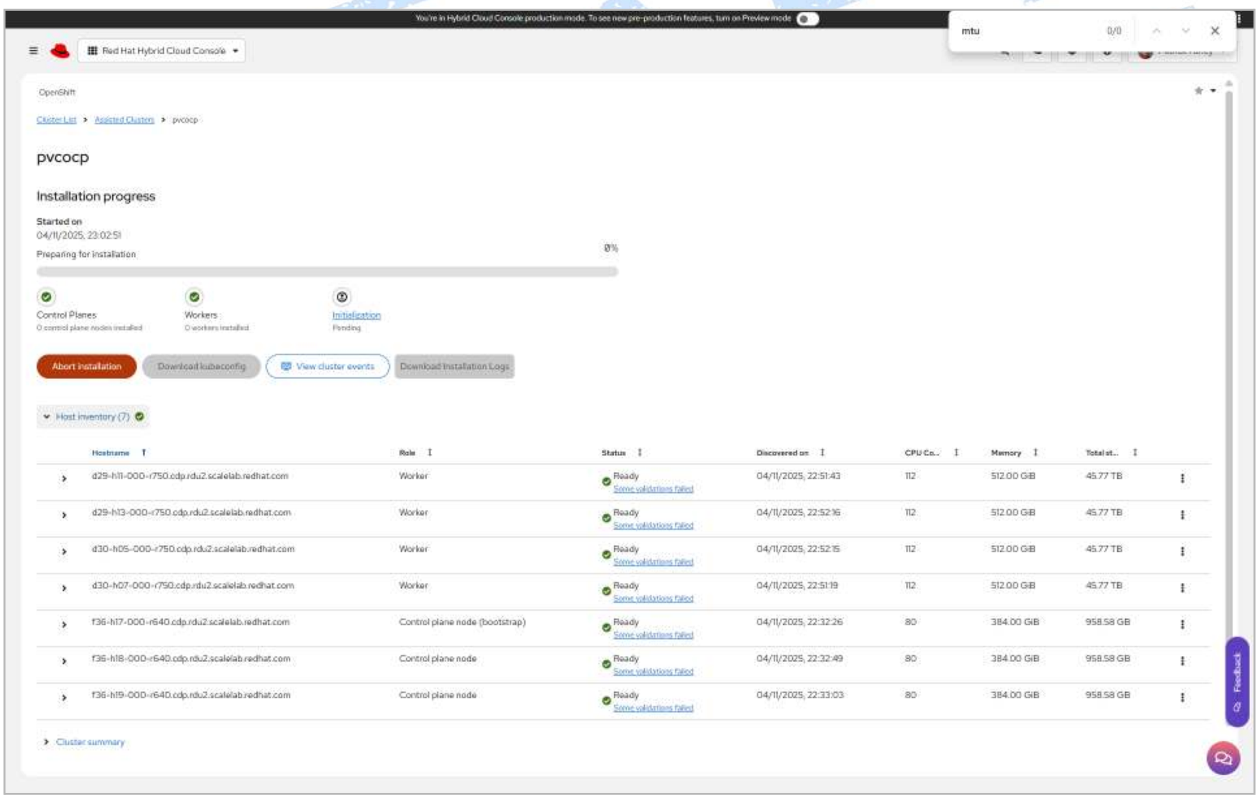



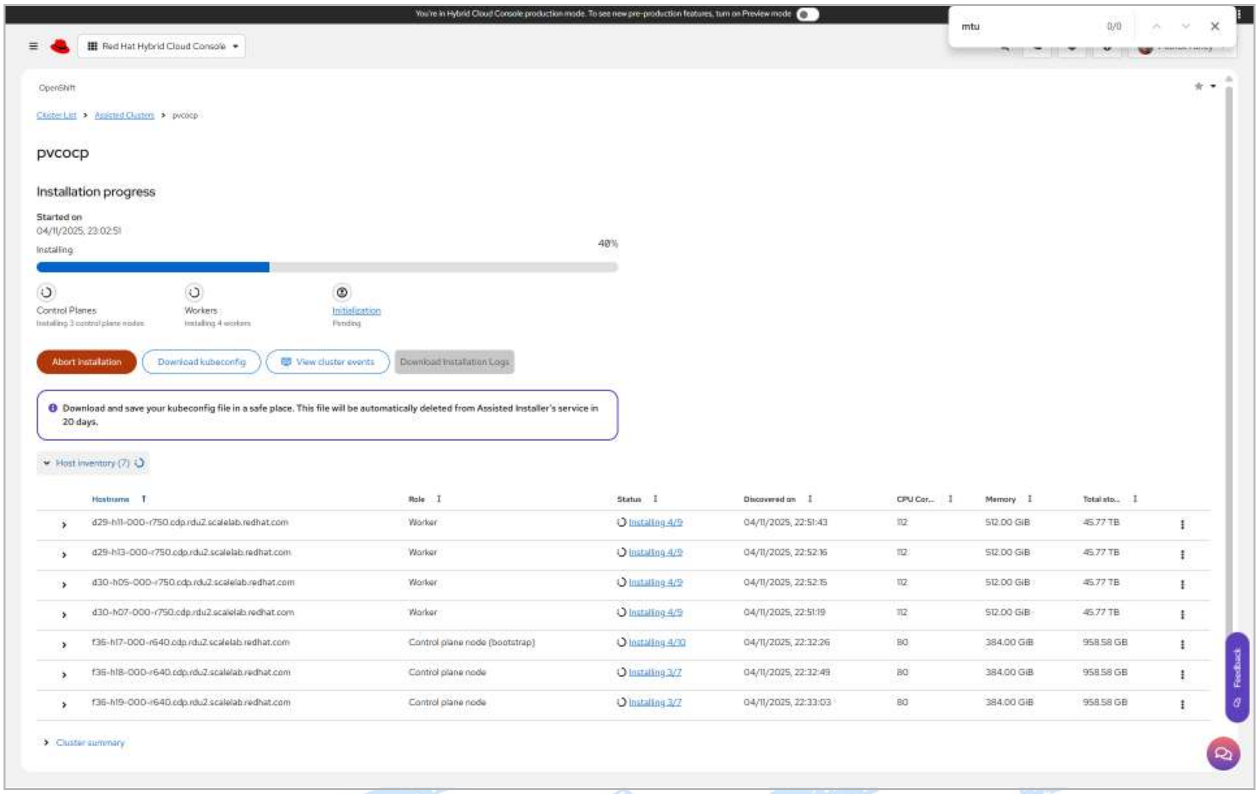


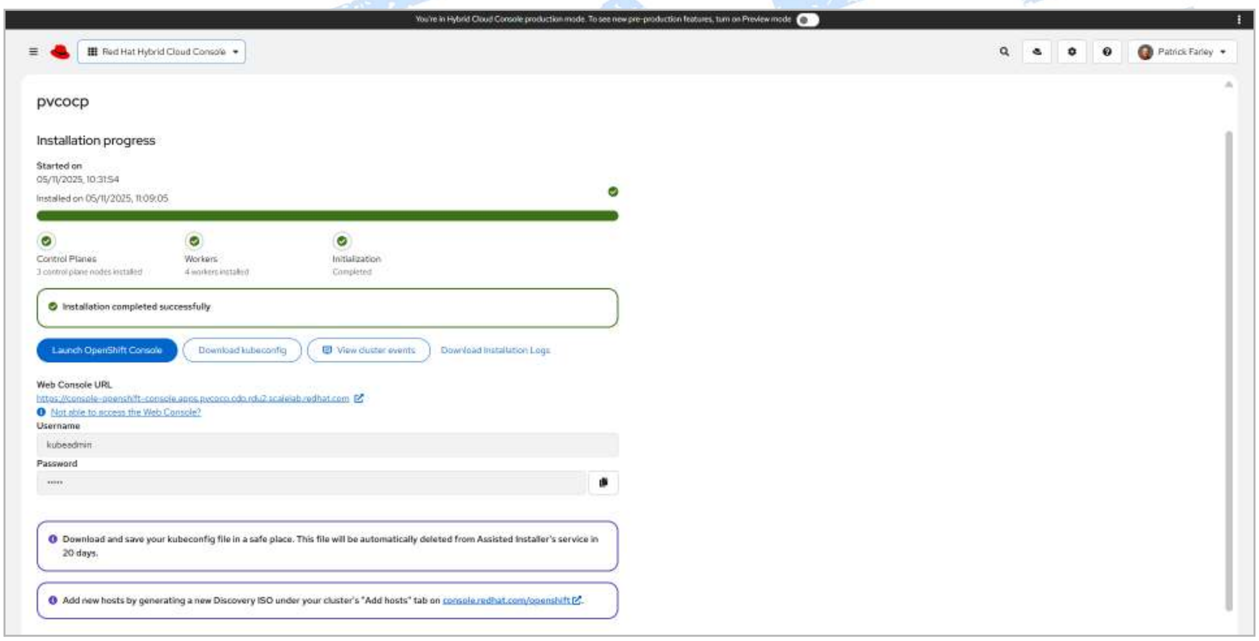

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode
Red Hat Hybrid Cloud Console
mtu
0/0
OpenShift
Cluster List
Assisted Clusters
pvcocp
Install OpenShift with the Assisted Installer
Assisted Installer documentation
What's new in Assisted Installer?
1 Cluster details
2 Static network configurations
3 Operators
4 Host discovery
5 Storage
6 Networking
7 Review and create
Networking
Network Management
Cluster-Managed Networking
User-Managed Networking
Networking stack type
IPv4
Dual-stack
Machine network *
10.1.48.0/23 (10.1.48.0 - 10.1.49.255)
API IP *
10.1.49.5
Ingress IP *
10.1.49.6
Use advanced networking
Configure advanced networking properties (e.g. CIDR ranges)
Host SSH Public Key for troubleshooting after installation
Use the same host discovery SSH key

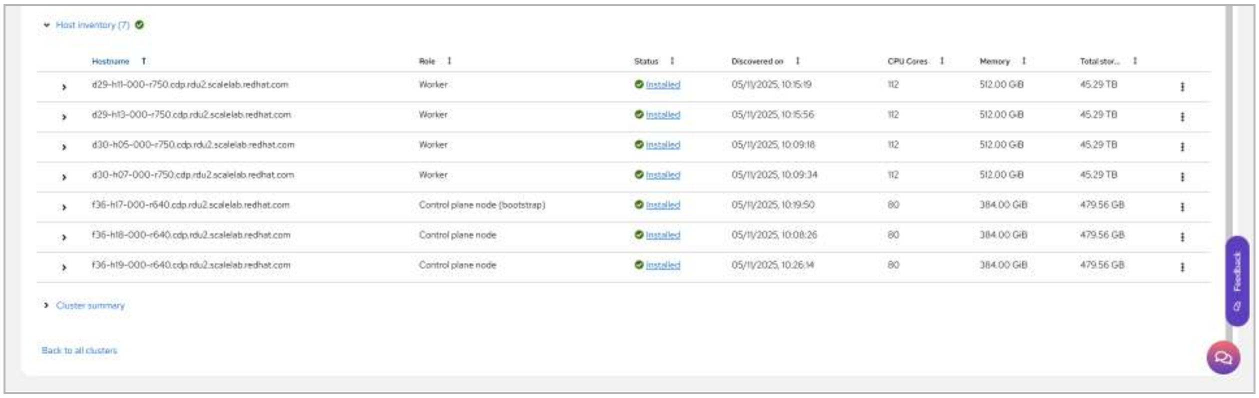

Host inventory
Hostname
Role
Status
Active ...
IPv4 add...
IPv...
MAC address
d29-h11-000-r750.cdp.rdu2.scalelab.redhat.com
Worker
Ready
Some validations failed
eno12409np1
10.1.49.219/23
00:62:0b:1c:07:61
d29-h13-000-r750.cdp.rdu2.scalelab.redhat.com
Worker
Ready
Some validations failed
eno12409np1
10.1.49.220/23
00:62:0b:1d:d3:31
d30-h05-000-r750.cdp.rdu2.scalelab.redhat.com
Worker
Ready
Some validations failed
eno12409np1
10.1.49.221/23
00:62:0b:1d:9a:a1
d30-h07-000-r750.cdp.rdu2.scalelab.redhat.com
Worker
Ready
Some validations failed
eno12409np1
10.1.49.222/23
00:62:0b:1d:d1:a1
f36-h17-000-r640.cdp.rdu2.scalelab.redhat.com
Control plane node
Ready
Some validations failed
eno2np1
10.1.49.216/23
bc:97:e1:78:dd:91
f36-h18-000-r640.cdp.rdu2.scalelab.redhat.com
Control plane node
Ready
Some validations failed
eno2np1
10.1.49.217/23
bc:97:e1:78:d7:91
f36-h19-000-r640.cdp.rdu2.scalelab.redhat.com
Control plane node
Ready
Some validations failed
eno2np1
10.1.49.218/23
bc:97:e1:69:d2:41
Next
Back
Cancel
View cluster events
Feedback


Red Hat Hybrid Cloud Console
4 Host discovery
5 Storage
6 Networking
7 Review and create
Cluster details
Cluster address pvcocp.cdp.rdu2.scalelab.redhat.com
OpenShift version 4.17.42
CPU architecture x86_64
Hosts' network configuration Static IP
Host inventory
Hosts 7
Total cores 688
Total memory 3.13 TiB
Total storage 185.95 TB
Networking
Networking management type Cluster-managed networking
Stack type IPv4
Machine networks CIDR 10.1.48.0/23
API IP 10.1.49.5
Ingress IP 10.1.49.6
Advanced networking settings
Cluster network CIDR 10.128.0.0/14
Cluster network host prefix 23
Service network CIDR 172.30.0.0/16
Networking type Open Virtual Network (OVN)
Install cluster
Back
Cancel
View cluster events


Red Hat Hybrid Cloud Console
pvcocp
Installation progress
Started on
04/11/2025, 23:02:51
Preparing for installation
0%
Control Planes
Workers
Initialization
Pending
Abort installation
Download kubeconfig
View cluster events
Download Installation Logs
Host inventory (7)
Hostname
Role
Status
Discovered on
Memory
d29-h11-000-r750.cdp.rdu2.scalelab.redhat.com
d29-h13-000-r750.cdp.rdu2.scalelab.redhat.com
d30-h05-000-r750.cdp.rdu2.scalelab.redhat.com
d30-h07-000-r750.cdp.rdu2.scalelab.redhat.com
f36-h17-000-r640.cdp.rdu2.scalelab.redhat.com
f36-h18-000-r640.cdp.rdu2.scalelab.redhat.com
f36-h19-000-r640.cdp.rdu2.scalelab.redhat.com
Worker
Control plane node (bootstrap)
Control plane node
Ready
Some validations failed
Cluster summary


Red Hat Hybrid Cloud Console
pvcocp
Installation progress
Started on
04/11/2025, 23:02:51
Installing
40%
Control Planes
Workers
Initialization
Pending
Abort installation
Download kubeconfig
View cluster events
Download Installation Logs
Download and save your kubeconfig file in a safe place. This file will be automatically deleted from Assisted Installer's service in 20 days.
Host inventory (7)
Hostname
Role
Status
Discovered on
Memory
d29-h11-000-r750.cdp.rdu2.scalelab.redhat.com
d29-h13-000-r750.cdp.rdu2.scalelab.redhat.com
d30-h05-000-r750.cdp.rdu2.scalelab.redhat.com
d30-h07-000-r750.cdp.rdu2.scalelab.redhat.com
f36-h17-000-r640.cdp.rdu2.scalelab.redhat.com
f36-h18-000-r640.cdp.rdu2.scalelab.redhat.com
f36-h19-000-r640.cdp.rdu2.scalelab.redhat.com
Worker
Control plane node (bootstrap)
Control plane node
Installing 4/9
Installing 4/10
Installing 3/7
512.00 GiB
384.00 GiB
45.77 TB
958.58 GB
Cluster summary


Red Hat Hybrid Cloud Console
Patrick Farley
pvcocp
Installation progress
Started on
05/11/2025, 10:31:54
Installed on 05/11/2025, 11:09:05
Control Planes
3 control plane nodes installed
Workers
4 workers installed
Initialization
Completed
Installation completed successfully
Launch OpenShift Console
Download kubeconfig
View cluster events
Download Installation Logs
Web Console URL
Not able to access the Web Console?
Username
kubeadmin
Password
Download and save your kubeconfig file in a safe place. This file will be automatically deleted from Assisted Installer's service in 20 days.
Add new hosts by generating a new Discovery ISO under your cluster's "Add hosts" tab on console.redhat.com/openshift



https://console.redhat.com/openshift/assisted-installer/clusters/415eb0de-ca80-47d8-a96c-151f60f58d20

- - -

Create OpenShift cluster with the Agent Based Installer (ABI)

The Agent Based Installer is used to create an OpenShift cluster for CDP Data Services.
This method of OpenShift installation is used in place of the Assisted Installer.

- Chapter 1. Preparing to install with the Agent-based Installer
- 4.2. Installing OpenShift Container Platform with the Agent-based Installer
- 4.2.1. Downloading the Agent-based Installer
- 4.2.3. Creating the preferred configuration inputs

STEP 1: Navigate to the Red Hat Hybrid Cloud Console using your login credentials.

STEP 2: Click ***Download Installer*** to download and extract the install program.

For this installation, we download the OpenShift installer for the latest stable release of **4.17**. Using this version of the installer will install the same version of OpenShift.

Download and extract the OpenShift installer using the following commands:

```
[root@ipaserver ~]# wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable-4.17/
openshift-install-linux.tar.gz
[root@ipaserver ~]# tar xvf openshift-install-linux.tar.gz
[root@ipaserver ~]# chmod +x openshift-install
[root@ipaserver ~]# mv openshift-install /usr/local/bin/openshift-install
```

STEP 3: Download or copy the pull secret by clicking on ***Download pull secret*** or ***Copy pull secret.*** This will be used when the install-config.yaml file is created.

STEP 4: Click Download command-line tools and place the openshift-install binary in a directory that is on your PATH.

```
[root@ipaserver ~]# wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable/openshift-client-li
nux.tar.gz
[root@ipaserver ~]# tar xvf openshift-client-linux.tar.gz
[root@ipaserver ~]# chmod +x oc
[root@ipaserver ~]# mv oc /usr/local/sbin/oc
```

STEP 5: Install the nmstate dependency by running the following command:

```
[root@ipaserver ~]# dnf install /usr/bin/nmstatectl -y
```

install-config.yaml

STEP 6: Create the install-config.yaml file

```
[root@ipaserver ~]# cat > install-config.yaml << EOF
apiVersion: v1
baseDomain: rdu2.scalelab.redhat.com
compute:
- architecture: amd64
  name: worker
  replicas: 13
controlPlane:
  architecture: amd64
  name: master
  replicas: 3
metadata:
  name: cdp
```



```
networking:
  clusterNetwork:
  - cidr: 10.128.0.0/14
    hostPrefix: 23
  machineNetwork:
  - cidr: 10.1.48.0/23
  networkType: OVNKubernetes
  serviceNetwork:
  - 172.30.0.0/16
platform:
  baremetal:
    hosts:
      - name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
        role: master
        bootMACAddress: 02:00:e4:00:03:00
      - name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
        role: master
        bootMACAddress: 02:00:e4:00:03:01
      - name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
        role: master
        bootMACAddress: 02:00:e4:00:03:02
      - name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:03
      - name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:04
      - name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:05
      - name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:06
      - name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:07
      - name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:08
      - name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:09
      - name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:0a
      - name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
```

```
        role: worker
        bootMACAddress: 02:00:e4:00:03:0b
      - name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:0c
      - name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:0d
      - name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:0e
      - name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:0f
    apiVIP: "10.1.49.3"
    ingressVIP: "10.1.49.4"
pullSecret:
```

```
'{"auths":{"cloud.openshift.com":{"auth":"b3BlbnNoaWZ0LXJlbGVhc2UtZGV2K29jbV9
hY2Nlc3NfMzU4NDUyN2M1MWNiNDYzMDljNDJhYjI0NTcxZDhiOTE6V0YzRFZOSVlVS1BET0RKT0JI
SVNWSkJTV0tOWVNYVlJBOTVRSEo3NU5DVlJMS0s4RldJRE5GQ09TVldET1VYUQ==","email":"pf
arley@redhat.com"},"quay.io":{"auth":"b3BlbnNoaWZ0LXJlbGVhc2UtZGV2K29jbV9hY2N
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SkJTV0tOWVNYVlJBOTVRSEo3NU5DVlJMS0s4RldJRE5GQ09TVldET1VYUQ==","email":"pfarle
y@redhat.com"},"registry.connect.redhat.com":{"auth":"fHVoYy1wb29sLTcxZWM2OWM
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aU9pSmlPVEU0WXpjNU1ESXlNalkwTTJOa09HTTROemMzWXpSallqTmhaRGhoTnlKOS5DZjZCdU1td
VN2aldjUHVDZmlxVmwxaWRTSUJhSlZ3V0JIdnRfU05STEFhRW9MeXFtUloxSW1ROGllaE9KaXhZSn
oybnlOdkVza0Mwak1UTTY5YzA1QVNBR0puQTdsTXlHZUhpdVhUdzZTUnJDRFdBTGZWYjhTOFNyOXQ
tZFp5Z1lndm84WHFBNk5ubEl1RWlzQkxuZWRaZW12bGVud2VJdXBUYTNvM2U0THJPM3lfV2J0MzVu
R1NhZ0Y5SklmRGUtR1ZsbGNHa3p1WlFieXJQZldyZmZjVkF1V3dqZUpXLUpvdkdKZ25LajB1V2ZqM
nFTY19sMWlWU3lHeGVYVV96S2tmaUF6dmFuY0NsZ09wT01TRS1BVmx6OFprRDVsNFBGZzNzdmU2Sl
FhbXVaLTJaeDM3bHJUajF0WEJVOHB2dEZYSGV3QzczN2k2LWJkeEZTWTBvT3JlS21ydFdpT3lpTml
DekR2YUl1X1BjSkx4cXNFUEJRZUdPeWd4aDVORHNjRVB5QmtZYTdlOHYtdVFXV3VldzNYSXo5d28y
ZjhsSlc1cGVPamJzZi1jcWxmX19INUZ4UWI5M24zOWw1X29lYkh1NHFPcW9IM2U2Z2FCWERhZmx1U
E1kYzRENkU2MDNHZkxXY1NQcFFMNVE1QTFaa3Nhc183NUl1ejVkYU5UM1padTVoOER3YVY2dTQ2TG
szNU9OblkybzRoUWhXYkxPUmptalo5LVplVnN2STkxaXIweVk5YkdyZ3MyQlZUN252MFowU3pNdEh
pR0o0aWpzYXpjY3dMbVUtM2poSUdBaXREaEN2eEJiY0plUGVWaENwZ3dGTkstMFdZT0V1cElHN1Rm
SjE5WHRlV1FVbDNzaTlQVlg5Q2RHdU93RHViV1NMQXJ5RXAtTVlZdw==","email":"pfarley@re
dhat.com"},"registry.redhat.io":{"auth":"fHVoYy1wb29sLTcxZWM2OWMwLTBiMDAtNDg5
Yi1hOTYyLTE4NzExNTkyOTA2NjpleUpoYkdjaU9pSlNVelV4TWlKOS5leUp6ZFdJaU9pSmlPVEU0W
XpjNU1ESXlNalkwTTJOa09HTTROemMzWXpSallqTmhaRGhoTnlKOS5DZjZCdU1tdVN2aldjUHVDZm
lxVmwxaWRTSUJhSlZ3V0JIdnRfU05STEFhRW9MeXFtUloxSW1ROGllaE9KaXhZSnoybnlOdkVza0M
wak1UTTY5YzA1QVNBR0puQTdsTXlHZUhpdVhUdzZTUnJDRFdBTGZWYjhTOFNyOXQtZFp5Z1lndm84
WHFBNk5ubEl1RWlzQkxuZWRaZW12bGVud2VJdXBUYTNvM2U0THJPM3lfV2J0MzVuR1NhZ0Y5SklmR
GUtR1ZsbGNHa3p1WlFieXJQZldyZmZjVkF1V3dqZUpXLUpvdkdKZ25LajB1V2ZqMnFTY19sMWlWU3
lHeGVYVV96S2tmaUF6dmFuY0NsZ09wT01TRS1BVmx6OFprRDVsNFBGZzNzdmU2SlFhbXVaLTJaeDM
```

```
3bHJUajF0WEJVOHB2dEZYSGV3QzczN2k2LWJkeEZTWTBvT3JlS21ydFdpT3lpTmlDekR2YUl1X1Bj
Skx4cXNFUEJRZUdPeWd4aDVORHNjRVB5QmtZYTdlOHYtdVFXV3VldzNYSXo5d28yZjhsSlc1cGVPa
mJzZi1jcWxmX19INUZ4UWI5M24zOWw1X29lYkh1NHFPcW9IM2U2Z2FCWERhZmx1UE1kYzRENkU2MD
NHZkxXY1NQcFFMNVE1QTFaa3Nhc183NUl1ejVkYU5UM1padTVoOER3YVY2dTQ2TGszNU9OblkybzR
oUWhXYkxPUmptalo5LVplVnN2STkxaXIweVk5YkdyZ3MyQlZUN252MFowU3pNdEhpR0o0aWpzYXpj
Y3dMbVUtM2poSUdBaXREaEN2eEJiY0plUGVWaENwZ3dGTkstMFdZT0V1cElHN1RmSjE5WHRlV1FVb
DNzaTlQVlg5Q2RHdU93RHViV1NMQXJ5RXAtTVlZdw==","email":"pfarley@redhat.com"}}}'
sshKey: |
  ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABgQC5f46fvSfwLSXxOzCC52qFa21AS0kvJoolggoGL+CUWGdq6
Kf4ExWCJ2d1iDkAcPt6y7jGshujhfUR6DE1kuFpozTOgqBjwZl7ICtA5lmkhb7sxHuYYNFAZNli6D
38IbdMSfth23I2ypBpcQAOWweCzfPnjsjaniLG5EfTWw3K75QQOMxLvE4W0uKbA/EKg9F8Eq2ImKL
HZekB8nCFoi2mVgwEYGTSMSD5zAz56tuSbuppbkXfgf+4FFb/LFp3TJyCPQvu396ZRbEF64EWvbMC
01aAYNM3NpRqSqsdGWWJoTnRzXSRKb096tOxWmYTyRIfOs/nuRAlSb3kU0EN4qC5YQs9wobpnLQ5T
3Um8QP4pQxWhWIHEAW3SWMOlIfeBjT8TPS8OCG3EfLRVxJFdNZc/IfD3CGnevW8ovlSek//HfspwM
MMauXQfY57OfXhn9BxVtnC6ymGlDBv9hoa82qtvigfkMU1uOxD9Ep7rl14vprcQ5PFl8ypO9DPsKj
4Etc= root@ipaserver.cdp.rdu2.scalelab.redhat.com
```
EOF

agent-config.yaml

STEP 7: Create the agent-config.yaml file

Hostname	CPU	RAM	Disk	MAC Address	IP Address
pvcocp-master01.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:00	10.1.49.200/23
pvcocp-master02.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:01	10.1.49.201/23
pvcocp-master03.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:02	10.1.49.202/23
pvcocp-worker01.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:03	10.1.49.203/23
pvcocp-worker02.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:04	10.1.49.204/23
pvcocp-worker03.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:05	10.1.49.205/23
pvcocp-worker04.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:06	10.1.49.206/23
pvcocp-worker05.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:07	10.1.49.207/23
pvcocp-worker06.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:08	10.1.49.208/23
pvcocp-worker07.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:09	10.1.49.209/23
pvcocp-worker08.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0a	10.1.49.210/23
pvcocp-worker09.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0b	10.1.49.211/23

pvcocp-worker10.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0c	10.1.49.212/23
pvcocp-infra01.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0d	10.1.49.213/23
pvcocp-infra02.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0e	10.1.49.214/23
pvcocp-infra03.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0f	10.1.49.215/23

<domain_name> for the RH scale lab will be: cdp.rdu2.scalelab.redhat.com

[root@ipaserver ~]# **cat > agent-config.yaml << EOF**

```
apiVersion: v1alpha1
kind: AgentConfig
metadata:
  name: cdp
rendezvousIP: 10.1.49.200
hosts:
  - hostname: master01.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:00 # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:00 # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.200 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
```

```
          table-id: 254
  - hostname: master02.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:01 # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:01 # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.201 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
  - hostname: master03.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:02 # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:02 # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.202 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
```

```
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
  - hostname: worker01.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:03 # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:03 # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.203 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
  - hostname: worker02.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:04 # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
```

```yaml
          mac-address: 02:00:e4:00:03:04 # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.204 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
  - hostname: worker03.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:05 # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:05 # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.205 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
  - hostname: worker04.cdp.rdu2.scalelab.redhat.com
    interfaces:
```

```
      - name: ens192
        macAddress: 02:00:e4:00:03:06 # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:06 # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.206 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
  - hostname: worker05.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:07 # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:07 # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.207 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
```

```yaml
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: worker06.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:08 # Change for each host
  networkConfig:
    interfaces:
      - name: ens192
        type: ethernet
        mtu: 9000
        state: up
        mac-address: 02:00:e4:00:03:08 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.49.208 # Change for each host
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: worker07.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:09 # Change for each host
  networkConfig:
    interfaces:
      - name: ens192
        type: ethernet
        mtu: 9000
        state: up
        mac-address: 02:00:e4:00:03:09 # Change for each host
        ipv4:
          enabled: true
```

```
          address:
            - ip: 10.1.49.209 # Change for each host
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: worker08.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:0a # Change for each host
  networkConfig:
    interfaces:
      - name: ens192
        type: ethernet
        mtu: 9000
        state: up
        mac-address: 02:00:e4:00:03:0a # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.49.210 # Change for each host
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: worker09.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:0b # Change for each host
  networkConfig:
```

```yaml
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:0b # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.211 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
  - hostname: worker10.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:0c # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:0c # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.212 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
```



```
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: infra01.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:0d # Change for each host
  networkConfig:
    interfaces:
      - name: ens192
        type: ethernet
        mtu: 9000
        state: up
        mac-address: 02:00:e4:00:03:0d # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.49.213 # Change for each host
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: infra02.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:0e # Change for each host
  networkConfig:
    interfaces:
      - name: ens192
        type: ethernet
        mtu: 9000
        state: up
        mac-address: 02:00:e4:00:03:0e # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.49.214 # Change for each host
              prefix-length: 23
```

```
          dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
  - hostname: infra03.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:0f # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:0f # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.215 # Change for each host
                prefix-length: 23
            dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
```
EOF

STEP 8: Create Install Dir

[root@ipaserver ~]# **mkdir ocp-install-dir**

STEP 9: Create backup copies of the yaml manifests

```
[root@ipaserver ~]# cp agent-config.yaml ocp-install-dir/agent-config.yaml
[root@ipaserver ~]# cp install-config.yaml
ocp-install-dir/install-config.yaml
```

STEP 10: Create ISO agent image

```
[root@ipaserver ~]# openshift-install --dir ocp-install-dir agent create
image
INFO Configuration has 3 master replicas and 13 worker replicas
WARNING hosts from install-config.yaml are ignored
INFO The rendezvous host IP (node0 IP) is 10.1.49.200
INFO Extracting base ISO from release payload
INFO Verifying cached file
INFO Using cached Base ISO /root/.cache/agent/image_cache/coreos-x86_64.iso
INFO Consuming Install Config from target directory
INFO Consuming Agent Config from target directory
INFO Generated ISO at ocp-install-dir/agent.x86_64.iso.
```

STEP 11: Upload the Discovery ISO image to a PVC using the virtctl utility:

```
[root@ipaserver ~]# oc project cdppvcds

[root@ipaserver ~]# virtctl image-upload dv discovery-image-cdp --size=2Gi
--image-path=ocp-install-dir/agent.x86_64.iso --insecure
--storage-class=ocs-storagecluster-ceph-rbd
PVC cdp-ds/discovery-image-cdp not found
DataVolume cdp-ds/discovery-image-cdp created
Waiting for PVC discovery-image-cdp upload pod to be ready...
Pod now ready
Uploading data to https://cdi-uploadproxy-openshift-cnv.apps.vlan601.rdu2.scalelab.redhat.com

1.24 GiB / 1.24 GiB
[---------------------------------------------------------------------------------------
-------------------------------------------------------------] 100.00% 161.43 MiB p/s
8.1s

Uploading data completed successfully, waiting for processing to complete, you can hit ctrl-c
without interrupting the progress
Processing completed successfully
Uploading ocp-install-dir/agent.x86_64.iso completed successfully
```


Create VMs for OpenShift Cluster (OpenShift Virtualization)

The following table is a reference of the VirtualMachines that will be created for the OpenShift cluster for CDP Data Services:

Hostname	CPU	RAM	Disk	MAC Address	IP Address
pvcocp-master01.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:00	10.1.49.200/23
pvcocp-master02.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:01	10.1.49.201/23
pvcocp-master03.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:02	10.1.49.202/23
pvcocp-worker01.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:03	10.1.49.203/23
pvcocp-worker02.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:04	10.1.49.204/23
pvcocp-worker03.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:05	10.1.49.205/23
pvcocp-worker04.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:06	10.1.49.206/23
pvcocp-worker05.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:07	10.1.49.207/23
pvcocp-worker06.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:08	10.1.49.208/23
pvcocp-worker07.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:09	10.1.49.209/23
pvcocp-worker08.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0a	10.1.49.210/23
pvcocp-worker09.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0b	10.1.49.211/23
pvcocp-worker10.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0c	10.1.49.212/23
pvcocp-infra01.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0d	10.1.49.213/23
pvcocp-infra02.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0e	10.1.49.214/23
pvcocp-infra03.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0f	10.1.49.215/23

<domain_name> for the RH scale lab will be: cdp.rdu2.scalelab.redhat.com

STEP 13: Create the VM's for the OpenShift cluster using the following VM YAML files:

```
[root@bastion ~]# oc new-project cdp-ds
[root@bastion ~]# oc create -f <nodename>-vm.yaml
```

```
for i in {100..199}; do oc create -f <nodename>-vm.yaml; done
```

pvcocp-master01.cdp.rdu2.scalelab.redhat.com-vms.yaml


```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
```



```
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 8
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
            name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
            name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:00'
            name: eth0
        rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
      firmware:
        bootloader:
          efi: {}
      machine:
        type: pc-q35-rhel9.4.0
      memory:
        guest: 16Gi
      resources: {}
    networks:
      - multus:
          networkName: default/localnet-network
        name: eth0
```

```yaml
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-master02.cdp.rdu2.scalelab.redhat.com-vms.yaml

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
```

```yaml
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
        cpu:
          cores: 1
          sockets: 8
          threads: 1
        devices:
          disks:
            - bootOrder: 1
              disk:
                bus: virtio
              name: rootdisk
            - bootOrder: 2
              cdrom:
                bus: sata
              name: installation-cdrom
          interfaces:
            - bridge: {}
              model: virtio
              macAddress: '02:00:e4:00:03:01'
              name: eth0
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
```

```
          type: pc-q35-rhel9.4.0
        memory:
          guest: 16Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-master03.cdp.rdu2.scalelab.redhat.com-vms.yaml

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
```

```
            storage: 120Gi
  - metadata:
      name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com-cdrom
    spec:
      source:
        pvc:
          name: discovery-image-cdp
          namespace: cdppvcds
      storage:
        resources:
          requests:
            storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 8
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
            name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
            name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:02'
            name: eth0
        rng: {}
```

```yaml
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 16Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-worker01.cdp.rdu2.scalelab.redhat.com-vms.yaml

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
```

```
      creationTimestamp: null
      name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
    spec:
      source:
        blank: {}
      storage:
        resources:
          requests:
            storage: 120Gi
  - metadata:
      name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com-cdrom
    spec:
      source:
        pvc:
          name: discovery-image-cdp
          namespace: cdppvcds
      storage:
        resources:
          requests:
            storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
            name: rootdisk
          - bootOrder: 2
            cdrom:
```



```yaml
              bus: sata
            name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:03'
            name: eth0
        rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
      firmware:
        bootloader:
          efi: {}
      machine:
        type: pc-q35-rhel9.4.0
      memory:
        guest: 64Gi
      resources: {}
    networks:
      - multus:
          networkName: default/localnet-network
        name: eth0
    terminationGracePeriodSeconds: 180
    volumes:
      - dataVolume:
          name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
        name: rootdisk
      - dataVolume:
          name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com-cdrom
        name: installation-cdrom
```

pvcocp-worker02.cdp.rdu2.scalelab.redhat.com-vms.yaml

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
```

```yaml
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
        cpu:
          cores: 1
          sockets: 32
          threads: 1
```

```
        devices:
          disks:
            - bootOrder: 1
              disk:
                bus: virtio
              name: rootdisk
            - bootOrder: 2
              cdrom:
                bus: sata
              name: installation-cdrom
          interfaces:
            - bridge: {}
              model: virtio
              macAddress: '02:00:e4:00:03:04'
              name: eth0
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-worker03.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
```

```
  labels:
    app: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
```

```yaml
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
        cpu:
          cores: 1
          sockets: 32
          threads: 1
        devices:
          disks:
            - bootOrder: 1
              disk:
                bus: virtio
              name: rootdisk
            - bootOrder: 2
              cdrom:
                bus: sata
              name: installation-cdrom
          interfaces:
            - bridge: {}
              model: virtio
              macAddress: '02:00:e4:00:03:05'
              name: eth0
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
```

```
          name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com-cdrom
        name: installation-cdrom
```

pvcocp-worker04.cdp.rdu2.scalelab.redhat.com-vms.yaml


```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
```

```
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
            name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
            name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:06'
            name: eth0
        rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
      firmware:
        bootloader:
          efi: {}
      machine:
        type: pc-q35-rhel9.4.0
      memory:
        guest: 64Gi
      resources: {}
    networks:
      - multus:
```

```
          networkName: default/localnet-network
        name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-worker05.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
```

```
          name: discovery-image-cdp
          namespace: cdppvcds
      storage:
        resources:
          requests:
            storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
            name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
            name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:07'
            name: eth0
        rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
      firmware:
        bootloader:
```

```
          efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-worker06.cdp.rdu2.scalelab.redhat.com-vms.yaml


```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
```

```
        resources:
          requests:
            storage: 120Gi
  - metadata:
      name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com-cdrom
    spec:
      source:
        pvc:
          name: discovery-image-cdp
          namespace: cdppvcds
      storage:
        resources:
          requests:
            storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
            name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
            name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:08'
```

```yaml
              name: eth0
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-worker07.cdp.rdu2.scalelab.redhat.com-vms.yaml

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
```

```
      annotations:
        cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
      creationTimestamp: null
      name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
    spec:
      source:
        blank: {}
      storage:
        resources:
          requests:
            storage: 120Gi
  - metadata:
      name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com-cdrom
    spec:
      source:
        pvc:
          name: discovery-image-cdp
          namespace: cdppvcds
      storage:
        resources:
          requests:
            storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
            name: rootdisk
```

```yaml
            - bootOrder: 2
              cdrom:
                bus: sata
              name: installation-cdrom
          interfaces:
            - bridge: {}
              model: virtio
              macAddress: '02:00:e4:00:03:09'
              name: eth0
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-worker08.cdp.rdu2.scalelab.redhat.com-vms.yaml

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
```

```yaml
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
        cpu:
          cores: 1
```

```
          sockets: 32
          threads: 1
        devices:
          disks:
            - bootOrder: 1
              disk:
                bus: virtio
              name: rootdisk
            - bootOrder: 2
              cdrom:
                bus: sata
              name: installation-cdrom
          interfaces:
            - bridge: {}
              model: virtio
              macAddress: '02:00:e4:00:03:0a'
              name: eth0
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-worker09.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
```

```yaml
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
```



```yaml
      labels:
        kubevirt.io/domain: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
        cpu:
          cores: 1
          sockets: 32
          threads: 1
        devices:
          disks:
            - bootOrder: 1
              disk:
                bus: virtio
              name: rootdisk
            - bootOrder: 2
              cdrom:
                bus: sata
              name: installation-cdrom
          interfaces:
            - bridge: {}
              model: virtio
              macAddress: '02:00:e4:00:03:0b'
              name: eth0
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
```

```yaml
          name: rootdisk
        - dataVolume:
            name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-worker10.cdp.rdu2.scalelab.redhat.com-vms.yaml

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
```

```yaml
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
        cpu:
          cores: 1
          sockets: 32
          threads: 1
        devices:
          disks:
            - bootOrder: 1
              disk:
                bus: virtio
              name: rootdisk
            - bootOrder: 2
              cdrom:
                bus: sata
              name: installation-cdrom
          interfaces:
            - bridge: {}
              model: virtio
              macAddress: '02:00:e4:00:03:0c'
              name: eth0
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
```

```
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
```

pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-cdrom
```

```
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
    - metadata:
        name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-disk1
      spec:
        source:
          blank: {}
        storage:
          accessModes:
            - ReadWriteMany
          resources:
            requests:
              storage: 2000Gi
          storageClassName: ocs-storagecluster-ceph-rbd
          volumeMode: Block
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
        cpu:
          cores: 1
          sockets: 32
          threads: 1
        devices:
          disks:
            - bootOrder: 1
              disk:
                bus: virtio
              name: rootdisk
```

```
          - bootOrder: 2
            cdrom:
              bus: sata
            name: installation-cdrom
          - bootOrder: 3
            disk:
              bus: virtio
            name: disk-1
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:0d'
            name: eth0
        rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
      firmware:
        bootloader:
          efi: {}
      machine:
        type: pc-q35-rhel9.4.0
      memory:
        guest: 64Gi
      resources: {}
    networks:
      - multus:
          networkName: default/localnet-network
        name: eth0
    terminationGracePeriodSeconds: 180
    volumes:
      - dataVolume:
          name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
        name: rootdisk
      - dataVolume:
          name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-cdrom
        name: installation-cdrom
      - dataVolume:
          name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-disk1
        name: disk-1
```

pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
```

```
metadata:
  labels:
    app: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
    - metadata:
        name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-disk1
      spec:
        source:
          blank: {}
        storage:
          accessModes:
            - ReadWriteMany
          resources:
```

```
          requests:
            storage: 2000Gi
        storageClassName: ocs-storagecluster-ceph-rbd
        volumeMode: Block
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
            name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
            name: installation-cdrom
          - bootOrder: 3
            disk:
              bus: virtio
            name: disk-1
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:0e'
            name: eth0
        rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
```

```yaml
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
          name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
        - dataVolume:
            name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
        - dataVolume:
            name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-disk1
          name: disk-1
```

pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-vms.yaml

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
```



```yaml
        name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
    - metadata:
        name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-disk1
      spec:
        source:
          blank: {}
        storage:
          accessModes:
            - ReadWriteMany
          resources:
            requests:
              storage: 2000Gi
          storageClassName: ocs-storagecluster-ceph-rbd
          volumeMode: Block
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
```

```
    cpu:
      cores: 1
      sockets: 32
      threads: 1
    devices:
      disks:
        - bootOrder: 1
          disk:
            bus: virtio
          name: rootdisk
        - bootOrder: 2
          cdrom:
            bus: sata
          name: installation-cdrom
        - bootOrder: 3
          disk:
            bus: virtio
          name: disk-1
      interfaces:
        - bridge: {}
          model: virtio
          macAddress: '02:00:e4:00:03:0f'
          name: eth0
      rng: {}
    features:
      acpi: {}
      smm:
        enabled: true
    firmware:
      bootloader:
        efi: {}
    machine:
      type: pc-q35-rhel9.4.0
    memory:
      guest: 64Gi
    resources: {}
  networks:
    - multus:
        networkName: default/localnet-network
      name: eth0
  terminationGracePeriodSeconds: 180
  volumes:
    - dataVolume:
        name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
      name: rootdisk
    - dataVolume:
```

```
            name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom
        - dataVolume:
            name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-disk1
          name: disk-1
---
```

STEP 14: Track the installation progress.

[root@bastion ~]# **openshift-install --dir ocp-install-dir agent wait-for install-complete --log-level=debug**

```
[root@ipaserver ocp-agent-install]# openshift-install --dir ocp-install-dir agent wait-for install-complete --log-level=debug
DEBUG OpenShift Installer 4.17.42
DEBUG Built from commit f4a8c8c1eead4b425c434d2300c1a9a387789ed9
DEBUG asset directory: ocp-install-dir
DEBUG Loading Agent Config...
DEBUG Using Agent Config loaded from state file
DEBUG Loading Agent Manifests...
DEBUG   Loading Agent PullSecret...
DEBUG     Loading Agent Workflow...
DEBUG     Using Agent Workflow loaded from state file
DEBUG     Loading Agent Installer ClusterInfo...
DEBUG       Loading Agent Workflow...
DEBUG       Loading AddNodes Config...
DEBUG       Using AddNodes Config loaded from state file
DEBUG     Using Agent Installer ClusterInfo loaded from state file
DEBUG     Loading Install Config...
DEBUG     Using Install Config loaded from state file
DEBUG   Using Agent PullSecret loaded from state file
DEBUG   Loading InfraEnv Config...
DEBUG     Loading Agent Workflow...
DEBUG     Loading Agent Installer ClusterInfo...
DEBUG     Loading Install Config...
DEBUG     Loading Agent Config...
DEBUG   Using InfraEnv Config loaded from state file
DEBUG   Loading NMState Config...
DEBUG     Loading Agent Workflow...
DEBUG     Loading Agent Installer ClusterInfo...
DEBUG     Loading Agent Hosts...
DEBUG       Loading Agent Workflow...
DEBUG       Loading AddNodes Config...
DEBUG       Loading Install Config...
DEBUG       Loading Agent Config...
DEBUG     Using Agent Hosts loaded from state file
DEBUG     Loading Install Config...
DEBUG   Using NMState Config loaded from state file
DEBUG   Loading AgentClusterInstall Config...
DEBUG     Loading Agent Workflow...
DEBUG     Loading Install Config...
DEBUG     Loading Agent Hosts...
DEBUG   Using AgentClusterInstall Config loaded from state file
DEBUG   Loading ClusterDeployment Config...
```

```
DEBUG     Loading Agent Workflow...
DEBUG     Loading Install Config...
DEBUG   Using ClusterDeployment Config loaded from state file
DEBUG   Loading ClusterImageSet Config...
DEBUG     Loading Agent Workflow...
DEBUG     Loading Agent Installer ClusterInfo...
DEBUG     Loading Release Image Pull Spec...
DEBUG     Using Release Image Pull Spec loaded from state file
DEBUG     Loading Install Config...
DEBUG   Using ClusterImageSet Config loaded from state file
DEBUG Using Agent Manifests loaded from state file
DEBUG Loading Install Config...
DEBUG   Loading SSH Key...
DEBUG   Loading Base Domain...
DEBUG     Loading Platform...
DEBUG   Loading Cluster Name...
DEBUG     Loading Base Domain...
DEBUG     Loading Platform...
DEBUG   Loading Pull Secret...
DEBUG   Loading Platform...
DEBUG Loading Agent Hosts...
DEBUG RendezvousIP from the AgentConfig 10.1.49.200
DEBUG Loading Agent Installer API Auth Config...
DEBUG   Loading Agent Installer InfraEnv ID...
DEBUG   Using Agent Installer InfraEnv ID loaded from state file
DEBUG   Loading Agent Workflow...
DEBUG   Using Agent Workflow loaded from state file
DEBUG   Loading AddNodes Config...
DEBUG   Using AddNodes Config loaded from state file
DEBUG Using Agent Installer API Auth Config loaded from state file
DEBUG Agent Rest API Initialized
INFO Cluster is not ready for install. Check validations
WARNING Cluster validation: The cluster has hosts that are not ready to install.
DEBUG Cluster validation: The cluster has the exact amount of dedicated control plane nodes.
DEBUG Cluster validation: API virtual IPs are defined.
DEBUG Cluster validation: api vips 10.1.49.3 belongs to the Machine CIDR and is not in use.
DEBUG Cluster validation: The Cluster Network CIDR is defined.
DEBUG Cluster validation: The base domain is defined.
DEBUG Cluster validation: Ingress virtual IPs are defined.
DEBUG Cluster validation: ingress vips 10.1.49.4 belongs to the Machine CIDR and is not in use.
DEBUG Cluster validation: The Machine Network CIDR is defined.
DEBUG Cluster validation: The Cluster Machine CIDR is equivalent to the calculated CIDR.
DEBUG Cluster validation: The Cluster Network prefix is valid.
DEBUG Cluster validation: The cluster has a valid network type
DEBUG Cluster validation: Same address families for all networks.
DEBUG Cluster validation: No CIDRS are overlapping.
DEBUG Cluster validation: No ntp problems found
DEBUG Cluster validation: The Service Network CIDR is defined.
DEBUG Cluster validation: cnv is disabled
DEBUG Cluster validation: lso is disabled
DEBUG Cluster validation: lvm is disabled
DEBUG Cluster validation: mce is disabled
DEBUG Cluster validation: odf is disabled
DEBUG Cluster validation: Platform requirements satisfied
DEBUG Cluster validation: The pull secret is set.
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Hostname worker08.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Hostname worker08.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
```

```
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
93bed28b-df8f-558c-81f9-6422039241dd
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role master
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role master
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Hostname master02.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Hostname master02.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host master02.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
64c50e5e-5bfd-5a20-b59b-a03a9a1458a1
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
```

```
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Hostname worker02.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Hostname worker02.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
00089653-134d-5a4c-bc25-e3b77da8fbec
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role master
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role master
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Hostname master01.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Hostname master01.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host master01.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
```

```
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
462a303e-0cb3-5fe4-a266-a580b4fcfa76
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Hostname worker06.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Hostname worker06.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
eb194bfd-9bbb-5e78-852b-b9649fe65211
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Hostname worker05.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Hostname worker05.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
```

```
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
f7b7f1b9-817a-5e9e-85ee-4d05d0d8f3be
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Hostname worker04.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Hostname worker04.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
2ec0e61a-c7ec-5f37-917c-88a8a486ace9
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
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DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Hostname worker10.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Hostname worker10.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
dfd6e101-e113-5938-9c9f-b5598c386584
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role master
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role master
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Hostname master03.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Hostname master03.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host master03.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
```

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DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
3833b526-a7bf-5672-a028-f59d7d680ec8
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Hostname worker01.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Hostname worker01.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
b1cfd435-1098-50fe-a95f-ab3765e345e9
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Hostname infra01.cdp.rdu2.scalelab.redhat.com is unique in
cluster
```



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DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Hostname infra01.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
WARNING Host infra01.cdp.rdu2.scalelab.redhat.com validation: No connectivity to the majority of hosts in the cluster
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WARNING Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
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DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled successfully
or no attempt was made to pull them
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
1ddf58df-3d52-50e0-bd87-9391b5b9f1dc
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Hostname worker09.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Hostname worker09.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
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DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
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cluster
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DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
422bcd9f-8ff0-57a3-9c2e-e575ff094535
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
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DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Hostname worker03.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Hostname worker03.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
def36ddf-7a48-5a98-860a-1ca00fe0ffec
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
4ae3ae26-a3ef-5845-8a63-f0481cb9bff4
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DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Hostname worker07.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Hostname worker07.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: calculated role is worker
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
WARNING Host infra02.cdp.rdu2.scalelab.redhat.com validation: No connectivity to the majority of hosts in the cluster
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled successfully
or no attempt was made to pull them
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
fd61237d-f691-5b68-8def-e0e42f360e98
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role auto-assign
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role auto-assign
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Hostname infra02.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Hostname infra02.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
INFO Host infra02.cdp.rdu2.scalelab.redhat.com: calculated role is worker
INFO Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the cluster
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: updated status from insufficient to known (Host is ready to be installed)
INFO Host 590a24c7-af35-5db9-bddc-03da5310c1f0: Successfully registered
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role auto-assign
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DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role auto-assign
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Hostname infra03.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Hostname infra03.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
WARNING Host infra03.cdp.rdu2.scalelab.redhat.com validation: No connectivity to the majority of hosts in the cluster
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled successfully
or no attempt was made to pull them
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
590a24c7-af35-5db9-bddc-03da5310c1f0
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
INFO Host infra03.cdp.rdu2.scalelab.redhat.com: calculated role is worker
INFO Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the cluster
INFO Host infra02.cdp.rdu2.scalelab.redhat.com: updated status from insufficient to known (Host is ready to be installed)
INFO Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the cluster
INFO Host infra03.cdp.rdu2.scalelab.redhat.com: updated status from insufficient to known (Host is ready to be installed)
INFO Cluster is ready for install
INFO Cluster validation: All hosts in the cluster are ready to install.
INFO Preparing cluster for installation
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: updated status from known to preparing-for-installation (Host finished
successfully to prepare for installation)
INFO Host worker05.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.38 seconds; size: 447.71 Megabytes; download rate: 338.98 MBps
INFO Host worker10.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.05 seconds; size: 447.71 Megabytes; download rate: 449.24 MBps
INFO Host worker08.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.73 seconds; size: 447.71 Megabytes; download rate: 271.09 MBps
INFO Host worker03.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.25 seconds; size: 447.71 Megabytes; download rate: 375.09 MBps
INFO Host worker01.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.16 seconds; size: 447.71 Megabytes; download rate: 404.46 MBps
INFO Host worker04.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Host infra02.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.13 seconds; size: 447.71 Megabytes; download rate: 413.80 MBps
INFO Host worker08.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Host worker06.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
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INFO Host worker03.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.06 seconds; size: 447.71 Megabytes; download rate: 442.97 MBps
INFO Host master02.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.28 seconds; size: 447.71 Megabytes; download rate: 366.99 MBps
INFO Host worker07.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Cluster installation in progress
INFO Host master02.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Host infra02.cdp.rdu2.scalelab.redhat.com: updated status from preparing-successful to installing (Installation is
in progress)
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 6%
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 14%
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 21%
INFO Host: worker09.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk
INFO Host: worker09.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 7%
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 45%
INFO Host: worker05.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 16%
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 67%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 10%
INFO Host: worker03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 14%
INFO Host: worker03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 20%
INFO Host: worker06.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 22%
INFO Host: worker03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 30%
INFO Host: master03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 6%
INFO Host: worker10.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 81%
INFO Host: worker10.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 93%
INFO Host: infra03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 41%
INFO Host: worker01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 74%
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 32%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 43%
INFO Host: infra02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 71%
INFO Host: infra02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 79%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 60%
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 61%
INFO Host: worker07.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 58%
INFO Host: infra01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 69%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 77%
INFO Host: infra01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 86%
INFO Host: infra01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 97%
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 93%
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 100%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 100%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Waiting for control plane
INFO Bootstrap Kube API Initialized
INFO Host: master03.cdp.rdu2.scalelab.redhat.com, reached installation stage Rebooting
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Rebooting
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Waiting for control plane: Waiting for
masters to join bootstrap control plane
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Configuring
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Waiting for bootkube
INFO Host: master03.cdp.rdu2.scalelab.redhat.com, reached installation stage Done
INFO Node master03.cdp.rdu2.scalelab.redhat.com has been rebooted 1 times before completing installation
INFO Node master02.cdp.rdu2.scalelab.redhat.com has been rebooted 1 times before completing installation
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Waiting for bootkube: waiting for ETCD
bootstrap to be complete
INFO Bootstrap configMap status is complete
INFO Bootstrap is complete
INFO cluster bootstrap is complete
DEBUG Still waiting for the cluster to initialize: Working towards 4.17.42: 791 of 901 done (87% complete)
DEBUG Still waiting for the cluster to initialize: Multiple errors are preventing progress:
DEBUG * Cluster operators authentication, image-registry, ingress, insights, kube-apiserver, machine-api, monitoring,
openshift-apiserver, openshift-samples, operator-lifecycle-manager-packageserver are not available
DEBUG * Could not update imagestream "openshift/driver-toolkit" (611 of 901): resource may have been deleted
DEBUG * Could not update oauthclient "console" (545 of 901): the server does not recognize this resource, check extension
API servers
```

```
DEBUG * Could not update role "openshift-console-operator/prometheus-k8s" (818 of 901): resource may have been deleted
DEBUG * Could not update role "openshift-console/prometheus-k8s" (822 of 901): resource may have been deleted
DEBUG Still waiting for the cluster to initialize: Multiple errors are preventing progress:
DEBUG * Cluster operators authentication, console, image-registry, ingress, kube-apiserver, machine-api, monitoring,
openshift-samples are not available
DEBUG * Could not update role "openshift-console-operator/prometheus-k8s" (818 of 901): resource may have been deleted
DEBUG * Could not update role "openshift-console/prometheus-k8s" (822 of 901): resource may have been deleted
DEBUG Route found in openshift-console namespace: console
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DEBUG Route found in openshift-console namespace: console
DEBUG Route found in openshift-console namespace: console
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DEBUG Route found in openshift-console namespace: console
DEBUG Still waiting for the cluster to initialize: Cluster operators authentication, console, ingress, kube-apiserver,
machine-api, monitoring, openshift-samples are not available
DEBUG Route found in openshift-console namespace: console
DEBUG Still waiting for the cluster to initialize: Cluster operators authentication, console, ingress, kube-apiserver,
machine-api, monitoring, openshift-samples are not available
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DEBUG Route found in openshift-console namespace: console
DEBUG Still waiting for the cluster to initialize: Cluster operators authentication, console, ingress, machine-api,
monitoring are not available
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DEBUG Still waiting for the cluster to initialize: Cluster operators authentication, console, ingress, monitoring are not
available
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DEBUG Route found in openshift-console namespace: console
DEBUG Route found in openshift-console namespace: console
DEBUG Route found in openshift-console namespace: console
DEBUG Route found in openshift-console namespace: console
DEBUG Route found in openshift-console namespace: console
DEBUG OpenShift console route is admitted
DEBUG Still waiting for the cluster to initialize: Cluster operators authentication, console, monitoring are not
available
DEBUG Still waiting for the cluster to initialize: Cluster operators authentication, monitoring are not available
DEBUG Still waiting for the cluster to initialize: Cluster operator authentication is not available
INFO Cluster is installed
INFO Install complete!
INFO To access the cluster as the system:admin user when using 'oc', run
INFO     export KUBECONFIG=/root/ocp-agent-install/ocp-install-dir/auth/kubeconfig
INFO Access the OpenShift web-console here:
https://console-openshift-console.apps.cdp.rdu2.scalelab.redhat.com
INFO Login to the console with user: "[your_openshift_username]", and password:
"[your_openshift_password]"
```

NTP

Since we didn't use the OpenShift Assisted Installer, we will need to configure the NTP server that will be used. Until then, there may be NodeClockNotSynchronising messages logged.

Create the following .bu file and use the butane utility to convert it to the format used for the MachineConfig.

vi 99-worker-chrony.bu

```
variant: openshift
version: 4.16.0
metadata:
  name: 99-worker-chrony
  labels:
    machineconfiguration.openshift.io/role: worker
storage:
  files:
  - path: /etc/chrony.conf
    mode: 0644
```

```
    overwrite: true
    contents:
      inline: |
        pool foreman.rdu2.scalelab.redhat.com iburst
        driftfile /var/lib/chrony/drift
        makestep 1.0 3
        rtcsync
        logdir /var/log/chrony
```

./butane 99-worker-chrony.bu -o 99-worker-chrony.yaml

cat 99-worker-chrony.yaml


```
# Generated by Butane; do not edit
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: worker
  name: 99-worker-chrony
spec:
  config:
    ignition:
      version: 3.4.0
    storage:
      files:
        - contents:
            compression: ""
            source:
data:,pool%20foreman.rdu2.scalelab.redhat.com%20iburst%20%0Adriftfile%20%2Fva
r%2Flib%2Fchrony%2Fdrift%0Amakestep%201.0%203%0Artcsync%0Alogdir%20%2Fvar%2Fl
og%2Fchrony%0A
          mode: 420
          overwrite: true
          path: /etc/chrony.conf
```


Apply the MachineConfig using the following command:
oc create -f 99-worker-chrony.yaml

This will initiate a rolling reboot to update each worker node in the cluster. You can use the following commands to view the status of the nodes during the update:

```
oc get mcp
oc get nodes

oc describe machineconfig 99-worker-chrony
```

vi 99-master-chrony.bu

```
variant: openshift
version: 4.16.0
metadata:
  name: 99-master-chrony
  labels:
    machineconfiguration.openshift.io/role: master
storage:
  files:
  - path: /etc/chrony.conf
    mode: 0644
    overwrite: true
    contents:
      inline: |
        pool foreman.rdu2.scalelab.redhat.com iburst
        driftfile /var/lib/chrony/drift
        makestep 1.0 3
        rtcsync
        logdir /var/log/chrony
```

./butane 99-worker-chrony.bu -o 99-worker-chrony.yaml

cat 99-worker-chrony.yaml

```
# Generated by Butane; do not edit
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: master
  name: 99-master-chrony
spec:
  config:
    ignition:
      version: 3.4.0
    storage:
      files:
        - contents:
            compression: ""
            source:
data:,pool%20foreman.rdu2.scalelab.redhat.com%20iburst%20%0Adriftfile%20%2Fva
r%2Flib%2Fchrony%2Fdrift%0Amakestep%201.0%203%0Artcsync%0Alogdir%20%2Fvar%2Fl
og%2Fchrony%0A
          mode: 420
          overwrite: true
          path: /etc/chrony.conf
```

```
oc create -f 99-master-chrony.yaml
```

This will initiate a rolling reboot to update each master node in the cluster. You can use the following commands to view the status of the nodes during the update:

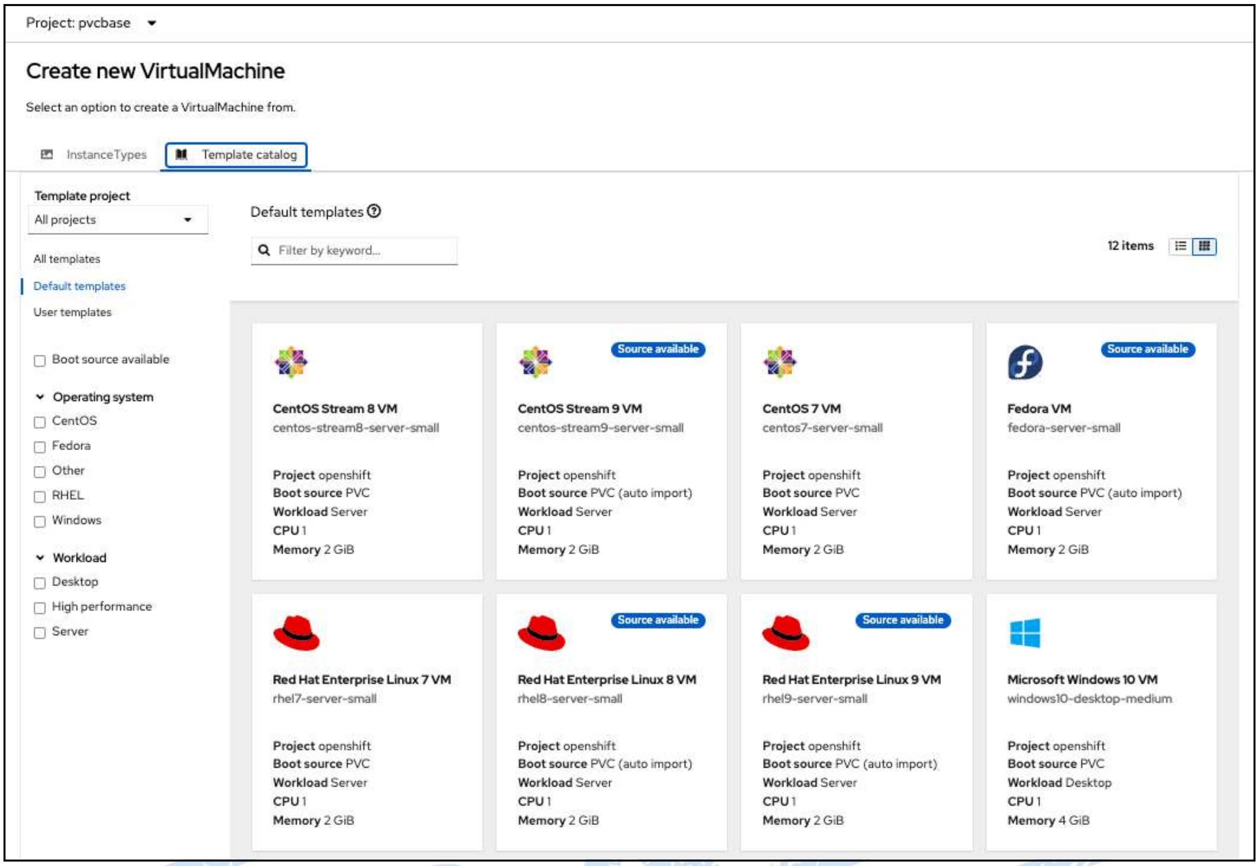
```
oc get mcp
oc get nodes

oc describe machineconfig 99-master-chrony
```

--

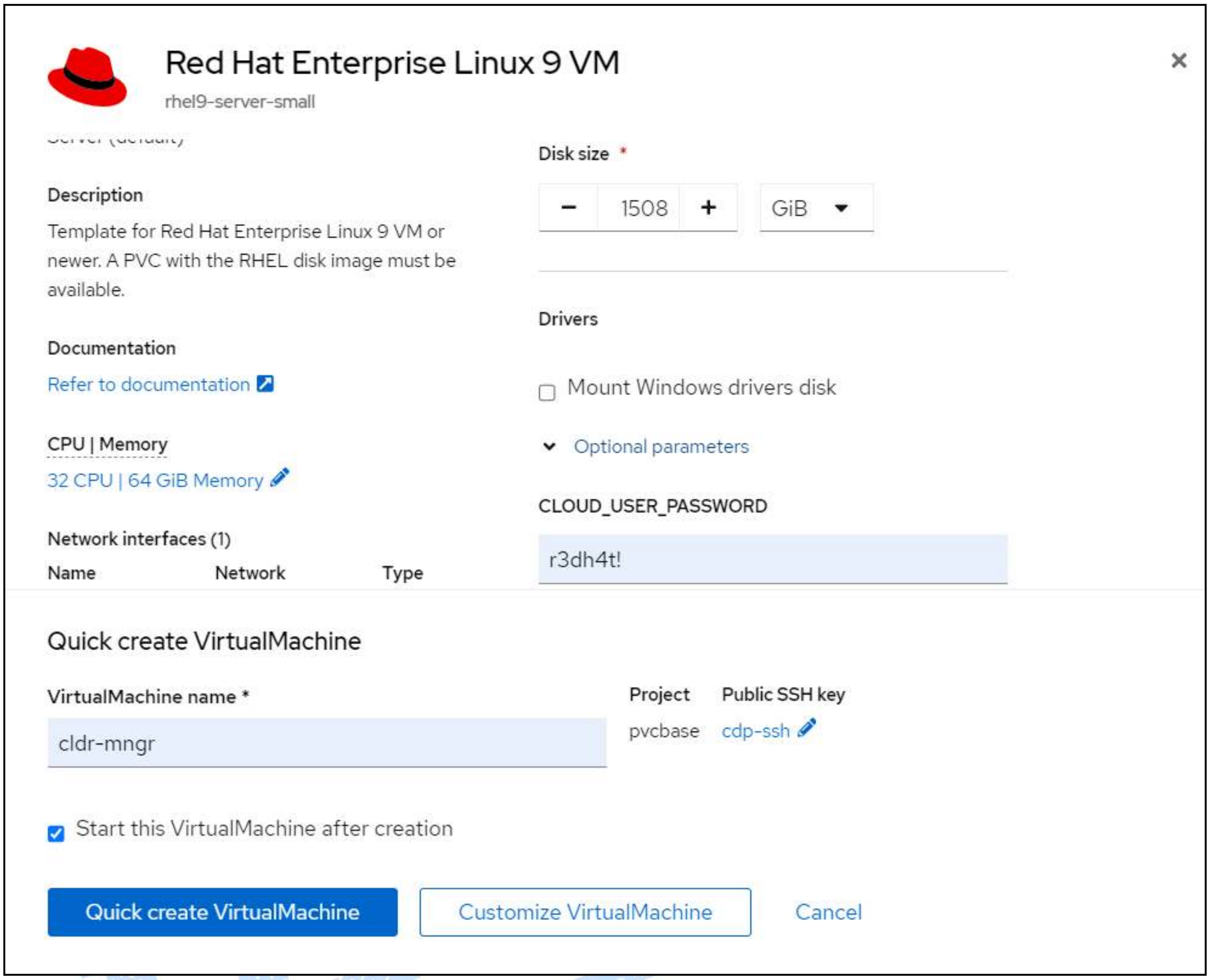
ALTERNATIVE CONFIGURATION STEPS

OPTIONAL Method to create VMs

STEP 1. Select **Virtualization > Catalog > Template Catalog > Red Hat Enterprise Linux 9 VM**



STEP 2. Edit the following parameters: **VirtualMachine name, CPU | Memory, Disk size, Optional Parameters-CLOUD_USER_PASSWORD:**



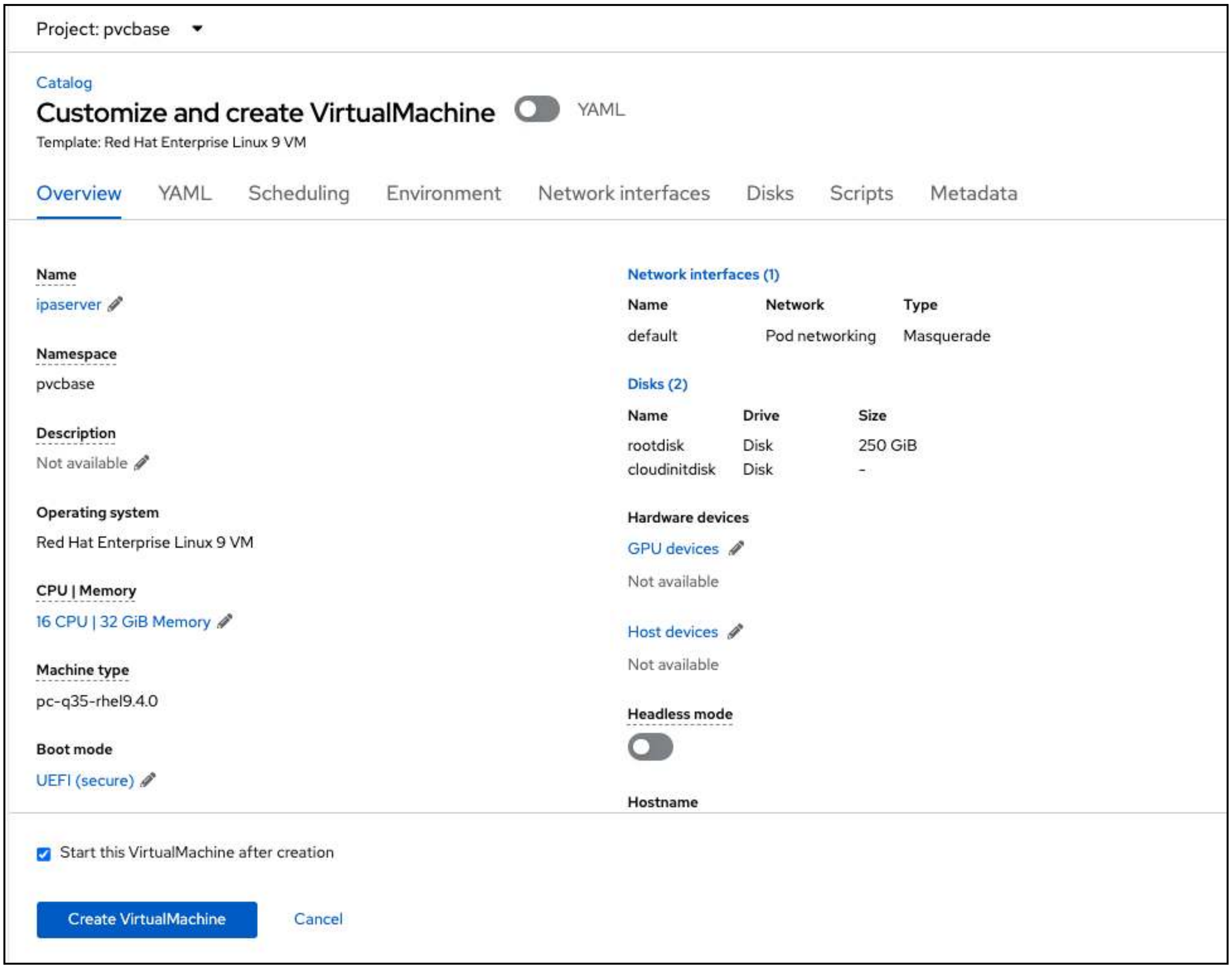
VM configuration parameters:

Parameter	Setting/Value
VirtualMachine name	cldr-mngr.redhat.local
Project name	pvc-base
Operating System	Red Hat Enterprise Linux 9 VM
CPU	32 CPU
Memory	64 GiB
Storage:	
Disk source	Blank
Disk Size	1500 GiB

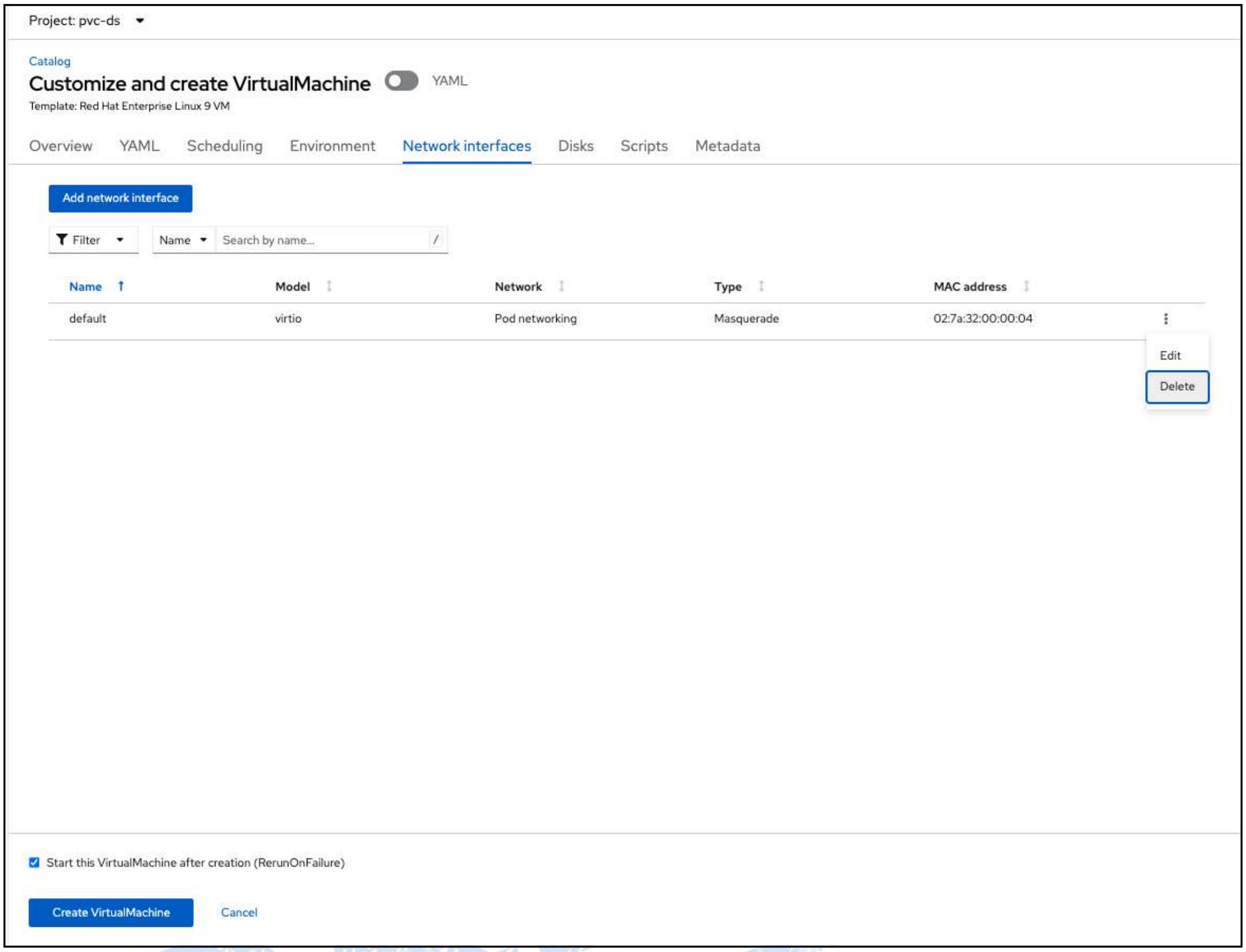
CLOUD_USER_PASSWORD	r3dh4t!

STEP 3. Select **Customize Virtualmachine.**

IMPORTANT: From the screen that follows, make sure that you edit several settings ***before*** you select **Create Virtual Machine.**

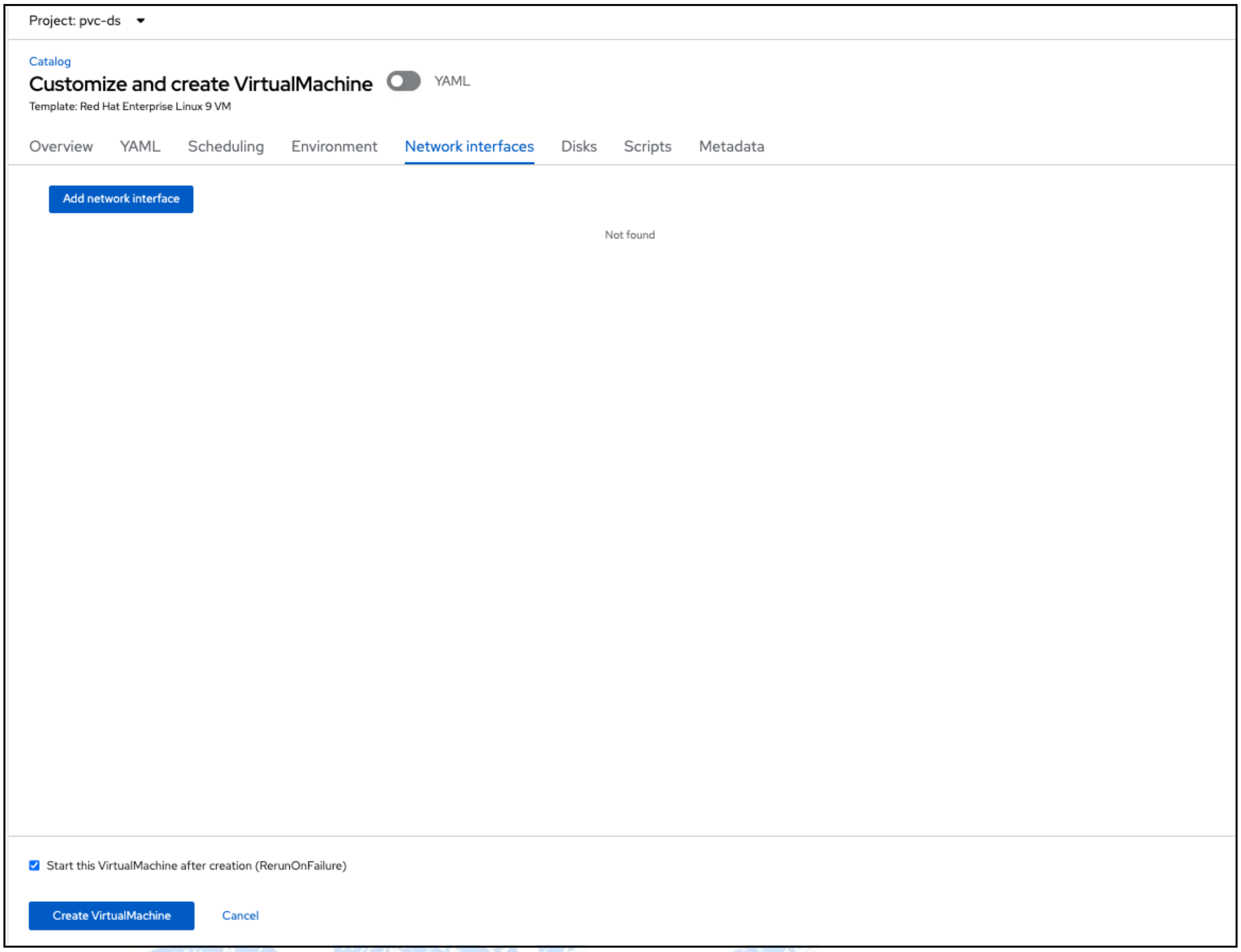


STEP 4. From the “***Network Interfaces***” screen, click on the three vertical dots to the far right of the “***default***” network interface, and click on “***Delete***”.



STEP 5. Then, confirm by clicking on “Delete”.

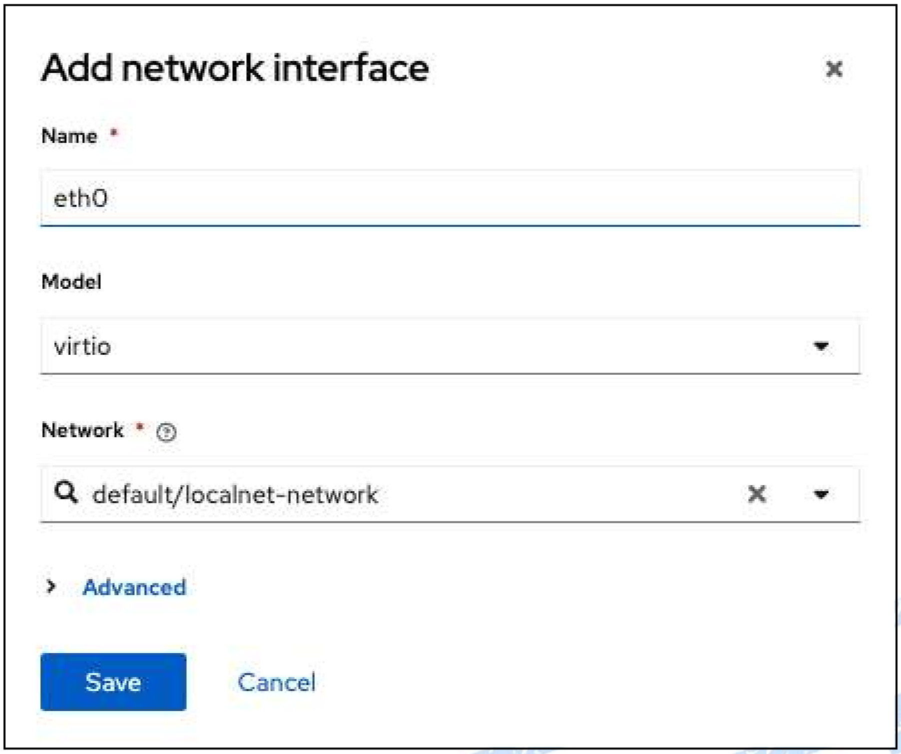
STEP 6. From the ***“Network Interfaces”*** screen, click on ***“Add network interface”***.



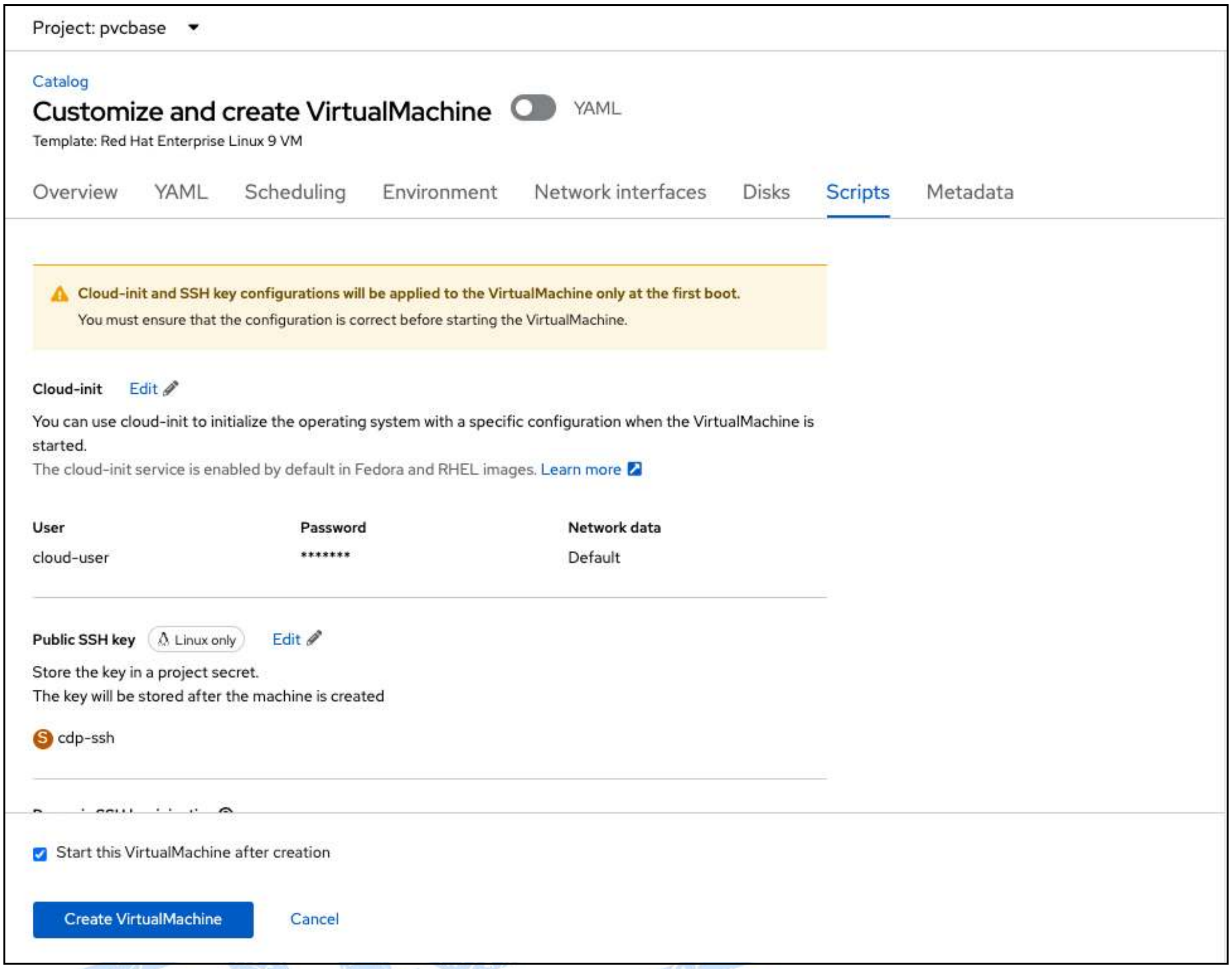
STEP 7. From the “Add network interface” screen, enter the following parameters and then click on “**Save**”.

Network interface configuration parameters:

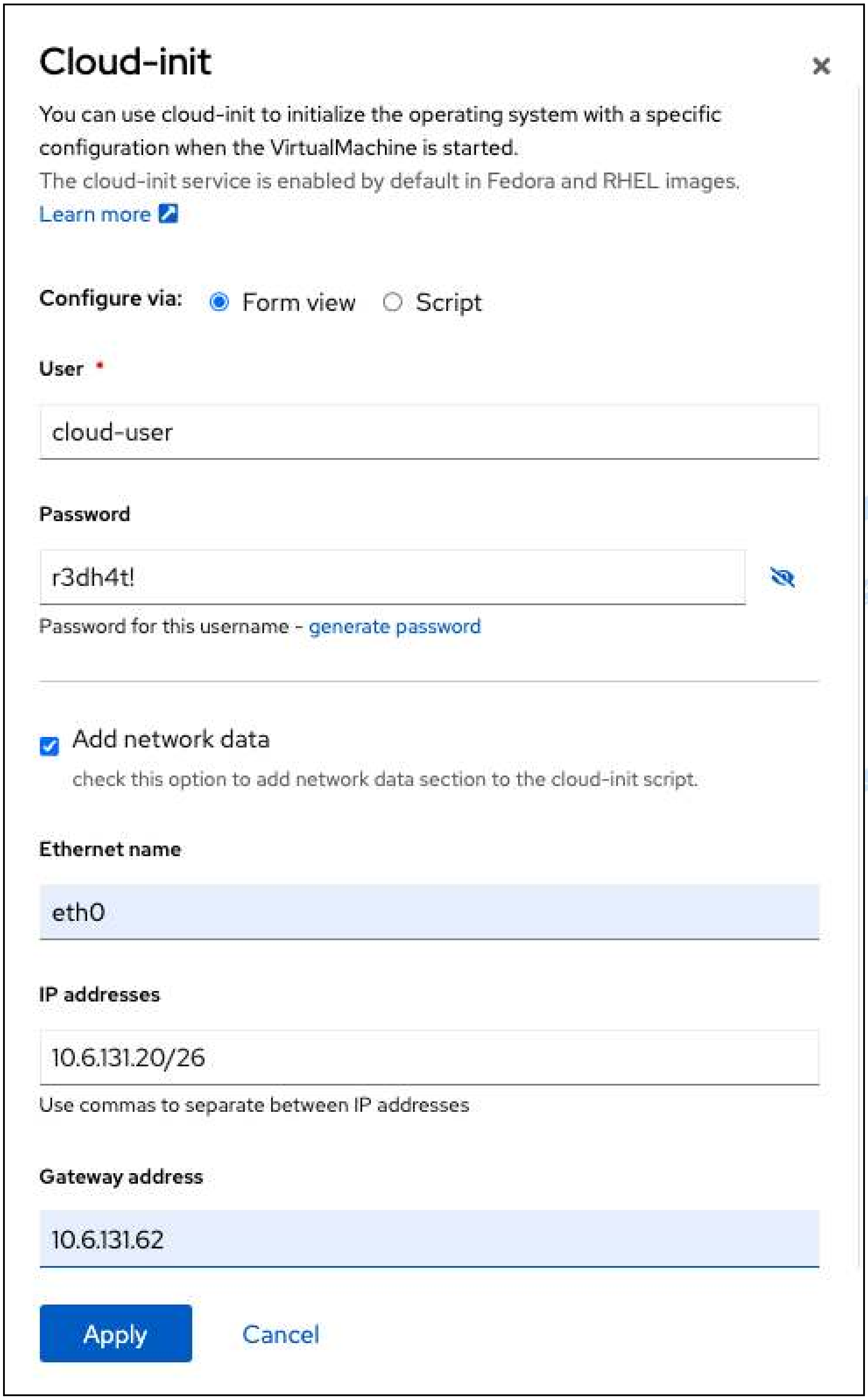
Parameter	Setting/Value
Network interface name	eth0
Model	virtio
Network	default/localnet-network



STEP 8. Next, select the **Scripts** tab**; and** click on **Edit** in the **Cloud-init** section:

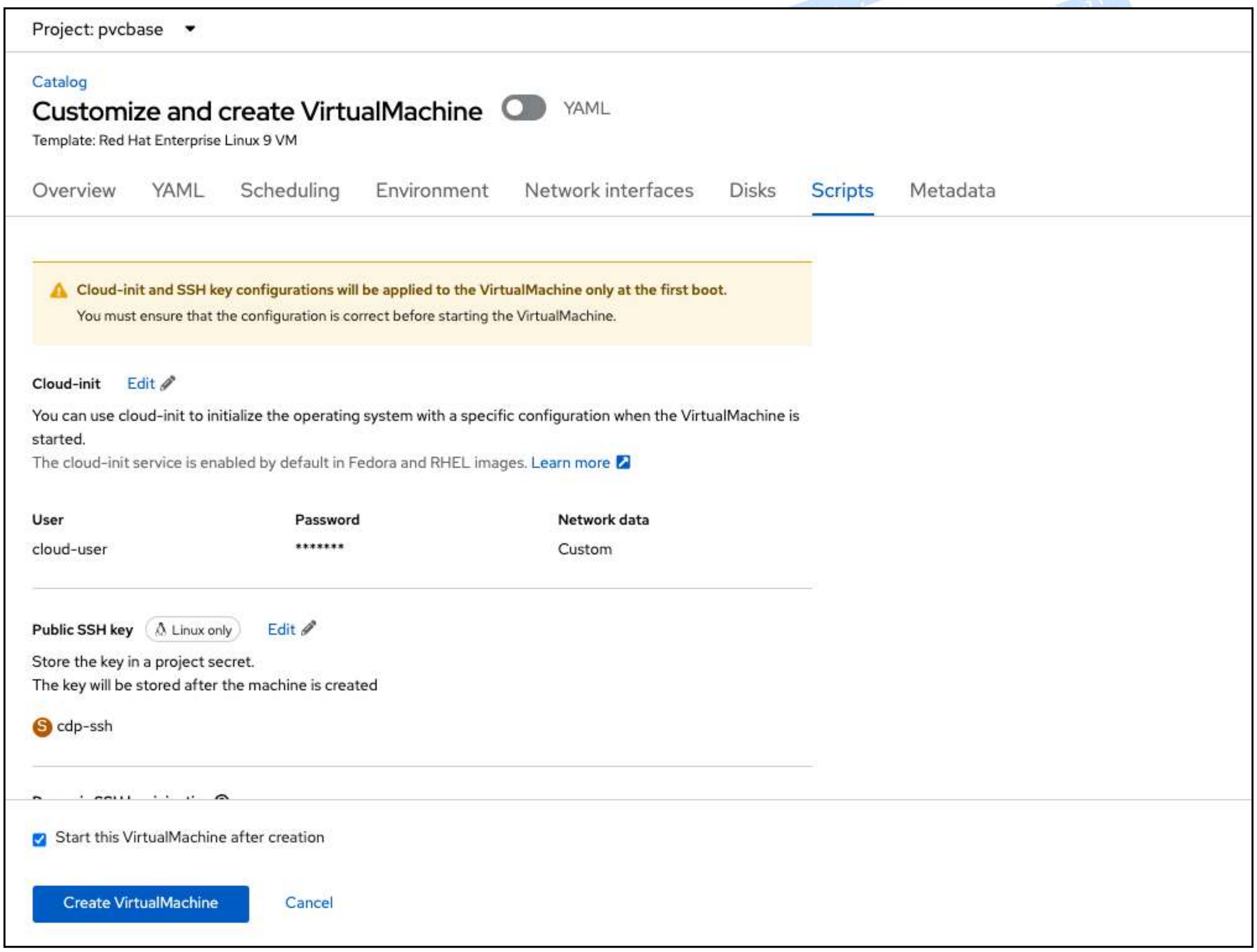


STEP 9. From the **Cloud-init** menu, select the checkbox next to **Add network data,** then enter the parameters as shown in the following screen for the **Ethernet name** and **Gateway address.**
Refer to the table provided at the beginning of this activity for the **IP address** to use for each VM. Select **Apply** when done.



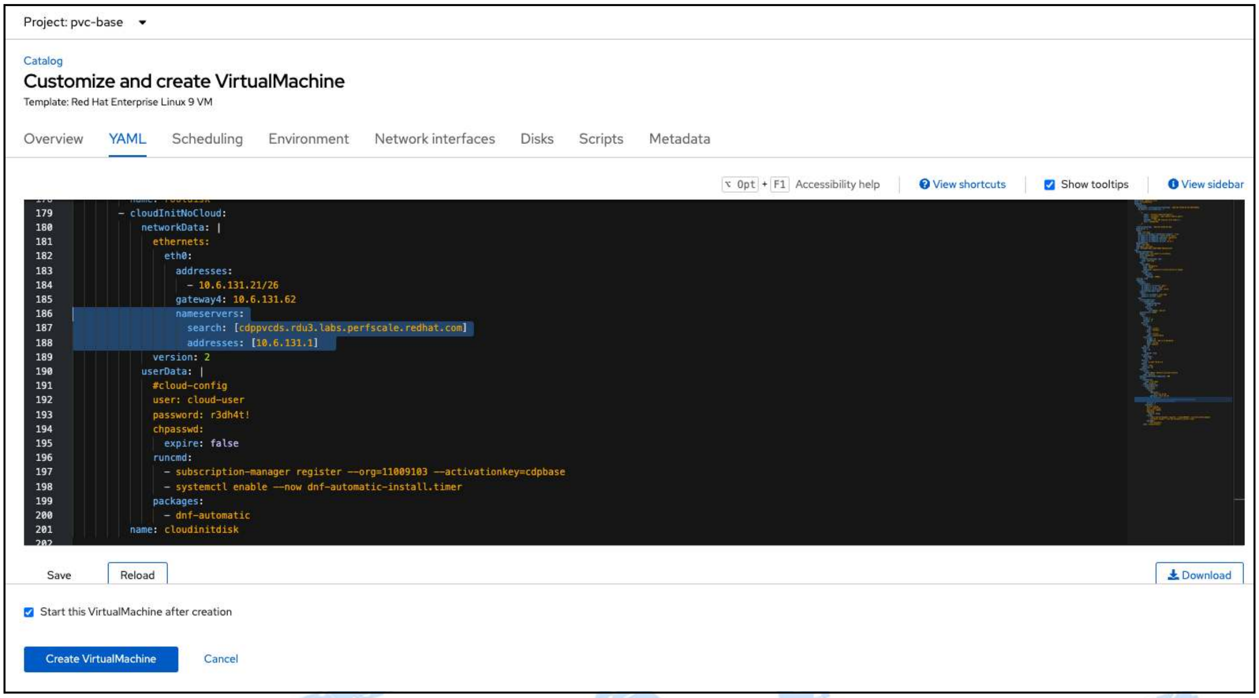
Use the following table as a reference to create the VirtualMachines for CDP Base:

Hostname	CPU	RAM	Disk	MAC Address	IP Address
cldr-mngr.redhat.local	32	64G	1.5T	02:7c:71:05:00:01	192.168.2.190/24
pvcbase-master.redhat.local	32	64G	1T	02:7c:71:05:00:02	192.168.2.191/24
pvcbase-worker1.redhat.local	32	64G	1T	02:7c:71:05:00:03	192.168.2.192/24
pvcbase-worker2.redhat.local	32	64G	1T	02:7c:71:05:00:04	192.168.2.193/24
pvcbase-worker3.redhat.local	32	64G	1T	02:7c:71:05:00:05	192.168.2.194/24
pvcbase-worker4.redhat.local	32	64G	1T	02:7c:71:05:00:06	192.168.2.195/24
pvcbase-worker5.redhat.local	32	64G	1T	02:7c:71:05:00:07	192.168.2.196/24



STEP 10. Select the **YAML** tab to edit the YAML definition for the VM. Add the highlighted ***nameservers*** section for DNS servers and search domains to match the provided section,

IMPORTANT: Make sure you click on ***Save***, before you click on ***Create VirtualMachine***. Otherwise, the changes made to the YAML will NOT be saved.



Edit the YAML to include the following, under the **spec.template.spec.volumes.cloudInitNoCloud** section for the DNS servers and search domains:

```
spec:
  template:
    spec:
      volumes:
        - dataVolume:
            name: cldr-mngr
          name: rootdisk
        - cloudInitNoCloud:
            networkData: |
              ethernets:
                eth0:
                  addresses:
                    - 192.168.2.190/26
                  gateway4: 192.168.2.1
                  nameservers:
                    search: [redhat.local]
                    addresses: [192.168.1.210]
              version: 2
            userData: |
              #cloud-config
              user: cloud-user
```

```
              password: r3dh4t!
              chpasswd:
                expire: false
              runcmd:
                - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
                - systemctl enable --now dnf-automatic-install.timer
              packages:
                - dnf-automatic
          name: cloudinitdisk
```

NOTE: In case the VM does not register with Red Hat Insights, you can use the following command to manually register the VM: rhc connect

Example:
rhc connect --activation-key [your_activation_key] --organization [your_org_#]

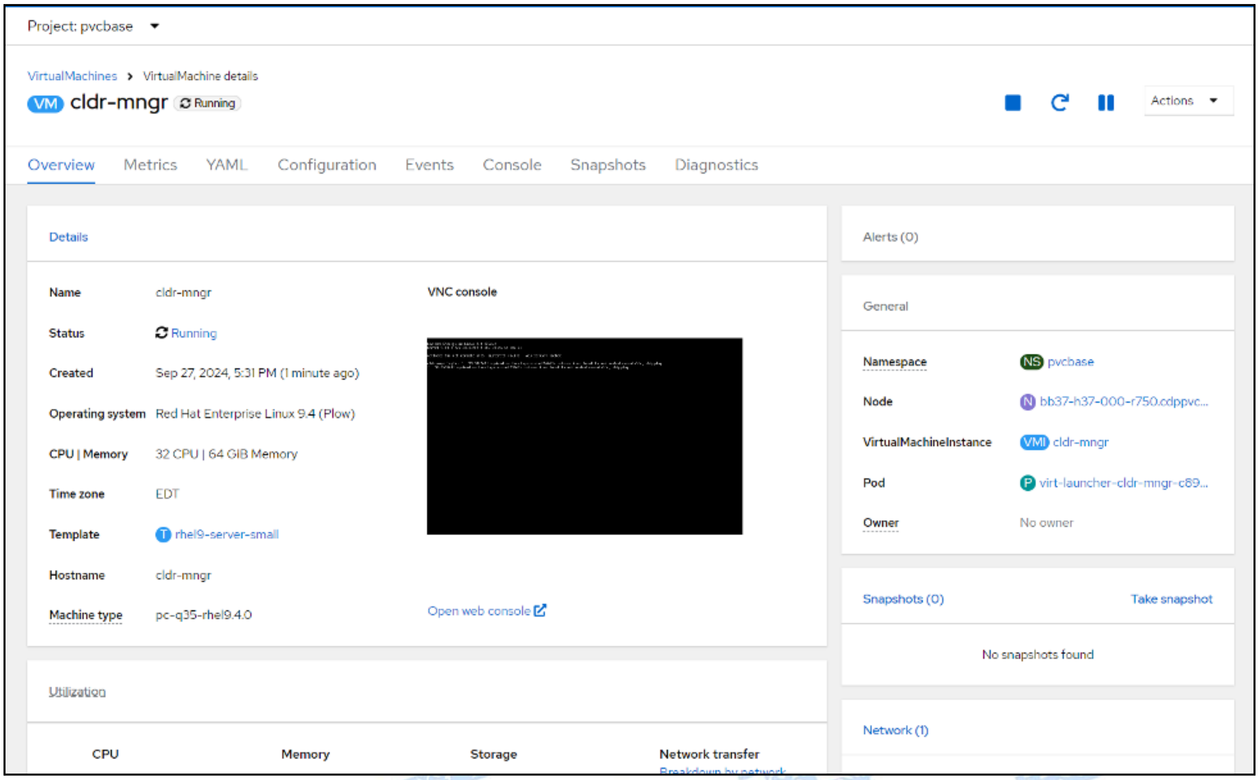
STEP 11. When you have finished editing the VM, select the ***Create VirtualMachine*** button.

IMPORTANT: When editing the YAML, make sure you click on the ***Save*** button before you click on ***Create VirtualMachine***. Otherwise, the changes made in the YAML editor will NOT be saved.

IMPORTANT: Some of the parameters, such as CloudInit only take effect after the ***first*** boot of the virtual machine.

If there are settings that still need to be made before this, then make sure to unselect the option to ***'Start this VirtualMachine after creation'***.

STEP 12. From the **Overview** tab, you can view the VM provisioning status.



ADD NODES TO THE CLUSTER

[6.4.3. Adding nodes to your cluster]

vi nodes-config.yaml file

```
hosts:
- hostname: f36-h03-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:79:19:f1 # Change for each host
  networkConfig:
    interfaces:
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:79:19:f1 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.48.113 # Change for each host
              prefix-length: 23
          dhcp: false
```

```yaml
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: eno2np1
          table-id: 254
- hostname: f36-h05-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:7c:f6:31 # Change for each host
  networkConfig:
    interfaces:
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:7c:f6:31 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.48.114 # Change for each host
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: eno2np1
          table-id: 254
- hostname: f36-h07-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:79:2c:11 # Change for each host
  networkConfig:
    interfaces:
```

```
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:79:2c:11 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.48.115 # Change for each host
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: eno2np1
          table-id: 254
- hostname: f36-h09-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:78:e8:61 # Change for each host
  networkConfig:
    interfaces:
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:78:e8:61 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.48.116 # Change for each host
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
```

```
        next-hop-interface: eno2np1
        table-id: 254
- hostname: f36-h10-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:78:e7:a1 # Change for each host
  networkConfig:
    interfaces:
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:78:e7:a1 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.48.117 # Change for each host
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: eno2np1
          table-id: 254
- hostname: f36-h11-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:69:92:31 # Change for each host
  networkConfig:
    interfaces:
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:69:92:31 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.48.118 # Change for each host
```



```
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: eno2np1
          table-id: 254
```

```
[root@ipaserver]# oc adm node-image create nodes-config.yaml

[root@ipaserver]# ll *.iso
-rw-r--r--. 1 root root 1326184448 Nov  4 18:01 node.x86_64.iso

cp node.x86_64.iso /var/www/html/.
```

Verify that the discovery image is available at:
http://10.1.49.1/node.x86_64.iso

BADFISH

Configure the servers to boot from the Discovery image:

```
export USER=quads
export PASS=rdu2@4618

dnf install -y podman

vi workerlist
10.1.64.196
10.1.65.226
10.1.65.96
10.1.65.139
10.1.64.17
10.1.65.3
10.1.64.139
10.1.65.220
```

```
10.1.65.210

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-off;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --unmount-virtual-media;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --mount-virtual-media
http://10.1.49.1/node.x86_64.iso;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --boot-to-virtual-media;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-on;done

for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-state;done

- - -
```

NodeNetworkConfigurationPolicy

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h03-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h03-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---
```

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h05-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h05-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h07-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h07-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h09-000-r640
```

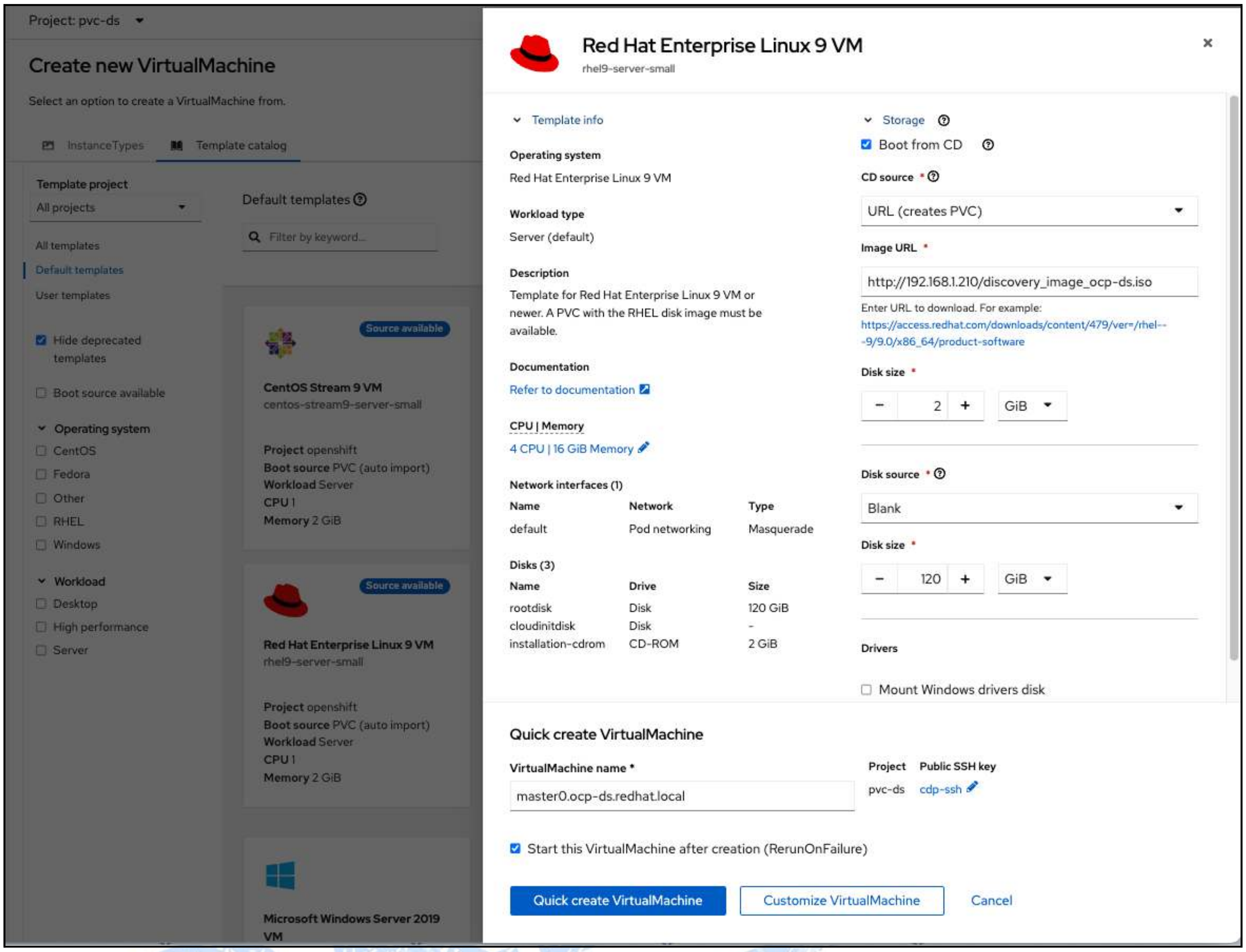
```
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h09-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h10-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h10-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h11-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
```

```
    kubernetes.io/hostname: f36-h11-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h13-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h13-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h14-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h14-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
```

```
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h15-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h15-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---
```

OPTIONAL METHOD TO CREATE VM YAML

STEP 6. Switch to the OpenShift console, and navigate to Virtualization > Catalog > Create a VM > from Catalog > RHEL 9. Enter the VM configuration parameters from the table provided below, and then select ***"Customize VirtualMachine"*** to further customize the VM.

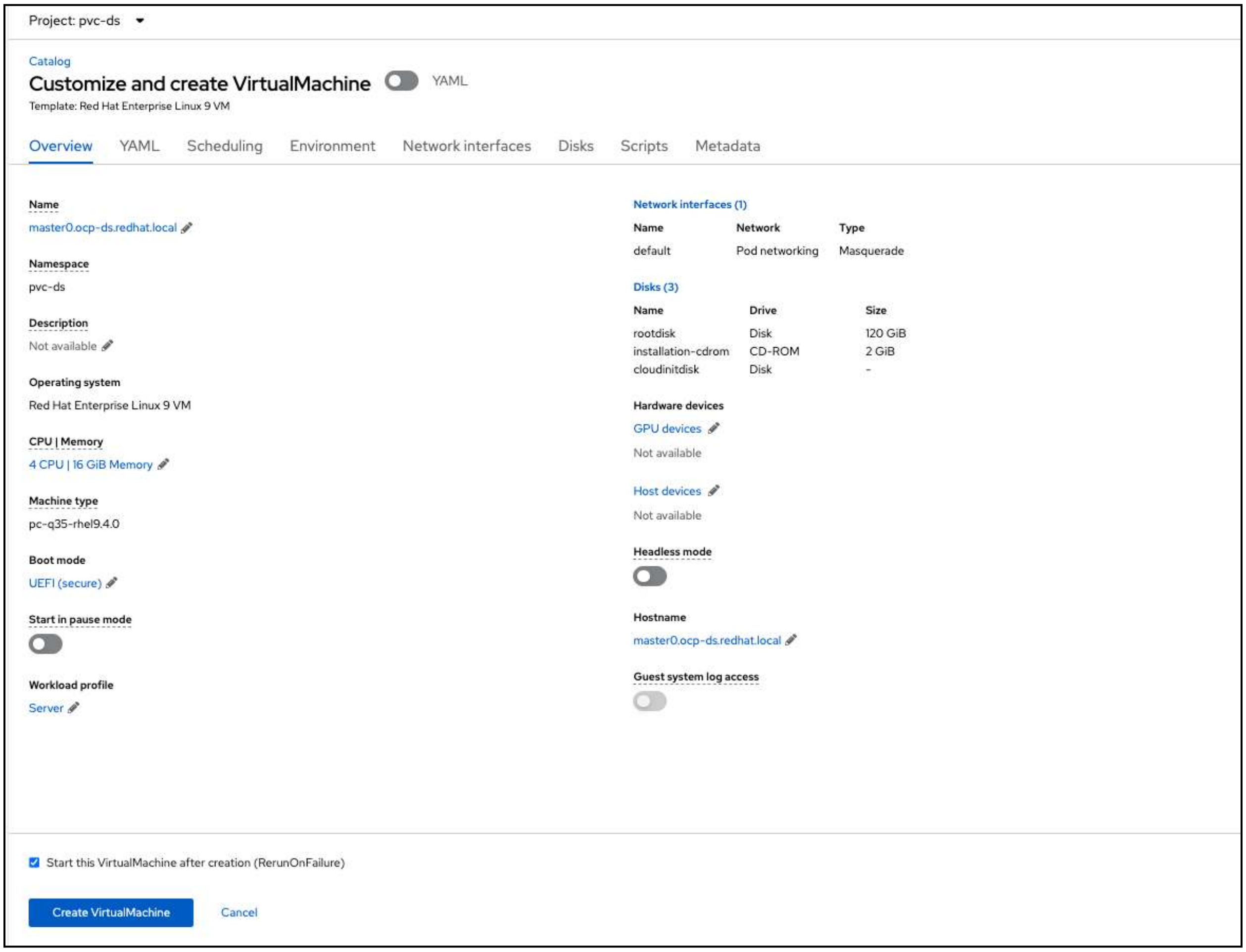


VM configuration parameters:

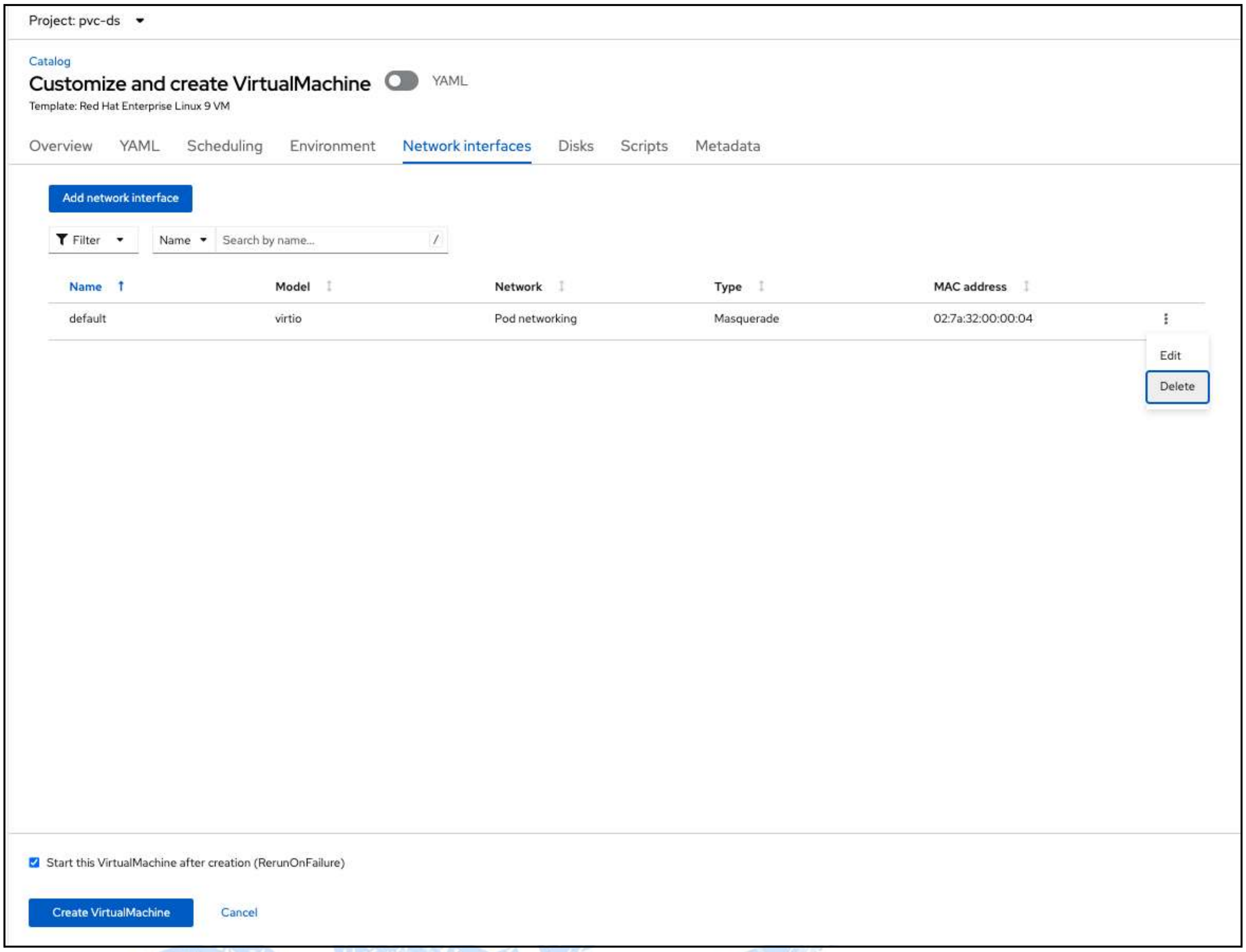
Parameter	Setting/Value
VirtualMachine name	master0.cdp.rdu2.scalelab.redhat.com
Project name	cdp-ds
Operating System	Red Hat Enterprise Linux 9 VM
CPU	4 CPU
Memory	16 GiB
Storage:	
Boot from CD	Selected (enabled)
CD source	URL (creates PVC)
Image URL	http://10.1.49.1/discovery_image_cdpds.iso

Disk Size	2 GiB
Disk source	Blank
Disk Size	120 GiB

STEP 7. From the Customize and create VirtualMachine screen, select the ***"Network Interfaces"*** tab.

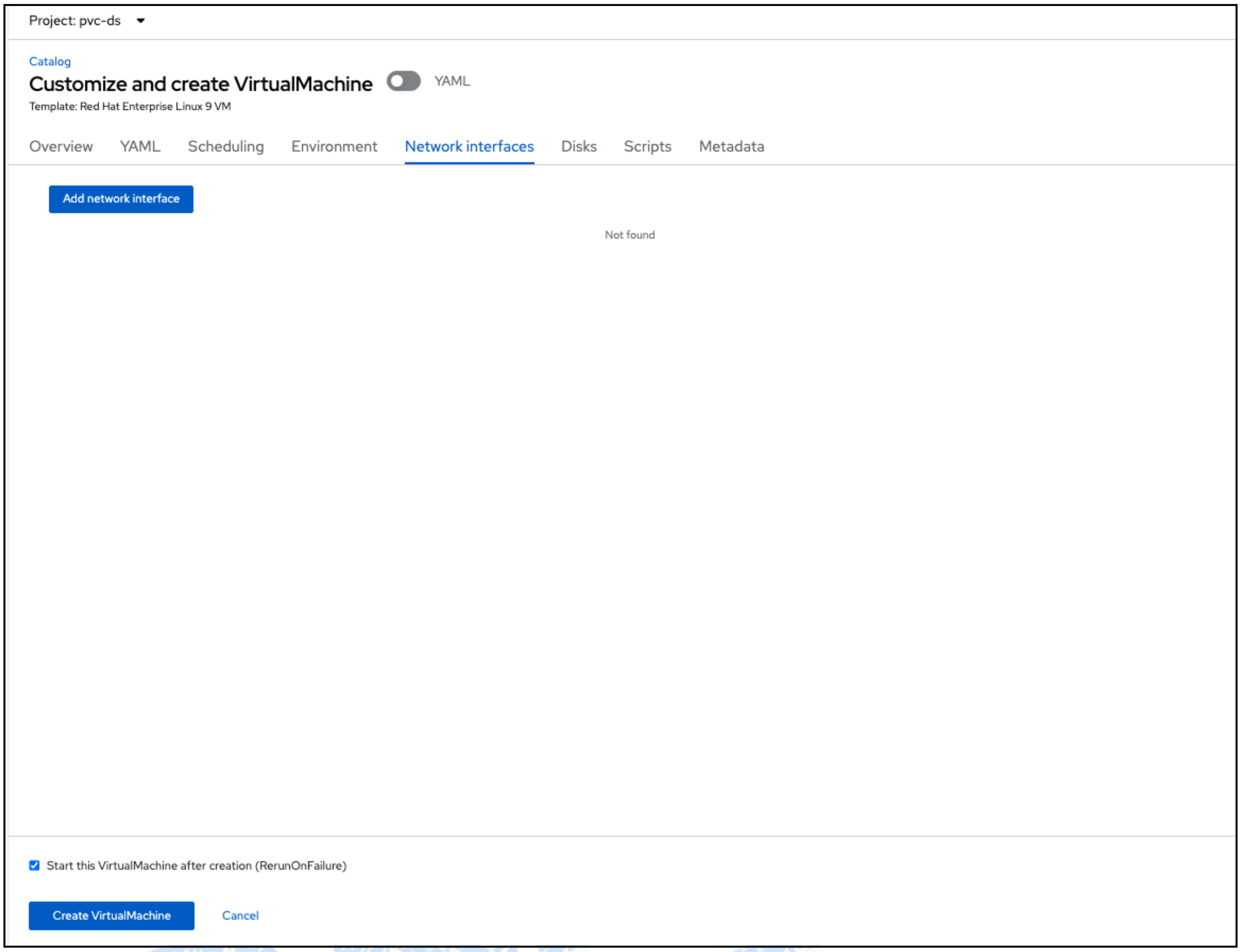


STEP 8. From the ***"Network Interfaces"*** screen, click on the three vertical dots to the far right of the ***"default"*** network interface, and click on ***"Delete"***.



STEP 9. Then, confirm by clicking on "Delete".

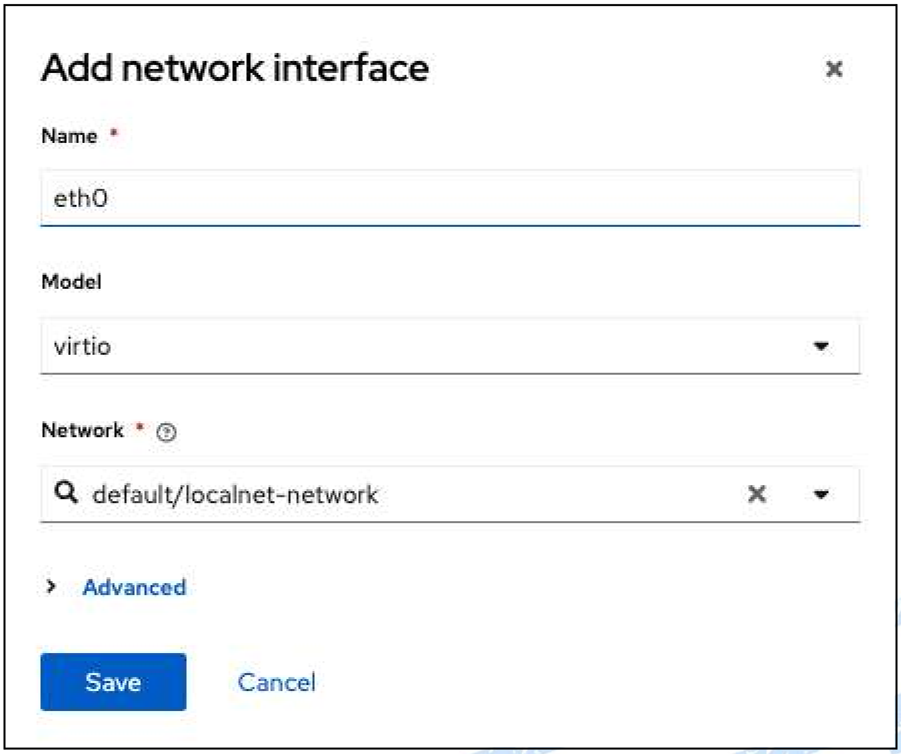
STEP 10. From the "***Network Interfaces***" screen, click on "***Add network interface***".



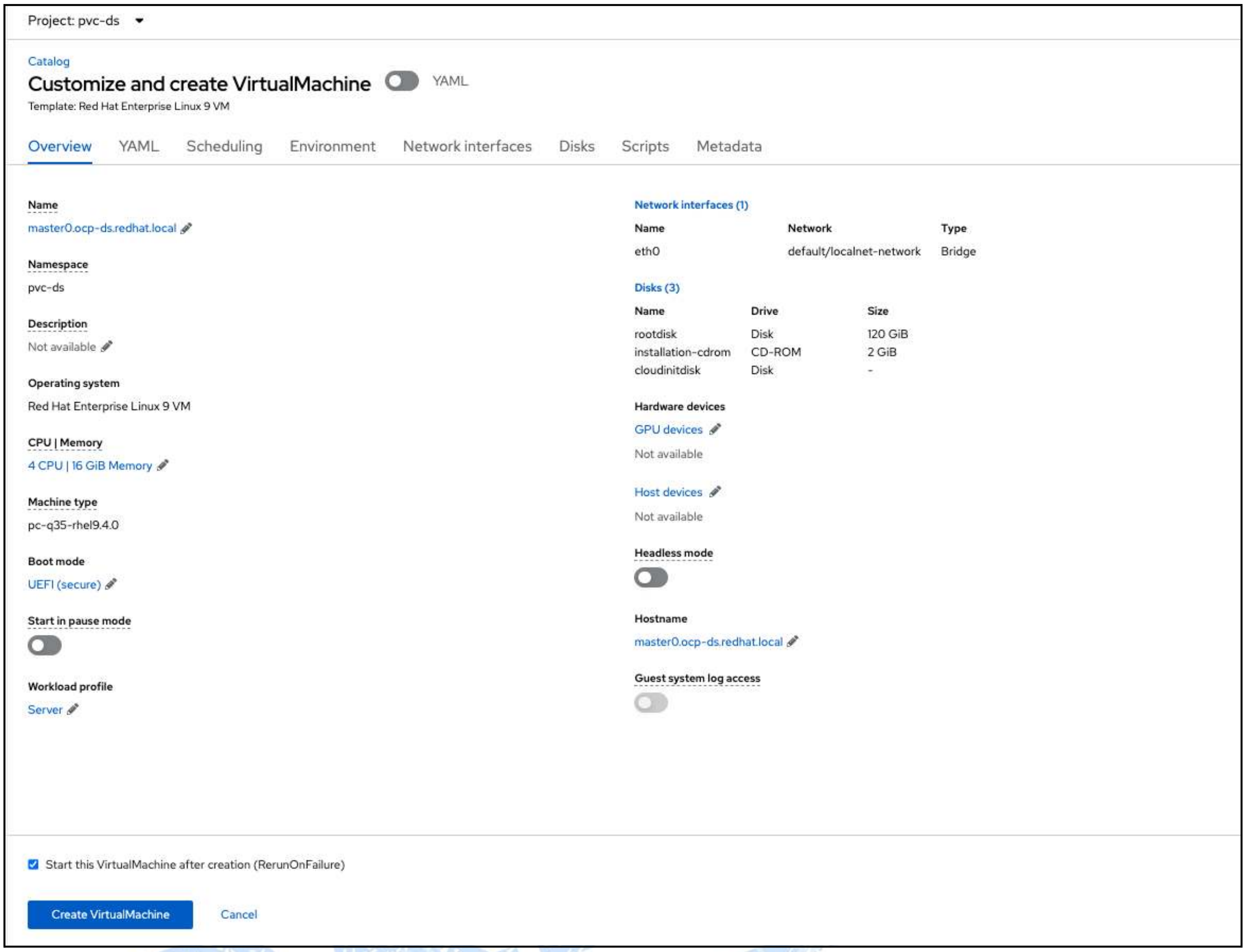
STEP 11. From the “Add network interface” screen, enter the following parameters and then click on “**Save**”.

Network interface configuration parameters:

Parameter	Setting/Value
Network interface name	eth0
Model	virtio
Network	default/localnet-network



STEP 12. Select the “Overview” tab, and confirm the configuration settings. Then click on “**Create VirtualMachine**”.


Project: pvc-ds
Catalog
Customize and create VirtualMachine
YAML
Template: Red Hat Enterprise Linux 9 VM
Overview
YAML
Scheduling
Environment
Network interfaces
Disks
Scripts
Metadata
Name
master0.ocp-ds.redhat.local
Namespace
pvc-ds
Description
Not available
Operating system
Red Hat Enterprise Linux 9 VM
CPU | Memory
4 CPU | 16 GiB Memory
Machine type
pc-q35-rhel9.4.0
Boot mode
UEFI (secure)
Start in pause mode
Workload profile
Server
Network interfaces (1)
Name
Network
Type
eth0
default/localnet-network
Bridge
Disks (3)
Name
Drive
Size
rootdisk
Disk
120 GiB
installation-cdrom
CD-ROM
2 GiB
cloudinitdisk
Disk
-
Hardware devices
GPU devices
Not available
Host devices
Not available
Headless mode
Hostname
master0.ocp-ds.redhat.local
Guest system log access
Start this VirtualMachine after creation (RerunOnFailure)
Create VirtualMachine
Cancel